

# WHEN A BLANK IS NOT A CONTROL: ABLATION CHOICE, GENERATION BUDGET, AND THE ATTRIBUTION OF GROUNDING GAINS

**Alexander Mueller**

Department of Computer Sciences  
University of Wisconsin–Madison  
Madison, WI 53706, USA  
amueller.mco@gmail.com

**Jae-Ho Lee**

Visual Intelligence and Graphics Lab  
Korea University  
Seoul, South Korea

**Gyeong-Moon Park**

Visual Intelligence and Graphics Lab  
Korea University  
Seoul, South Korea

## ABSTRACT

Object-hallucination benchmarks read a better score as better grounding, and the standard check removes the image to see whether a gain survives. We show that the answer depends on what replaces the image. On standard benchmarks a blank cannot run the check: given a grey image, LLAVA-1.5-7B names no objects in CHAIR captions and answers “no” to every POPE question. A mismatched image can, and under it a published method’s gain is tied to the image on both benchmarks, though mostly as a threshold shift. To see when the choice matters, we prefill the model’s description with captions of other images from one scene or five. A present-minus-absent score rises by 0.49 although only the text changed. A grey field leaves the whole rise intact (1.03 times its sighted size); a mismatched image leaves 0.67, both as registered (0.68 on a post-hoc extension to 3,500 images), so about a third needs the correct image. Scoring only generations that end before the token budget lowers the surviving share to 0.47. On QWEN3-VL-8B the two ablations agree. Registered versions with natural prose or a yes/no question show no rise. We recommend reporting both error rates and choosing the ablation deliberately.

## 1 INTRODUCTION

When a vision-language model scores better on an object-hallucination benchmark, the improvement is usually read as better visual grounding. Training-free mitigation methods report better scores as reduced hallucination (Liu et al., 2024b; Li et al., 2025b; Zhong et al., 2026; Xu et al., 2026; Leng et al., 2024), some explicitly as better grounding (Lei et al., 2026; Zhang et al., 2026). Yet the scores also move with caption length and generation budget (Li et al., 2023; Xu et al., 2026), and lower CHAIR can accompany less informative output (Fazli et al., 2026). The natural check is to remove the image and ask whether a gain survives: a gain that survives did not come from looking. This paper shows that the answer to that check depends on what replaces the image.

The check needs the model to keep responding about the scored objects once the image is gone. POPE (Li et al., 2023) names each candidate object in a question, so its hit rate  $H$  on present objects and false-alarm rate  $F$  on absent ones stay measurable whatever the model sees. CHAIR (Rohrbach et al., 2018) scores only the objects the model chooses to mention; its instance score is a false-discovery rate (Kaul et al., 2024) whose denominator the model chooses. A fixed candidate list is necessary but not sufficient: a model that names nothing sets both rates to zero, one that affirms

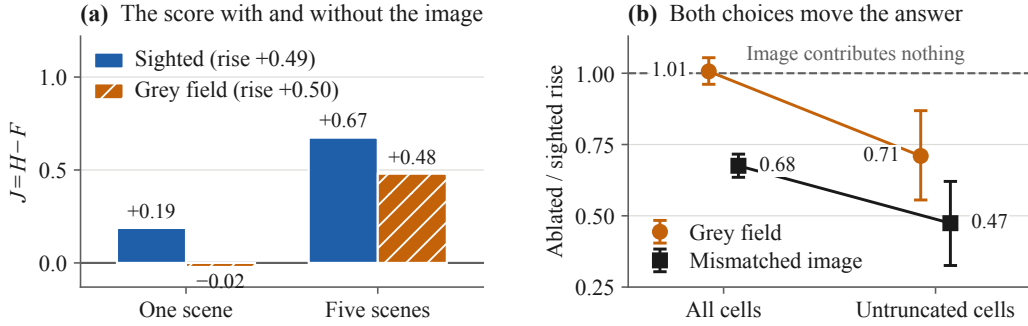


Figure 1: **(a)**  $J = H - F$  on LLAVA-1.5-7B with the target image and with a uniform grey field in its place, on the registered 500 images; every token of text is identical across each pair. The image raises  $J$  by about 0.2 in each condition and by nearly the same amount in both, so against *this* ablation the one-to-five-scene rise does not need it. **(b)** The ablated rise as a fraction of the sighted one on 3,500 images (post hoc), for two ablations, on all 10,500 cells and on the 2,089 cells whose sighted continuations both end before the budget; 95% intervals clustered on the image. A ratio of 1 means the image contributes nothing. The all-cells mismatched ratio replicates the registered 0.67; the untruncated cells are a selected subset, so that column is secondary (Section 4.4).

everything sets both to one, and either corner sends  $J = H - F$  to zero. On standard benchmarks the usual blank lands in exactly these corners. Given a grey image, LLAVA-1.5-7B names none of CHAIR’s 80 categories in 500 captions, so  $\text{CHAIR}_i$  is 0/0, and it answers “no” to all 9,000 POPE questions — correctly, so both rates are zero by construction. A mismatched real image keeps the model answering, and under it the gain of a published method, PAI (Liu et al., 2024b), is tied to the image on both benchmarks while decomposing mainly into a threshold shift (Section 3). None of this depends on a special stimulus.

What standard benchmarks cannot show is when the two ablations *disagree*, because no naturally occurring gain comes with a known cause. We therefore build one. We ask LLAVA-1.5-7B to describe a COCO image and prefill the start of its answer with a short passage cut from human captions of *other* images, varying only whether those captions describe one other scene or five (Figure 3). The endpoint is how often the continuation repeats an object the passage names, split by whether the object is in the image ( $H$ ) or not ( $F$ ). Only the text differs between the conditions, yet the score rises by 0.49 on its  $[-1, 1]$  scale (0.36 with continuation lengths matched), mainly because  $F$  falls from 0.49 to 0.11: the profile a benchmark reader would take as improved grounding.

Does that rise need the image? Our registered grey-field control found it intact, at 1.03 times its sighted size (Figure 1a). A registered mismatched-image control on the same data leaves 0.67 of it, and a post-hoc extension to 3,500 images gives 0.68: about a third of the rise depends on the image being the right one (Figure 1b). The cause of the rise is textual; how much of its effect runs through the image depends on the control, because a grey field removes the model’s ability to answer about any scene along with the evidence. Restricting to generations that end before the token budget lowers the surviving share to 0.47, a secondary and post-hoc result. The disagreement is shown on LLAVA-1.5-7B; on QWEN3-VL-8B, which is already at chance on the one-scene passages, the two ablations agree, which is consistent with that explanation. Our contributions in this paper can be summarized as follows:

- A statement of what an image-ablated control must do to attribute a score change to the image: keep the model responding about the scored objects, and change image content rather than remove the image (Section 2).
- Evidence on two standard benchmarks that the blank fails that test while a mismatched image passes it, with a published method’s gain decomposed into its rates (Section 3).
- A controlled case in which the choice of ablation changes the answer, with the reason and the hit and false-alarm rates that show how the image’s contributions cancel inside the score (Section 4), and a reporting format that exposes what a single score hides (Section 5).

## 2 WHAT A PRESENT-MINUS-ABSENT SCORE CAN IDENTIFY

Write  $H$  for the rate at which a model names a candidate object that is present in the image and  $F$  for the rate at which it names one that is absent. Scores built on a fixed candidate set are functions of  $(H, F)$ ; the simplest is  $J = H - F$ , which has the form of Youden’s  $J$  (Youden, 1950). We borrow the correspondence for its decomposition, not to claim that captioning is a diagnostic test (Appendix C). CHAIR has a different form. With  $P$  present and  $A$  absent candidates, counts pooled over captions, and counting mention occurrences as our CHAIR port does (Appendix G.3),  $\text{CHAIR}_i = \kappa F A / (H P + \kappa F A)$ , where  $\kappa$  is the number of mentions per named absent object divided by the number per named present object. It falls whenever  $\kappa F$  falls proportionally faster than  $H$ , including when  $H$  falls too, so CHAIR can improve while  $J$  worsens.

Under an equal-variance Gaussian formulation the two rates give a separation  $d' = Z(H) - Z(F)$  and a threshold  $c = -\frac{1}{2}[Z(H) + Z(F)]$ , with  $Z = \Phi^{-1}$  (Snodgrass & Corwin, 1988). Inverting,

$$J = \Phi\left(\frac{d'}{2} - c\right) - \Phi\left(-\frac{d'}{2} - c\right), \quad (1)$$

and  $\partial J / \partial c = \varphi(-d'/2 - c) - \varphi(d'/2 - c)$  vanishes only where  $c d' = 0$ : for any non-zero  $d'$  the score depends on where the threshold sits. A reader who sees  $J$  move cannot tell whether  $d'$  moved,  $c$  moved, or both, and, since  $J$  is a difference, cannot tell whether an unchanged  $J$  hides opposite movements in  $H$  and  $F$ . Nor does  $d'$  settle the question. It measures how well behaviour separates the two populations using whatever information the model has, and the text in the prompt, including which objects it puts up for scoring, is part of that information. Nothing in Equation 1 distinguishes separability achieved by looking at the image from separability achieved through the text.

Let  $\Delta(x)$  be the change in  $J$  that an intervention produces with image  $x$ . Its image-attributable part is defined against a counterfactual input  $\tilde{x}$  that replaces the image:  $\Delta_{\text{image}} = \Delta(x) - \Delta(\tilde{x})$ . Two choices of  $\tilde{x}$  answer different questions. A blank asks whether the change needs *an* image; another image from the same distribution asks whether it needs *this* image. A claim of better grounding is of the second kind, since it says the output now tracks what the image shows, and the two coincide only if removing the image leaves the model’s behaviour on the scored objects otherwise intact. A blank can fail that in two ways: it can leave nothing to score, and it can move the model to an operating point where it no longer answers about any scene, so that the image effect it removes is not the one the intervention acts on. A mismatched image keeps the model answering about a real scene, the counterfactual that measures of cross-modal interaction use (Hessel & Lee, 2020; Gat et al., 2021); its cost is that the substitute can contain some of the scored objects, which biases  $\Delta_{\text{image}}$  towards zero (Section 4.4).

## 3 THE BLANK ON STANDARD BENCHMARKS

We first test the requirement where it matters most: on CHAIR and POPE, with LLAVA-1.5-7B and PAI (Liu et al., 2024b), a training-free method that amplifies attention to image tokens during decoding. These analyses were registered in advance (Appendices G, I and J). On CHAIR the scored units are all 80 COCO categories per image, so  $H$  and  $F$  are defined, and because  $F$  is about 0.01,  $J$  is essentially  $H$ .

Under a grey field, and equally under a black one, the model names none of the 80 categories in any of 500 captions, with or without PAI, so  $\text{CHAIR}_i$  is 0/0. On official POPE it answers “no” to all 9,000 questions, correctly, since a grey field contains none of the queried objects, so  $H = F = 0$  by construction. Graded degradations leave CHAIR’s denominator intact (378 to 500 of 500 captions still name a category), so the failure belongs to the blank, not to ablation as such.

With another COCO image in place of the target, PAI’s  $\text{CHAIR}_i$  gain falls to +0.4 [−1.2, +2.0], leaving +5.2 [+2.9, +7.4] of its 5.6-point gain attributable to the image, which the registered rule classifies as image-attributable (Appendix I). On POPE, PAI’s attention-only configuration — which reproduces its published gain, +1.10 accuracy points against their +1.06 — raises  $J$  by +0.022 [+0.011, +0.034] sighted and +0.001 [−0.010, +0.012] with the mismatched image, so  $\Delta_{\text{image}} = +0.021 [+0.004, +0.037]$ . The gain needs the image; the registered rule certifies it only as partly image-attributable because the lower bound falls just below half the gain. The released `-use-attn -use-cfg` configuration shows no POPE gain at all, because it suppresses the amplification at the prefix position that carries a one-token answer (Appendix J).

Table 1: PAI against the unmodified model on standard CHAIR (LLAVA-1.5-7B, 500 COCO images). Lower CHAIR<sub>i</sub> is better. The change row gives PAI minus vanilla, with 95% intervals beneath; here  $d'$  and  $c$  were registered in advance.

| Arm                    | CHAIR <sub>i</sub> | $H$            | $F$              | $J$            | $d'$           | $c$            |
|------------------------|--------------------|----------------|------------------|----------------|----------------|----------------|
| Vanilla                | 12.8               | 0.78           | 0.010            | 0.77           | 3.10           | 0.77           |
| PAI ( $\alpha = 0.5$ ) | 7.2                | 0.71           | 0.005            | 0.70           | 3.12           | 1.01           |
| Change                 | −5.6               | −0.07          | −0.005           | −0.07          | +0.02          | +0.24          |
|                        | [−7.1, −3.9]       | [−0.09, −0.05] | [−0.006, −0.004] | [−0.09, −0.05] | [−0.06, +0.09] | [+0.20, +0.28] |

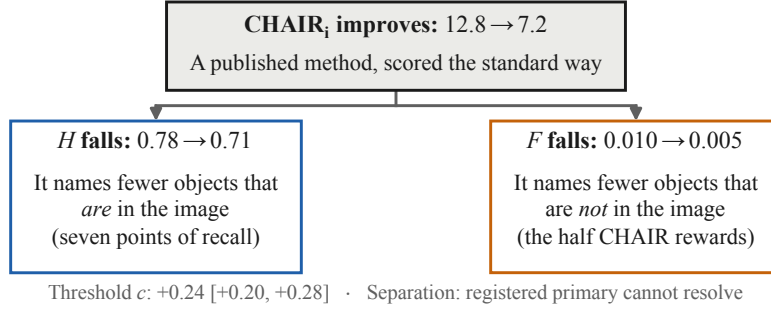


Figure 2: What the scalar score hides, on standard CHAIR: PAI’s CHAIR<sub>i</sub> improvement comes from naming fewer objects of *both* kinds. Numbers from Table 1.

At PAI’s prescribed strength, which we use throughout (at the repository’s default the method has no CHAIR gain,  $\Delta\text{CHAIR}_i -0.26 [-1.42, +0.91]$ ), PAI lowers CHAIR<sub>i</sub> from 12.8 to 7.2 (Table 1). That PAI buys precision with recall is already established: Jung et al. (2025) report precision rising from 84.70 to 90.64 and recall falling from 79.46 to 72.44 for PAI on LLAVA-1.5 over 500 COCO images. Our  $H$  is that recall, and it falls by 0.07 against their 0.0702, a check on the recoding rather than a finding. What the decomposition adds is that the rate of naming absent objects also halves (Figure 2), so CHAIR improves while  $J$  worsens by 0.07, the two summaries ordering the same captions oppositely. Probit  $d'$  does not move and the threshold does: moving  $c$  alone accounts for a change in  $J$  of  $-0.07$ , moving  $d'$  alone for  $+0.003$  (Appendix C.5). Shortening the unmodified model’s own captions until its  $F$  matches PAI’s gives CHAIR<sub>i</sub> of 8.0 [6.9, 9.3], a difference from PAI of  $-0.8 [-1.8, +0.4]$ ; truncation is a standard length baseline (Yue et al., 2024), but shortening is not a pure threshold change, so this places PAI near the unmodified model’s trajectory rather than on it. Whether a small separation gain accompanies the shift the registered test could not decide: a log-odds measure rises ( $+0.30 [+0.13, +0.48]$ ) while probit  $d'$  and the matched- $F$  comparison do not. PAI’s own published POPE table decomposes the same way: on balanced splits  $F = H + 1 - 2\text{acc}$  and  $H$  follows from accuracy and F1, giving  $\Delta d' = +0.04$  against  $\Delta c = +0.18$  (Appendix J).

On both benchmarks, then, the blank cannot run the control, a content-changing ablation can, and PAI’s gain is tied to the image and is mainly a threshold shift whose separation component the registered test could not resolve. Nothing here says the improvement is spurious, only that CHAIR<sub>i</sub> alone cannot say which kind of improvement it is, and that the rates which can are already computed wherever CHAIR is.

#### 4 A CONTROLLED CASE: WHEN THE ABLATION CHANGES THE ANSWER

Standard benchmarks cannot show when the two ablations disagree, because no naturally occurring gain comes with a known cause. We construct one whose cause is textual by design.

---

|                                                                             |                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Image 146155 (COCO). Annotated objects: bottle, <b>person</b> , wine glass. |                                                                                                                                                                                                                            |
| <hr/>                                                                       |                                                                                                                                                                                                                            |
| <b>One scene</b>                                                            | <i>Passage:</i> A <u>dog</u> and its owner playing <u>Frisbee</u> , in the dirt. A <b>man</b> and two <u>dogs</u> playing with an object. A <b>man</b> and two <u>dogs</u> on                                              |
| Sighted                                                                     | a beach. A <b>man</b> and two <u>dogs</u> playing with an object. [repeated to the budget]                                                                                                                                 |
| Grey field                                                                  | a field. A <u>dog</u> and its owner playing <u>Frisbee</u> . A <u>dog</u> and its owner playing <u>Frisbee</u> in the dirt.                                                                                                |
| Mismatched                                                                  | a field. A <b>woman</b> and a <u>dog</u> playing with a <u>frisbee</u> . [repeated to the budget]                                                                                                                          |
| <hr/>                                                                       |                                                                                                                                                                                                                            |
| <b>Five scenes</b>                                                          | <i>Passage:</i> <b>People</b> with <u>ski</u> and <u>snowboard</u> equipment shaking hands. A little <b>boy</b> wearing a baseball hat holding a <u>baseball bat</u> . A <b>person</b> stands between two <u>dogs</u> on a |
| Sighted                                                                     | leash. A <b>man</b> and a <b>woman</b> standing in front of a wine rack. [repeated to the budget]                                                                                                                          |
| Grey field                                                                  | leash. A <b>person</b> with a backpack on their back. [repeated to the budget]                                                                                                                                             |
| Mismatched                                                                  | leash. A <b>woman</b> wearing sunglasses and a green shirt. A <b>man</b> wearing a white shirt. [...]                                                                                                                      |

---

Figure 3: One cell of the stimulus, chosen for legibility, with the continuation of each arm. Objects the passage names are scored: **bold** if annotated in the image, underlined if not. With one scene every arm repeats absent objects; with five, every arm names only the present one, with or without the correct image. The sighted arm alone mentions the wine rack it sees.

#### 4.1 STIMULUS, ENDPOINT AND ABLATIONS

**Stimulus.** We ask the model to describe the image and prefill the start of its answer with a passage built from five human COCO captions of other images: in the one-scene condition all five describe one other image, in the five-scene condition five different images. They are shuffled, joined and cut to a per-cell token length — on LLAVA-1.5-7B the smaller of half the model’s own caption length and the length of a natural-prose description of the first source image, 44.6 tokens on average (Appendix A) — so a passage can end mid-word, and the model continues greedily up to a budget of 384 tokens. A *cell* is one image with its passage draw in both conditions, and each image contributes three. Both conditions have the same passage length; after the cut the five-scene passage covers about 2.7 distinct scenes against 1.0 and twice as many object types. A continuation that reaches the budget is *truncated*: 45% of one-scene and 68% of five-scene continuations are, many repeating one sentence (Figure 3).

**Endpoint.** A *unit* is an (image, object type named in the passage) pair, and the model carries a unit forward if its continuation names that object type.  $H$  and  $F$  are carry-forward rates over units whose object is and is not in the image’s COCO annotations, and  $\Delta_{\text{primary}}$  is the change in  $J$  from one scene to five. This endpoint was adopted after CHAIR proved unusable on these conditions (Appendix E).

**Ablations.** Each ablated arm reruns the pipeline with the target image replaced and every token of text unchanged, so it scores the same units with the same labels. The *grey field* is a flat mid-grey image of the target’s size. The *mismatched image* is another image of the same sample, assigned by a derangement: a random permutation of the images that maps none to itself. With  $\Delta_{\text{ablated}}$  the contrast in the ablated arm, the image-attributable part is  $\Delta_{\text{image}} = \Delta_{\text{primary}} - \Delta_{\text{ablated}}$ , and  $\Delta_{\text{ablated}}/\Delta_{\text{primary}}$  is the share of the rise that survives the ablation. We call each image condition — sighted, grey field, mismatched image — an *arm*.

**Samples.** The registered sample is 500 COCO val2017 images (1,500 cells), and a post-hoc extension uses 3,500 images that contain them (10,500 cells). The grey-field control and the mismatched image on the registered sample are registered, the latter by a dated amendment written before its data existed; everything on the 3,500-image sample is post hoc.

#### 4.2 A LARGE RISE WITH A TEXTUAL CAUSE

On LLAVA-1.5-7B the score rises from +0.19 to +0.67, a contrast of +0.49 [+0.43, +0.54]. It is the least controlled number we report: matching continuation lengths leaves +0.36 [+0.30, +0.41], and restricting to untruncated cells of the 3,500-image sample leaves +0.35 [+0.28, +0.41] (Appendix B.5). Units whose object is in the image are enriched for frequent objects the model

Table 2: The one-scene to five-scene contrast with the image and under the grey field, on truncated generations as registered.  $\Delta_{\text{image}} = \Delta_{\text{primary}} - \Delta_{\text{ablated}}$ ; bold marks intervals that exclude zero. The last column is the share of the rise that survives the grey field. QWEN3-VL-8B is documented in Appendix K; QWEN2.5-VL-7B, which is not interpreted, is in Appendix D.

| Architecture  | Sighted $\Delta_{\text{primary}}$ | Grey $\Delta_{\text{ablated}}$ | $\Delta_{\text{image}}$        | Grey/sighted |
|---------------|-----------------------------------|--------------------------------|--------------------------------|--------------|
| LLAVA-1.5-7B  | +0.486                            | +0.502                         | -0.016 [-0.070, +0.040]        | 1.03         |
| LLAVA-OV-0.5B | +0.378                            | +0.413                         | -0.035 [-0.091, +0.018]        | 1.09         |
| KOSMOS-2      | +0.215                            | +0.318                         | <b>-0.103</b> [-0.168, -0.036] | 1.48         |
| QWEN3-VL-8B   | +0.583                            | +0.523                         | <b>+0.060</b> [+0.024, +0.096] | 0.90         |

names anyway, a non-visual route to a higher score: stratifying by object frequency leaves +0.33 [+0.22, +0.44], and because every arm scores the same units, this composition effect cancels from  $\Delta_{\text{image}}$  to first order (Appendix B). With lengths matched,  $H$  barely moves (-0.03 [-0.08, +0.02]) while  $F$  falls from 0.46 to 0.08. The rise is monotone over a five-point grid in the number of scenes and replicates on LLAVA-ONEVISION-0.5B (+0.38 [+0.32, +0.43]; Appendix D).

**The rise is specific to assembled captions and to free generation.** A registered version that builds the passage from natural prose (Localized Narratives) instead of COCO captions was null (+0.01 [-0.04, +0.05]), under every specification, though it realised about half the headline’s dose; by that registration, the lead result is specific to the assembled-caption construct. An earlier registered natural-prose run did score five sources above one, by +0.07 [+0.02, +0.13] raw but only +0.02 [-0.03, +0.07] length-matched. A registered yes/no version of the probe on the same units does not move (-0.01 [-0.03, +0.01]), and by that registration every claim here concerns free generation (Appendix H).

#### 4.3 THE REGISTERED CONTROL: A GREY FIELD

Table 2 gives the contrast with the image and under the grey field on four architectures. On LLAVA-1.5-7B the grey rise is 1.03 [0.92, 1.16] times the sighted one. LLAVA-OV-0.5B agrees, although its grey continuations are short (medians of five and eleven tokens against 192 sighted). The other two depart in opposite directions: on KOSMOS-2 the rise is larger without the image, and on QWEN3-VL-8B about a tenth of it needs the image, though its arms repeat sentences at different rates (Appendix K). QWEN2.5-VL-7B produced four- or five-token continuations in every arm and is not interpreted, as its registration committed in advance (Appendix D).

The grey null is not a failure to use the image. Within each condition the image raises  $J$  by +0.21 [+0.16, +0.26] and +0.19 [+0.16, +0.23] on LLAVA-1.5-7B, and raises  $d'$  in every condition (Appendix D.4); because that effect is nearly equal in the two conditions, it cancels from  $\Delta_{\text{image}}$ . It is not equal in  $H$  and  $F$  separately (Table 4): under grey the image contributes -0.14 to the rise in  $H$  and -0.12 to the rise in  $F$ , and the two cancel in  $J$ . The grey null holds under three ground-truth constructions, every registered truncation rule and a length match; the registered mention-rate control fails on five of sixteen powered subsets, four of them with  $\Delta_{\text{image}}$  near +0.10 (Appendix B).

#### 4.4 THE ABLATION AND THE GENERATION REGIME

On the registered sample the mismatched image leaves 0.67 [0.57, 0.77] of the rise, so  $\Delta_{\text{image}} = +0.16$  [+0.11, +0.21], where grey leaves all of it (Appendix I). On the 3,500-image extension, run on all 10,500 cells with identical passages (Appendix L), the grey field again leaves all of it (ratio 1.01 [0.96, 1.05]) and the mismatched image 0.68 [0.64, 0.72], with  $\Delta_{\text{image}} = +0.15$  [+0.13, +0.17]; 3,000 of those images played no part in the registered result. The substitute image is not information-free: it also contains 46–48% of the present objects the passage names (10–12% of the absent ones), and excluding those units raises  $\Delta_{\text{image}}$  to +0.24 [+0.21, +0.27], so the substitution estimate is conservative. This all-cells ratio is our estimate of the image’s share: about a third.

Table 3: The ablation ladder on the prefix endpoint, LLAVA-1.5-7B. Severity is  $1 - \cos$  in the model’s own image-feature space.  $\Delta_{\text{image}}$  is the part of the one-to-five-scene rise attributable to the image. The two  $J$  columns show whether an ablation blinded the model: the five ablations above the substitution leave the one-scene  $J$  within 0.06 of its sighted +0.188, and the four below it collapse it to near zero. Only the substitution moves  $\Delta_{\text{image}}$  off zero. Intervals come from this experiment’s own resample, so the grey row differs in the third decimal from Tables 7 and 12.

| Ablation  | Severity | $J$ , one scene | $J$ , five scenes | $\Delta_{\text{ablated}}$ | $\Delta_{\text{image}}$        |
|-----------|----------|-----------------|-------------------|---------------------------|--------------------------------|
| identity  | 0.000    | +0.188          | +0.674            | +0.486                    | (reference)                    |
| noise16   | 0.008    | +0.177          | +0.670            | +0.493                    | −0.008 [−0.038, +0.022]        |
| noise32   | 0.016    | +0.176          | +0.669            | +0.492                    | −0.007 [−0.045, +0.030]        |
| noise64   | 0.034    | +0.164          | +0.648            | +0.485                    | +0.001 [−0.036, +0.039]        |
| blur4     | 0.044    | +0.154          | +0.652            | +0.498                    | −0.013 [−0.054, +0.031]        |
| shuffle24 | 0.079    | +0.137          | +0.635            | +0.498                    | −0.012 [−0.063, +0.041]        |
| mismatch  | 0.080    | +0.124          | +0.450            | +0.326                    | <b>+0.160</b> [+0.105, +0.213] |
| lowres16  | 0.089    | +0.025          | +0.514            | +0.489                    | −0.004 [−0.055, +0.048]        |
| black     | 0.094    | +0.007          | +0.485            | +0.477                    | +0.009 [−0.049, +0.067]        |
| grey      | 0.102    | −0.022          | +0.480            | +0.502                    | −0.016 [−0.073, +0.041]        |
| blur16    | 0.126    | +0.020          | +0.505            | +0.485                    | +0.001 [−0.055, +0.055]        |

Between 45% and 68% of continuations reach the budget, so we also scored the 2,089 untruncated cells, in which neither sighted continuation reaches it (Figure 1b). There the ratios fall to 0.71 [0.56, 0.87] under grey and 0.47 [0.33, 0.62] under the mismatched image,  $\Delta_{\text{image}} = +0.18$  [+0.12, +0.24]. We treat this as secondary. It is post hoc and rests on one derangement; it selects cells on the sighted arms’ truncation, which the passage itself influences, so the untruncated cells are a selected rather than a randomized subset; and inside them the ablated arms still reach the budget in 2–27% of continuations. It shows that truncation also moves the answer, not how far.

The disagreement between ablations is shown on LLAVA-1.5-7B. On QWEN3-VL-8B (post hoc) the two agree:  $\Delta_{\text{image}}$  is +0.065 [+0.031, +0.100] with a mismatched image and +0.060 with grey (ratios 0.89 and 0.90). That model’s sighted one-scene  $J$  is already near zero (+0.05), so grey has no level to remove, which is the condition the mechanism below identifies; its regime factor cannot be tested, since 94% of its continuations reach the budget (Appendix K). KOSMOS-2 was run only under grey, where it departs in the opposite direction. The general claim is therefore conditional: the ablation can change the answer, and whether it does depends on where the model sits without the image.

$\Delta_{\text{image}}$  is the difference between the image’s effect on  $J$  in the five-scene and the one-scene condition, and grey removes both. With one scene it drops  $H$  from 0.68 to 0.47, level with  $F$  (0.49), so  $J$  starts at zero; with five it lowers  $J$  by a similar amount, and the two removals cancel. The mismatched image leaves the model a scene to answer about: its one-scene  $H$  stays at 0.64, and what it removes is the five-scene rise in  $H$  that needs the correct image, since  $H$  falls from 0.64 to 0.60 where the sighted arm’s rises from 0.68 to 0.78. Table 4 shows the contribution sits in  $H$ , not  $F$ .

Of the ten ablations run on the registered sample (Table 3; Appendix I), five barely remove the image: under pixel noise, light blur and patch shuffling the one-scene  $J$  stays between +0.14 and +0.18, against +0.19 sighted, so their null  $\Delta_{\text{image}}$  says little. The four that do remove it — grey, black, heavy blur and 16×16 downsampling — collapse the one-scene  $J$  to between −0.02 and +0.03, and all four return  $\Delta_{\text{image}}$  near zero. Only the substitution removes the correct image while leaving an answerable one, and only it returns a positive  $\Delta_{\text{image}}$  (+0.16 [+0.11, +0.21]), which survives matching continuation lengths across conditions (+0.14 [+0.10, +0.18]).

Under either ablation the image changes the rates, but differently (Table 4). Against grey it lowers the rise in  $H$  and deepens the fall in  $F$  by similar amounts, which cancel in  $J$ ; against the mismatched image it adds to the rise in  $H$  and leaves  $F$  alone. In signal-detection terms the correct image contributes separation ( $d' + 0.56$  [+0.40, +0.72]) against the mismatched image, but mostly threshold ( $c + 0.41$ ) against grey, because the grey field moves the threshold itself. Separation still rises in every arm ( $d'$  gains 1.53 sighted, 1.35 under grey and 0.97 with the mismatched image),

Table 4: The rise on LLAVA-1.5-7B (registered sample), decomposed under each ablation. The first three columns are one-to-five-scene changes; the last two are the image-attributable parts, sighted minus ablated, with 95% intervals. Columns round independently. \*Post hoc for this contrast. This is the reporting format we recommend (Section 5).

| Quantity             | Change, one to five scenes |       |            | Image-attributable   |                      |
|----------------------|----------------------------|-------|------------|----------------------|----------------------|
|                      | Sighted                    | Grey  | Mismatched | Grey field           | Mismatched image     |
| Hit rate $H$         | +0.10                      | +0.24 | −0.05      | −0.14 [−0.19, −0.08] | +0.15 [+0.10, +0.20] |
| False-alarm rate $F$ | −0.38                      | −0.26 | −0.37      | −0.12 [−0.15, −0.10] | −0.01 [−0.03, +0.01] |
| Score $J$            | +0.49                      | +0.50 | +0.33      | −0.02 [−0.07, +0.04] | +0.16 [+0.11, +0.21] |
| Separation $d'^*$    | +1.53                      | +1.35 | +0.97      | +0.19 [+0.03, +0.35] | +0.56 [+0.40, +0.72] |
| Threshold $c^*$      | +0.45                      | +0.04 | +0.61      | +0.41 [+0.32, +0.50] | −0.16 [−0.24, −0.07] |

so on this construction much of it comes from the text. These coordinates recode the two rates exactly but were not registered for this contrast, and we do not claim the equal-variance model holds: one operating point cannot test it, and mixtures can mimic unequal variance (DeCarlo, 2002). The argument needs only Equation 1 and the rates.

## 5 RECOMMENDED REPORTING

1. **Report  $H$  and  $F$  with their decomposition, not a single score.** Report the image-attributable part of each rate, since a null on  $J$  can hide offsetting effects on the two, and  $d'$  and  $c$  where the question is separation against threshold (Table 4). For CHAIR, report the  $H$  and  $F$  its mention records already contain (Table 1).
2. **Ablate every endpoint whose candidates the input supplies, and choose the ablation for the question.** A question naming each object, as in POPE, or a passage naming it, as here, keeps the model answering about every scored object without the image. To test grounding, change the image’s content rather than remove it, and report the ablated arm’s levels in each condition: an ablation that leaves them unchanged has not blinded the model, and one that sends them to chance has removed the answerable content along with the evidence. A blank also empties CHAIR’s denominator and falsifies POPE’s labels.
3. **Do not read an endpoint whose denominator the model chooses as measuring grounding on its own** when the intervention can change how readily the model names objects — attention or activation interventions, contrastive or shortened decoding, instruction changes. Report its  $H$  and  $F$ , and say whether its attribution to the image was tested.

Published tables often already contain the rates: on POPE’s balanced splits  $F = H + 1 - 2 \text{acc}$ , which prices the two degenerate corners — both at accuracy 50.0 and  $J = 0$ , but F1 66.7 for an always-“yes” model against 0.0 for a blinded one.

## 6 RELATED WORK

**Object-hallucination endpoints.** CHAIR (Rohrbach et al., 2018) scores captions against COCO annotations (Lin et al., 2014) as the share of mentioned objects that are absent. POPE (Li et al., 2023) fixes the candidates and asks yes/no questions, drawing absent objects at random, by frequency, or by co-occurrence; its frequency-based splits are the natural template for separating the object-frequency effect of Section 4.2 from grounding. CHAIR’s authors found no correlation between caption length and hallucination in 2018 captioners (Rohrbach et al., 2018); later work finds CHAIR sensitive to length and budget, and rewarding short captions for want of a recall term (Li et al., 2023; Xu et al., 2026; Kaul et al., 2024). AMBER reports coverage beside its hallucination rate (Wang et al., 2023), and Fazli et al. (2026) find that across six methods lower CHAIR goes with lower recall; neither runs an ablated condition or decomposes an effect.

**Image-ablated controls.** Question-only baselines exposed language priors in VQA (Goyal et al., 2017; Agrawal et al., 2018); MMStar subtracts the image-free score (Chen et al., 2024); Lan et al.

(2026) degrade the image across four benchmarks, Kaduri et al. (2025) ablate the attention path instead of the pixels, Zafar et al. (2026) derive grounding metrics from blank and image-absent inputs, and PAI’s authors generate with the image excluded to diagnose text inertia (Liu et al., 2024b). These ask whether a score can be reached without looking; we ask whether a *change* in score is attributable to the image. Closest to us, Liao et al. (2026) find a training-free steering gain of 21 points with the image and 19 with a grey blank, and guard against input degeneracy by requiring five ablations, including a real but mismatched image, to agree; on their task they do. Ours is a case where they disagree, and Section 4.4 gives the reason. The mismatched-image control itself is not new: a model’s reliance on a modality has been measured with mismatched inputs (Hessel & Lee, 2020; Gat et al., 2021; Parcalabescu & Frank, 2023), and that an ablation’s answer depends on its baseline is known for feature attributions (Sturmfels et al., 2020) and activation patching (Zhang & Nanda, 2024). What we add is a registered case in which it and the blank disagree about an evaluation gain, the reason they disagree, and what each attributes to the image in  $H$ ,  $F$ ,  $d'$  and  $c$ . Likewise, PAI’s precision-recall trade-off is Jung et al.’s (2025) and the response-shift reading of decoding interventions is Yin et al.’s (2025); our addition there is the image-attribution test and the separation-threshold decomposition.

**Signal detection.** Separating sensitivity from threshold is the founding move of signal-detection theory, and reporting a corrected score without its components conflates them (Snodgrass & Corwin, 1988; Youden, 1950); applying it to language models is an active line (Cacioli, 2026b;a; Bogdan & de Valois-Franklin, 2026), with a text model’s own confidence as the decision variable. Yin et al. (2025) argue that contrastive decoding’s POPE gains are a response shift, since non-visual manipulations match or beat VCD (87.6 and 89.6 against VCD’s 88.6 and greedy’s 87.1 on LLAVA-1.5-7B); they report accuracy and yes-rate without decomposing them or ablating the image. We add a multimodal setting in which the sensitivity term is itself contaminated, because separability can come from the prompt. Unlike protocol artefacts in unfair comparisons (Lucic et al., 2018; Choi et al., 2019; Ferrari Dacrema et al., 2019; Sculley et al., 2018), ours is an endpoint that does not isolate its construct even under a fair one.

**Context-induced hallucination and mitigation.** Our stimulus follows Yang et al. (2023), who prompt a captioner with ground-truth captions of an unrelated image; we vary a caption property they hold fixed and add the ablated arms. Training-free methods that intervene on attention or activations (Liu et al., 2024b; Li et al., 2025b; Lei et al., 2026; Zhang et al., 2026; Xu et al., 2026; Zhong et al., 2026) or on the decoding distribution (Leng et al., 2024) are evaluated on the endpoints above; Wu et al. (2025) is the closest methodological neighbour in intervention evaluation.

## 7 LIMITATIONS

**What this establishes.** On CHAIR and POPE a blank image cannot run the check that separates grounding from naming propensity, while a mismatched image can, and a published method passes it. In a constructed case whose cause is textual, the grey field attributes none of a large rise to the image and a mismatched image attributes about a third. We do not show that models ignore images, that CHAIR or POPE are invalid, or that published gains are misattributed.

**Scope.** The stimulus is artificial: passages stitched from captions and cut mid-word, many continuations repeat one sentence to the budget, and the rise disappears with natural prose and in a yes/no format, so the constructed case is a demonstration, not a model of any benchmark. The disagreement between ablations is shown on one model; on QWEN3-VL-8B they agree, and KOSMOS-2 was run only under grey. The substitution uses one derangement, and the regime result is post hoc and selects on a post-treatment variable. The registered mention-rate control fails on five of sixteen powered subsets, the  $d'/c$  recoding is post hoc for the scene-count contrast, and PAI’s separation question could not be resolved. Appendix E lists every registered analysis with its outcome.

## 8 CONCLUSION

The check that removes the image to see whether a grounding gain survives is only as good as what replaces the image. On CHAIR and POPE a blank cannot run it at all: the model names nothing,

or answers “no” to every question, so the score returns no data rather than a null. A mismatched real image can run it, and under it a published method’s gain is tied to the image while decomposing mainly into a threshold shift. In a constructed case whose cause is textual, the two ablations disagree: the grey field attributes none of a large rise to the image and the mismatched image about a third. The disagreement follows from where the model sits without the image, because a blank removes the answerable content along with the evidence; on a model that is already near chance on the one-scene passages, the two agree.

Three practices follow. Report the hit and false-alarm rates with their image-attributable parts rather than a single score, because the image’s effects on the two can cancel inside it. Ablate with an image whose content differs rather than with a blank, and report the ablated arm’s levels, so that a reader can see whether the control blinded the model or silenced it. And state which ablation and which generation budget produced an attribution, since either choice can change the answer.

## REPRODUCIBILITY STATEMENT

An analysis is called registered when a written plan with its decision rules was committed to the authors’ git repository before the corresponding generation job; the timestamps are the authors’ own, not a third-party registry’s. Appendix E lists every registration with its commit hash and date, and the registrations with their dated amendments are released. The endpoint was adopted after CHAIR proved unusable on these conditions, the  $d'/c$  recoding of the scene-count contrast and the 3,500-image comparison of ablations are post hoc, and every element chosen after seeing data is marked where it appears. Each pipeline reproduces previously computed per-condition rates to within  $5 \times 10^{-3}$  before reporting a new quantity. Section 4.1 gives the stimulus and endpoint, Appendix A the prompts, decoding settings, model versions and unit counts, and Appendix F the implementation record. The analysis runs on CPU from the released artefacts, so every number can be checked without a GPU or the original checkpoints. Code and data: <https://github.com/alexmuel07/Grounding-Effect>.

## ETHICS STATEMENT

This work uses publicly available models and the public COCO and POPE benchmarks. No new data were collected and no human subjects were involved. The contribution is to evaluation practice: we show that whether a change in a grounding score is attributed to the image depends on how the image is removed, and that the usual blank can hide or fail to measure that contribution. Registered analyses whose outcomes cut against our claims are reported, and every registered analysis is listed with its outcome in Appendix E.

## AI USE STATEMENT

AI tools were used as supplementary aids during this work, primarily for developing an outline and structural scaffolding for the manuscript and for assisting with programming and debugging. The authors independently determined the research direction, methodology, experimental setup, analysis, and conclusions. Any AI assisted code or writing was critically reviewed, adapted as needed, and verified by the authors, who take full responsibility for the final manuscript and reported results.

## REFERENCES

- Aishwarya Agrawal, Dhruv Batra, Devi Parikh, and Aniruddha Kembhavi. Don’t just assume; look and answer: Overcoming priors for visual question answering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 4971–4980, 2018. doi: 10.1109/CVPR.2018.00522.
- Alex Bogdan and Adrian de Valois-Franklin. Machine psychometrics: A mathematical psychology of artificial intelligence. arXiv:2605.23952, 2026.
- Jon-Paul Cacioli. Do LLMs know what they know? measuring metacognitive efficiency with signal detection theory. arXiv:2603.25112, 2026a.

- Jon-Paul Cacioli. LLMs as signal detectors: Sensitivity, bias, and the temperature-criterion analogy. arXiv:2603.14893, 2026b.
- Lin Chen, Jinsong Li, Xiaoyi Dong, Pan Zhang, Yuhang Zang, Zehui Chen, Haodong Duan, Jiaqi Wang, Yu Qiao, Dahua Lin, and Feng Zhao. Are we on the right way for evaluating large vision-language models? In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, pp. 27056–27087, 2024. doi: 10.52202/079017-0850.
- Dami Choi, Christopher J. Shallue, Zachary Nado, Jaehoon Lee, Chris J. Maddison, and George E. Dahl. On empirical comparisons of optimizers for deep learning. arXiv:1910.05446, 2019.
- Lawrence T. DeCarlo. Signal detection theory with finite mixture distributions: Theoretical developments with applications to recognition memory. *Psychological Review*, 109(4):710–721, 2002. doi: 10.1037/0033-295X.109.4.710.
- Mehrdad Fazli, Sina Mansouri, Mohit Marvania, and Ziwei Zhu. Does playing it safe count as faithfulness? reassessing LVLm hallucination mitigation methods. arXiv:2609.01888, 2026.
- Maurizio Ferrari Dacrema, Paolo Cremonesi, and Dietmar Jannach. Are we really making much progress? a worrying analysis of recent neural recommendation approaches. In *Proceedings of the 13th ACM Conference on Recommender Systems (RecSys 2019)*, pp. 101–109. ACM, 2019. doi: 10.1145/3298689.3347058.
- Itai Gat, Idan Schwartz, and Alexander Schwing. Perceptual score: What data modalities does your model perceive? In *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*, 2021.
- Yash Goyal, Tejas Khot, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. Making the V in VQA matter: Elevating the role of image understanding in visual question answering. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 6325–6334, 2017. doi: 10.1109/CVPR.2017.670.
- Jack Hessel and Lillian Lee. Does my multimodal model learn cross-modal interactions? It’s harder to tell than you might think! In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 861–877, 2020. doi: 10.18653/v1/2020.emnlp-main.62.
- Mingi Jung, Saehyung Lee, Eunji Kim, and Sungroh Yoon. Visual attention never fades: Selective progressive attention ReCalibration for detailed image captioning in multimodal large language models. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*, pp. 28527–28551, 2025. arXiv:2502.01419.
- Omri Kaduri, Shai Bagon, and Tali Dekel. What’s in the image? A deep-dive into the vision of vision language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 14549–14558, 2025.
- Prannay Kaul, Zhizhong Li, Hao Yang, Yonatan Dukler, Ashwin Swaminathan, C. J. Taylor, and Stefano Soatto. THRONE: An object-based hallucination benchmark for the free-form generations of large vision-language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. arXiv:2405.05256.
- Zixuan Lan, Luzhe Sun, Matthew R. Walter, and Jiawei Zhou. Seeing without looking: Do vision-language benchmarks really test vision? In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pp. 11260–11273, June 2026. GRail-V Workshop; also arXiv:2605.22903.
- Yuqing Lei, Wenbo Lyu, Yingjun Du, Xiantong Zhen, Cees G. M. Snoek, and Ling Shao. See only when needed: Context-aware attention intervention for mitigating hallucinations in LVLms. arXiv:2606.29847, 2026.
- Sicong Leng, Hang Zhang, Guanzheng Chen, Xin Li, Shijian Lu, Chunyan Miao, and Lidong Bing. Mitigating object hallucinations in large vision-language models through visual contrastive decoding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 13872–13882, 2024. doi: 10.1109/CVPR52733.2024.01316.

- Bo Li, Yuanhan Zhang, Dong Guo, Renrui Zhang, Feng Li, Hao Zhang, Kaichen Zhang, Peiyuan Zhang, Yanwei Li, Ziwei Liu, and Chunyuan Li. LLaVA-OneVision: Easy visual task transfer. *Transactions on Machine Learning Research*, 2025a. ISSN 2835-8856. URL <https://openreview.net/forum?id=zKv8qULV6n>.
- Yifan Li, Yifan Du, Kun Zhou, Jinpeng Wang, Xin Zhao, and Ji-Rong Wen. Evaluating object hallucination in large vision-language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 292–305. Association for Computational Linguistics, 2023. doi: 10.18653/v1/2023.emnlp-main.20.
- Zhuowei Li, Haizhou Shi, Yunhe Gao, Di Liu, Zhenting Wang, Yuxiao Chen, Ting Liu, Long Zhao, Hao Wang, and Dimitris N. Metaxas. The hidden life of tokens: Reducing hallucination of large vision-language models via visual information steering. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*, pp. 35799–35819. PMLR, 2025b. URL <https://proceedings.mlr.press/v267/li25ca.html>.
- Chih-Ting Liao, Fei Shen, Xin Cao, and Tat-Seng Chua. Decodable is not grounded: A vision-ablation arbiter for VLM spatial reasoning. arXiv:2606.31257, 2026. Version v1, 30 June 2026.
- Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C. Lawrence Zitnick. Microsoft COCO: Common objects in context. In *Computer Vision – ECCV 2014*, Lecture Notes in Computer Science, pp. 740–755. Springer International Publishing, 2014. doi: 10.1007/978-3-319-10602-1\_48.
- Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. Improved baselines with visual instruction tuning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 26286–26296, 2024a. doi: 10.1109/CVPR52733.2024.02484.
- Shi Liu, Kecheng Zheng, and Wei Chen. Paying more attention to image: A training-free method for alleviating hallucination in LVLMs. In *Computer Vision – ECCV 2024*, Lecture Notes in Computer Science, pp. 125–140. Springer Nature Switzerland, 2024b. doi: 10.1007/978-3-031-73010-8\_8.
- Mario Lucic, Karol Kurach, Marcin Michalski, Sylvain Gelly, and Olivier Bousquet. Are GANs created equal? a large-scale study. In *Advances in Neural Information Processing Systems 31 (NeurIPS 2018)*, 2018. URL [https://papers.nips.cc/paper\\_files/paper/2018/hash/e46de7e1bcaaced9a54f1e9d0d2f800d-Abstract.html](https://papers.nips.cc/paper_files/paper/2018/hash/e46de7e1bcaaced9a54f1e9d0d2f800d-Abstract.html).
- Letitia Parcalabescu and Anette Frank. MM-SHAP: A performance-agnostic metric for measuring multimodal contributions in vision and language models & tasks. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 4032–4059, 2023. doi: 10.18653/v1/2023.acl-long.223.
- Zhiliang Peng, Wenhui Wang, Li Dong, Yaru Hao, Shaohan Huang, Shuming Ma, Qixiang Ye, and Furu Wei. Grounding multimodal large language models to the world. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024. URL <https://openreview.net/forum?id=1LmqxkfSIw>.
- Qwen Team. Qwen2.5-VL technical report. arXiv:2502.13923, 2025a.
- Qwen Team. Qwen3-VL technical report. arXiv:2511.21631, 2025b.
- Anna Rohrbach, Lisa Anne Hendricks, Kaylee Burns, Trevor Darrell, and Kate Saenko. Object hallucination in image captioning. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pp. 4035–4045. Association for Computational Linguistics, 2018. doi: 10.18653/v1/D18-1437.
- D. Sculley, Jasper Snoek, Ali Rahimi, and Alex Wiltschko. Winner’s curse? on pace, progress, and empirical rigor. In *International Conference on Learning Representations (ICLR) Workshop Track*, 2018. URL <https://openreview.net/forum?id=rJWF0Fywfw>.

- Joan G. Snodgrass and June Corwin. Pragmatics of measuring recognition memory: Applications to dementia and amnesia. *Journal of Experimental Psychology: General*, 117(1):34–50, 1988. doi: 10.1037/0096-3445.117.1.34.
- Pascal Sturmfels, Scott Lundberg, and Su-In Lee. Visualizing the impact of feature attribution baselines. *Distill*, 2020. doi: 10.23915/distill.00022.
- Junyang Wang, Yuhang Wang, Guohai Xu, Jing Zhang, Yukai Gu, Haitao Jia, Jiaqi Wang, Haiyang Xu, Ming Yan, Ji Zhang, and Jitao Sang. AMBER: An LLM-free multi-dimensional benchmark for MLLMs hallucination evaluation. arXiv:2311.07397, 2023.
- Zhengxuan Wu, Aryaman Arora, Atticus Geiger, Zheng Wang, Jing Huang, Dan Jurafsky, Christopher D. Manning, and Christopher Potts. AxBench: Steering LLMs? even simple baselines outperform sparse autoencoders. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*, pp. 67035–67080. PMLR, 2025. URL <https://proceedings.mlr.press/v267/wu25a.html>.
- Yuanzhi Xu, Qian Gao, Jun Fan, Guohui Ding, Zhenyu Yang, Sixue Lin, and Yuteng Xiao. Mitigating object hallucinations in vision-language models through region-aware attention recalibration. arXiv:2605.24957, 2026.
- Xu Yang, Yongliang Wu, Mingzhuo Yang, Haokun Chen, and Xin Geng. Exploring diverse in-context configurations for image captioning. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, pp. 40924–40943, 2023. doi: 10.52202/075280-1782.
- Hao Yin, Guangzong Si, and Zilei Wang. The mirage of performance gains: Why contrastive decoding fails to mitigate object hallucinations in MLLMs? arXiv:2504.10020, 2025.
- W. J. Youden. Index for rating diagnostic tests. *Cancer*, 3(1):32–35, 1950.
- Zihao Yue, Liang Zhang, and Qin Jin. Less is more: Mitigating multimodal hallucination from an EOS decision perspective. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024. arXiv:2402.14545.
- Anas Zafar, Leema Krishna Murali, Siddhant Bharadwaj, Ashish Vashist, and Jia Wu. Do medical vision language models actually see? A counterfactual grounding framework and hard-negative contrastive training for visually-reliant medical VLMs. arXiv:2607.03647, 2026. Version v1, 4 July 2026.
- Fred Zhang and Neel Nanda. Towards best practices of activation patching in language models: Metrics and methods. In *International Conference on Learning Representations (ICLR)*, 2024.
- Jiacheng Zhang, Feng Liu, Chao Du, and Tianyu Pang. SAVAA: Mitigating hallucinations in LVLMS via step-wise adaptive visual attention amplification. arXiv:2602.13600, 2026.
- Li’an Zhong, Ziqiang He, Jibin Zheng, Jin Li, Z. Jane Wang, and Xiangui Kang. AdaIAT: Adaptively increasing attention to generated text to alleviate hallucinations in LVLM. arXiv:2603.04908, 2026.

## A EXPERIMENTAL DETAILS

This appendix supplies everything needed to check the measurements in the main text: the stimulus construction, the scorer, the unit definition, the interval estimator, the integrity gates, the complete tables behind every figure, the signal-detection derivation the diagnosis rests on, and the pre-registration record with its decision rules. Figure 4 summarises the design.

**Terms.** The main text’s vocabulary is used throughout. A *cell* is one image with its passage draw in both conditions (`cap1`, one scene; `cap5`, five scenes); a *unit* is an (image, passage object) pair; an *arm* is one image condition; and a continuation is *truncated* when it reaches the 384-token budget.  $H$  and  $F$  are the carry-forward rates on present and absent units and  $J = H - F$ . One older term remains: *blind* denotes the grey-field arm, the originally registered ablation. Registered outcome labels are quoted as the registrations wrote them (Appendix E).

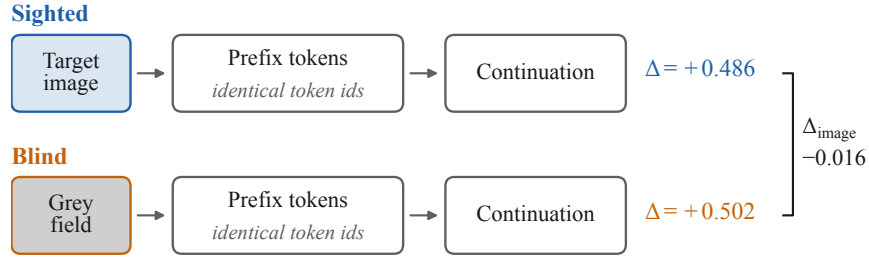


Figure 4: The design. The sighted and blind rows differ in the image alone; prefix token ids are bit-identical, so the difference between the two rises is attributable to replacing the image with a uniform grey field. On LLAVA-1.5-7B that difference is  $-0.016$ .

#### A.1 MODELS, DECODING AND PROMPT

Five checkpoints are used; the first two carry every primary number, and the fifth, QWEN3-VL-8B, is documented separately in Appendix K. LLAVA-1.5-7B is `llava-1.5-7b-hf` (Liu et al., 2024a) in `bfloat16` on a single consumer GPU. The second is `llava-onevision-qwen2-0.5b-ov-hf` (Li et al., 2025a), run with its own tokenizer and its own chat template. It differs in language-model family (Qwen2), vision tower (SigLIP), training recipe and resolution scheme (anyres), and is  $7.90\times$  smaller by parameter count. Where the two disagree, the disagreement is reported rather than pooled.

The two further models in this group were each run under their own registration. QWEN2.5-VL-7B is `Qwen/Qwen2.5-VL-7B-Instruct` (Qwen Team, 2025a) in `bfloat16` with its own chat template; its cut, chosen by a registered census rule before any generation, is  $f = 0.230$  (mean prefix 40.7 tokens) on all 500 images. KOSMOS-2 is `microsoft/kosmos-2-patch14-224` (Peng et al., 2024) in `float32`, its checkpoint’s own precision. It has no chat template, so its registered prompt is its trained task string without the grounding marker, `Describe this image in detail:`, which keeps its output in plain prose; its census cut is  $f = 0.460$  (mean prefix 36.1 tokens) on 473 images, after 27 whose free-running caption was under 32 tokens were excluded by the registered rule. Both use the same stimulus assembly, scorer, blinding constant and bootstrap as the first two; their unit counts and the full census tables are in the released registration records.

Decoding is greedy throughout (`do_sample=False`), with no sampling parameters of any kind. Free-running captions use `max_new_tokens = 256`; continuations from a supplied prefix use 384 in the two-point measurement on LLAVA-1.5-7B, with a 192-token reconstruction of the same measurement reported alongside it (Appendix B.5). Greedy decoding is load-bearing rather than convenient: because decoding is deterministic, the model’s own free-running caption *is* its own continuation from its own prefix at any cut, so the cut position is matched exactly rather than approximately.

The prompt is the LLaVA-1.5 v1 conversation format, verbatim, unchanged across every condition:

```

A chat between a curious human and an artificial intelligence assistant. The
assistant gives helpful, detailed, and polite answers to the human’s questions.
USER: <image>
Please describe this image in detail. ASSISTANT:
  
```

with one space after the system string and a newline after `<image>`. Image preprocessing is the stock LLaVA-1.5 processor, CLIP ViT-L/14-336, with no overrides. Generation is batched with left padding, batch 16 for captions and 8 for continuations.

LLAVA-OV-0.5B differs in one respect that matters for correctness rather than for style. Under its anyres tiling the prompt length varies per image, from 2,088 to 3,714 tokens, so the processor itself left-pads; Appendix A.7 gives the gate that catches a hand-rebuilt attention mask attending to that padding.

## A.2 IMAGES

All images are COCO val2017. An image is eligible if it has at least one instance annotation, at least one human caption and a readable file. Eligible ids are sorted, 500 are drawn without replacement with `default_rng(20260817)`, and the draw is re-sorted by image id. The identical 500 images are used by every LLaVA condition in the two-point measurement; KOSMOS-2 uses 473 of them (Appendix A.1), the dose grid 499, and the generation-regime control a larger sample of 3,500 that nests these 500. Images whose free-running caption is shorter than 32 tokens are excluded from the prefixed conditions; none of the 500 fell below this floor, and the observed minimum is 55 tokens.

One analysis — the generation-regime control of Appendix B.5 — extends the sample to 3,500 images with the original 500 nested inside, giving 10,500 cells and 21,000 rows.

## A.3 STIMULUS CONSTRUCTION

For image  $i$ , let  $L_i$  be the token length of the model’s own greedy caption; the cut is  $c_i(f) = \lfloor f \cdot L_i \rfloor$ . The primary cut is  $f = 0.5$  on LLaVA-1.5-7B, where  $c_i(0.5)$  has mean 58.24 tokens; the two-point measurement there caps each cell’s budget further at the token length of a Localized Narratives description of the first source image, so the realised budget has mean 44.6 (sd 13.7, range 12–87). On LLaVA-OV-0.5B  $f = 0.21$ , with mean prefix length 50.272 tokens. Prefix length is keyed to the image rather than to the condition, so it is identical across conditions at each cut by construction and the contrast is never a length contrast.

The two conditions of the primary contrast differ in exactly one property of the prefix:

- `cap1` — five human COCO captions of **one** other image.
- `cap5` — five human COCO captions of **five** different other images.

Both are built by the same procedure — strip terminal periods, shuffle under a per-image seed, join with ". ", tokenize, and slice the token id list to the budget — except that the one-scene prefixes use the doubling step described next and the five-scene prefixes do not need it (their doubling rate is zero by construction). The dose grid rebuilds the one-scene condition without doubling and lands at +0.1947 against +0.1881 (Appendix D.5). Four properties of that assembler are worth stating precisely, because each has been described loosely elsewhere. If the joined string tokenizes shorter than the budget it is concatenated to itself verbatim and re-checked, with a doubling bound of seven; the loop exits on the bound as well as on the budget, so a prefix can come out shorter than the target. Truncation is a slice of the token id list, so it never cuts inside a token, but because the tokenizer is subword the boundary can fall mid-word and routinely falls mid-sentence. The assembler never sees the image: it takes captions, a length and a seed. And the prefix ids are spliced after the prompt rather than re-encoded with it, so the model is shown a token sequence that was never produced by tokenizing the full string.

Each image contributes three cells, one per derangement of the partner assignment, giving 1,500 cells over 500 images in each condition.

**Realised rather than nominal dose.** The token budget truncates, so the model does not see all five assembled units. Across the whole dose grid, units-as-assembled is 5.000 at every grid point while units-as-seen is flat at 2.671–2.697; what moves is scenes-as-seen, which rises  $0.999 \rightarrow 1.718 \rightarrow 2.131 \rightarrow 2.414 \rightarrow 2.671$ . The manipulation is described as five-and-five throughout because that is what is held fixed; realised figures are stated wherever the contrast is quantified.

**The dose grid’s assembler.** The five-point grid varies the number of distinct source scenes  $S \in \{1, \dots, 5\}$  while holding five caption units at every grid point, under quotas  $[5]/[3, 2]/[2, 2, 1]/[2, 1, 1, 1]/[1, 1, 1, 1, 1]$ , with one assembler throughout, the same per-cell budget, the same join seed, and zero doubling by construction. Its census kept 1,361 of 1,500 cells across 499 images ( $f_{\text{keep}} = 0.907$ ).

**The blind conditions.** `cap1g` and `cap5g` run the identical pipeline with the target image replaced by a flat mid-grey rectangle at the target’s own pixel dimensions, and with prefix token

ids held bit-identical to the corresponding sighted condition. The image-attributable component is  $\Delta_{\text{image}} = \Delta_{\text{primary}} - \Delta_{\text{ablated}}$ , an algebraic identity in its components; what is empirical is its magnitude.

#### A.4 OBJECT EXTRACTION AND GROUND TRUTH

The scorer is a faithful port of the CHAIR object scorer. The 80-category synonym table is used verbatim, including its double-word merges (sports ball, hot dog, baby <animal> to <animal>, passenger jet to jet, bow tie to tie) and the canonical toilet-seat rule. The ground-truth object set  $G(i)$  is the union of image  $i$ 's segmentation category names and every object mined from that image's human captions with the same extractor. An object is *in the image* iff its canonical node lies in  $G(i)$ .

Two portability deviations from the reference implementation are applied identically to every condition, so their first-order bias cancels in every contrast reported: a regex word tokenizer (`[a-z]+` over lowercased text, retaining character offsets) replaces the NLTK tokenizer, and a rule-based singularizer replaces `pattern.singularize`. Absolute rates are therefore not comparable to published CHAIR numbers on these conditions; only within-paper contrasts are. Appendix G.2 bounds what the two substitutions actually touch at the standard free-captioning protocol, where the same port reproduces published levels closely; that is one configuration and does not widen this caveat.

The union construction is deliberately permissive, which makes  $F$  conservative: a permissive  $G(i)$  can only move objects out of the false class. It also has a consequence that determines the endpoint choice. A prefix built from an image's own human captions consists of members of  $G(i)$  by construction, so a prefix-copied mention is arithmetically incapable of being scored as a hallucination. Measured across the grounded conditions of both architectures, 10,608 copied mentions produced exactly 0 hallucinations. A per-mention hallucination rate is therefore structurally undefined on such prefixes, which is why the endpoint below classifies *prefix* objects and compares carry-forward rates within a condition.

**Three constructions.** Every number in the paper is scored under the union. Two further constructions are used for the sensitivity analysis of Appendix B.1: segmentation-only ( $G(i)$  restricted to instance-annotation category names) and caption-only ( $G(i)$  restricted to objects mined from the human captions). The estimator is unmodified across the three; only the ground-truth dictionary is swapped, and the switch asserts its own union branch equal to the published  $G(i)$  per image.

#### A.5 THE ENDPOINT AND ITS UNITS

For each (image, condition) pair we extract the object types named in the prefix,  $P$ , and in the continuation,  $C$ . The unit is the pair (image, object type in  $P$ ), deduplicated within a condition-image, so an object named five times in the prefix contributes exactly one unit, and carry-forward is the binary event that the type appears anywhere in the continuation. That deduplication makes the measurement immune to repetition loops in the continuation; it does not address repetition in the prefix, which is measured directly on the prefix string.

Pooled over units,  $H$  and  $F$  are the carry-forward rates for prefix objects in and not in  $G(i)$ , and  $J = H - F$ . Table 5 gives the realised unit counts. Because the unit set is fixed by the prefix, and the blind conditions hold prefix token ids bit-identical, the blind conditions carry the same unit counts as their sighted counterparts by construction.

The endpoint  $J$  is not pre-registered. It was adopted after CHAIR and the per-mention rate were found defective on these conditions, for the reason given in Appendix A.4; this is recorded in Appendix E among the specification elements selected in the presence of data.

#### A.6 INTERVAL ESTIMATION

All intervals are paired image-level bootstraps clustered on the image. A replicate resamples the image ids with replacement, carries *all* of that image's cells into the replicate together, recomputes both conditions on the resampled set, and the interval is the 2.5/97.5 percentile range over repli-

Table 5: Unit counts per condition. A unit is one (image, prefix object type) pair, deduplicated within a condition-image. The blind conditions share these counts by construction.

| architecture  | condition          | true units | false units | cells |
|---------------|--------------------|------------|-------------|-------|
| LLAVA-1.5-7B  | one scene (cap1)   | 528        | 2,332       | 1,500 |
| LLAVA-1.5-7B  | five scenes (cap5) | 892        | 4,988       | 1,500 |
| LLAVA-OV-0.5B | one scene (cap1)   | 554        | 2,452       | 1,500 |
| LLAVA-OV-0.5B | five scenes (cap5) | 1,020      | 6,308       | 1,500 |

cates. Clustering is required rather than optional, because each image contributes several rows and resampling rows would understate the interval. Multiplicity is preserved — an image drawn three times counts three times — and the resample size is asserted equal to  $n$  on every replicate.

$B = 8000$  for pooled estimates and  $B = 4000$  elsewhere. The two-point measurement uses `default_rng(20260830)`; the blind-baseline and signal-detection analyses use seed 20260912 at  $B = 4000$ . Alternate-seed replications are reported for every primary contrast (Appendix B.6).

#### A.7 INTEGRITY GATES

Six gates run before any number is reported. All are pass/abort rather than advisory.

**Verification gate.** No pipeline may report a new quantity unless it first reproduces previously published per-condition rates to within  $5 \times 10^{-3}$ . Realised margins: the blind analysis’s positive control reproduced the published  $\Delta_{\text{primary}}$  to  $2.9 \times 10^{-5}$  and  $1.8 \times 10^{-5}$  on the two architectures; the signal-detection analysis reproduced the unit counts of Table 5 exactly with rate deviations of 0.0 on all four conditions, and reproduced the published  $\Delta$  intervals to  $3.3 \times 10^{-5}$  and  $2.5 \times 10^{-5}$ .

**Mask-validity gate.** The free-running condition comes straight from the processor, while the prefixed conditions append prefix token ids and rebuild the batch — and its attention mask — by hand. An error in that hand-built mask would corrupt only the prefixed conditions, and would manufacture an effect that is large, clean, internally consistent and entirely spurious; nothing in the resulting numbers would look wrong. The gate is a longest-common-prefix check of the rebuilt encoding against the processor’s own, with a known expected value of 51, and a batch that fails it aborts the condition rather than emitting numbers. The risk is acute under LLAVA-OV-0.5B’s anyres tiling, where the processor itself left-pads.

**Attention gate.** Registered, and required to pass before any outcome can be declared. It compares object recall from the model’s own free caption with a real image against the same with flat grey: 0.839 and 0.781 on the two architectures against 0.000 grey, with an identical-caption rate of 0.000 and mean caption length 116.98 real against 73.01 grey on LLAVA-1.5-7B. The grey condition produces different, shorter, object-free captions rather than no captions.

**Construction and prefix-identity gates.** Every published prefix is rebuilt from its source captions under its per-row seed and compared bit-for-bit; realised drift is 0 of 3,000. Census checksum drift is 0. Per-architecture integrity gates on the blind-baseline analysis pass at 1500/1500 on all six checks (four conditions present, budget, prefix identity, five-scene zero doubling, cap, self).

**Budget and cell-count gates.** Fatal assertions on budget mismatch, cap exceedance, incomplete cells and cell count against the frozen census; none triggered on any reported run.

**Bootstrap multiplicity assertion.** Active in every analysis reported here (Appendix A.6, and see Appendix F.2).

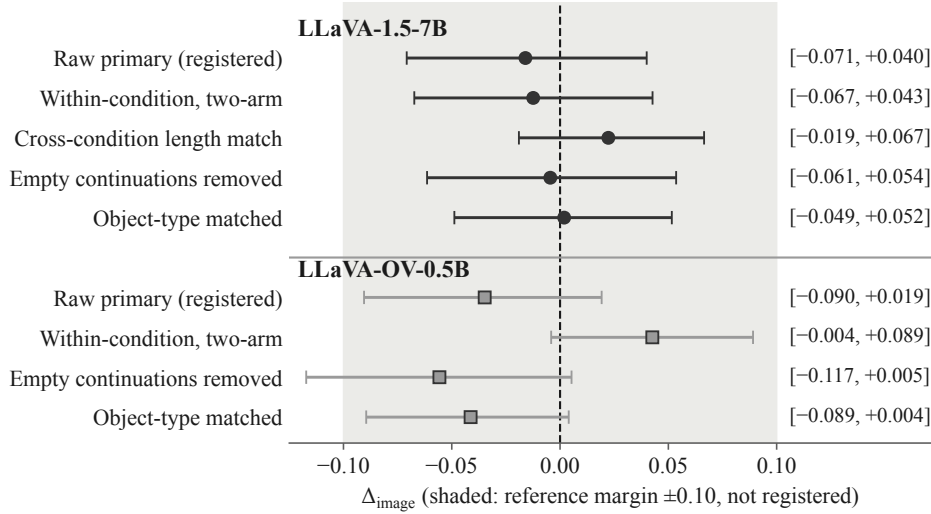


Figure 5: The image-attributable component on the two LLaVA-lineage architectures under every rule in Table 7, against the  $\pm 0.10$  reference margin, which is a reporting choice rather than a registered bound. One interval, LLaVA-OV-0.5B’s with empty continuations removed, extends beyond the margin.

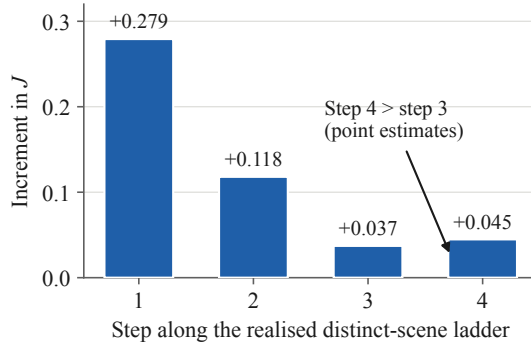


Figure 6: Increments in  $J$  along the realised distinct-scene ladder on LLaVA-1.5-7B. The first step carries 58% of the rise, so the curve is front-loaded; the fourth step’s point estimate slightly exceeds the third’s, so it shows no plateau within the grid. Distinct-type count and mentions-per-type track the manipulation with the same precision, so the ladder is graded in a quantity that co-varies with source-scene count rather than isolating it.

## B ADDITIONAL ROBUSTNESS ANALYSES

Every analysis in this section tests the same quantity:  $\Delta_{\text{image}} = \Delta_{\text{primary}} - \Delta_{\text{ablated}}$ , the part of the contrast attributable to the image. Figure 5 plots a subset of these rows; the full set is here, and Figure 6 shows the dose-response grid the contrast sits on.

### B.1 GROUND-TRUTH CONSTRUCTION

The null does not depend on how the ground truth was built. Re-scoring all four conditions under segmentation-only and caption-only definitions, with the estimator unmodified and only the ground-truth dictionary swapped, gives Table 6.

The largest pairwise spread in  $\Delta_{\text{primary}}$  across the three constructions is 0.0167, which is 3.4% of the effect and roughly a seventh of any one interval’s width, and the two architectures’ shifts point in opposite directions — the signature of noise at this scale rather than of a systematic construction bias.

Table 6: The contrast under three independent ground-truth constructions. The six  $\Delta_{\text{image}}$  point estimates lie between  $-0.042$  and  $-0.0156$ , and in all six  $\Delta_{\text{ablated}}$  exceeds  $\Delta_{\text{primary}}$ ; intervals are shown only for the union construction (Table 7).

| architecture  | quantity                  | union   | segmentation only | captions only |
|---------------|---------------------------|---------|-------------------|---------------|
| LLAVA-1.5-7B  | $\Delta_{\text{primary}}$ | +0.4859 | +0.5015           | +0.4848       |
| LLAVA-1.5-7B  | $\Delta_{\text{ablated}}$ | +0.5019 | +0.5172           | +0.5126       |
| LLAVA-1.5-7B  | $\Delta_{\text{image}}$   | -0.0160 | -0.0156           | -0.0279       |
| LLAVA-OV-0.5B | $\Delta_{\text{primary}}$ | +0.3780 | +0.3682           | +0.3743       |
| LLAVA-OV-0.5B | $\Delta_{\text{ablated}}$ | +0.4128 | +0.4101           | +0.4129       |
| LLAVA-OV-0.5B | $\Delta_{\text{image}}$   | -0.0348 | -0.0420           | -0.0386       |

The arithmetic reason is that reclassification is small and balanced across conditions: dropping the caption-mined half moves 91 of 8,740 units (1.04%) on LLAVA-1.5-7B and 109 of 10,334 (1.05%) on LLAVA-OV-0.5B, affecting 6.4% of union-true units in *both* conditions on LLAVA-1.5-7B and 7.4% and 6.7% on LLAVA-OV-0.5B. A construction that reclassified the two conditions at different rates is what would move a contrast mechanically, and this one does not.

The union row was cross-checked against a second, independently written scorer with its own bootstrap seed, which returns 0.48587082 / 0.50151970 / 0.48475570 against this analysis’s 0.485871 / 0.5015 / 0.4848.

Two constraints travel with this block. That analysis’s own registered decision on the segmentation-only runs is mechanical rather than substantive — its gate compares the recomputed contrast against a union-anchored reference, which a segmentation-only run legitimately misses — and the six paired cross-construction differences are point estimates without intervals (Appendix F.3). They say how far the constructions agree, not how confidently.

## B.2 TRUNCATION AND LENGTH-MATCHING RULES

“Length-matched  $\Delta_{\text{image}}$ ” names more than one estimand, and the distinction is consequential, so every rule we ran is reported. Let  $L$  be the per-cell truncation length at which continuations are re-scored. A *within-condition* rule sets  $L$  separately inside the sighted and the blind halves; a *cross-condition* rule sets one  $L$  per cell and applies it to both halves, so the sighted and blind continuations are evaluated at the same length. Only the cross-condition rule is length-matched in the sense the phrase implies.

The four truncation rules are the raw (unmatched) rule, the within-condition rule, the cross-condition rule and the empty-continuation rule; the type-matched row is a matching rule rather than a truncation rule and is discussed in Appendix B.3.

The empty-continuation rows run in opposite directions on the two architectures — closer to zero on LLAVA-1.5-7B, further from it on LLAVA-OV-0.5B, and containing zero in both — so the empty-rate asymmetry between conditions does not explain the null in either direction.

**A fifth rule, and why it is not in the table.** A replication analysis computed a within-condition length match in which  $L$  is minimised over *four* conditions, two of which are not in the contrast, and reports  $+0.0395$  [ $+0.0050$ ,  $+0.0754$ ]. We report it here because it is in the released artefacts and because it is the one length-matched estimate that excludes zero. It should not be read as a truncation rule for this contrast, for a reason that is checkable arithmetically. Under that rule the truncation depth is itself a function of the manipulation being contrasted: removing the image shortens continuations, so the blind half is truncated on 43.7% of cells against the sighted half’s 23.6%. The differential shrinkage is  $0.1020 - 0.0667 = 0.0353$ , against an observed discrepancy of  $+0.0395 - (+0.0042) = +0.0353$ , where  $+0.0042$  [ $-0.0335$ ,  $+0.0427$ ] is that analysis’s cells recomputed under the registered two-condition rule. The cross-condition version on those cells reads  $-0.0140$  [ $-0.0452$ ,  $+0.0172$ ], the narrowest of the three. The registered raw primary is untouched by any of this; the replication analysis computes it on its own cell set, at  $-0.0104$  against the registered  $-0.0160$ .

Table 7:  $\Delta_{\text{image}}$  under each truncation and matching rule. Every interval on the two LLAVA architectures contains zero. The full ladder was run only on those two; for the other two we give the registered primary, and for KOSMOS-2 the length-matched form as well. That match is within each condition only: blind continuations on KOSMOS-2 run about 1.7 times longer than sighted ones, no sighted-against-blind match was run, and the design cannot separate the image from length. The two KOSMOS-2 intervals exclude zero. Rows for the two LLAVA architectures come from the registration run; the last three, and Table 12, come from the run that measured all four architectures together. Point estimates agree between the runs; interval endpoints differ in the third decimal because the bootstrap seeds differ, and both runs are released.

| architecture  | rule                                          | $\Delta_{\text{image}}$ | 95% CI             |
|---------------|-----------------------------------------------|-------------------------|--------------------|
| LLAVA-1.5-7B  | raw, no truncation match (registered primary) | -0.0160                 | [-0.0708, +0.0400] |
| LLAVA-1.5-7B  | within-condition, two-condition $L$           | -0.0124                 | [-0.0673, +0.0427] |
| LLAVA-1.5-7B  | cross-condition $L$                           | +0.0223                 | [-0.0190, +0.0665] |
| LLAVA-1.5-7B  | empty continuations removed                   | -0.0045                 | [-0.0614, +0.0536] |
| LLAVA-1.5-7B  | type-matched (mention-rate control)           | +0.0019                 | [-0.0488, +0.0516] |
| LLAVA-OV-0.5B | raw, no truncation match (registered primary) | -0.0348                 | [-0.0905, +0.0192] |
| LLAVA-OV-0.5B | within-condition, two-condition $L$           | +0.0426                 | [-0.0041, +0.0891] |
| LLAVA-OV-0.5B | empty continuations removed                   | -0.0557                 | [-0.1172, +0.0053] |
| LLAVA-OV-0.5B | type-matched (mention-rate control)           | -0.0413                 | [-0.0895, +0.0040] |
| QWEN2.5-VL-7B | raw, no truncation match (registered primary) | +0.0157                 | [-0.0226, +0.0552] |
| KOSMOS-2      | raw, no truncation match (registered primary) | -0.1028                 | [-0.1677, -0.0360] |
| KOSMOS-2      | length-matched                                | -0.0873                 | [-0.1502, -0.0250] |

### B.3 THE MENTION-RATE CONTROL

The strongest competing explanation for the blind null is that a shift in overall mention rate, rather than blinding, drives it. The threat is real: the one-scene to five-scene increase in mention count is substantially larger in the blind condition than in the sighted one (sighted minus blind,  $-12.68$   $[-14.77, -10.56]$ , on LLAVA-1.5-7B).

The pre-registered control holds distinct object *types* fixed rather than mentions, because carry-forward is set membership and cannot depend on multiplicity. The continuations make the distinction concrete: on LLAVA-1.5-7B they average 26.59 object mentions across only 1.515 distinct types, so most mentions are re-namings that cannot change any carry outcome. Type-matched, the null holds on both architectures (Table 7, last row of each block).

Table 8: The adversarial re-analysis. The registered verdict passes on all fourteen powered blocks on LLAVA-OV-0.5B and fails on five of sixteen powered blocks on LLAVA-1.5-7B — four mention-count strata in the positive direction and one type-matched block in the negative direction; both architectures are reported.

| block                                           | estimate                      | reading      |
|-------------------------------------------------|-------------------------------|--------------|
| type-matched, LLAVA-1.5-7B                      | +0.0019 [-0.0488, +0.0516]    | passes       |
| type-matched, LLAVA-OV-0.5B                     | -0.0413 [-0.0895, +0.0040]    | passes       |
| mention-count strata, LLAVA-1.5-7B              | +0.093 to +0.107, four blocks | <b>fails</b> |
| type-matched block ( $\tau = 2$ ), LLAVA-1.5-7B | -0.0688 [-0.1343, -0.0057]    | <b>fails</b> |
| all fourteen powered blocks, LLAVA-OV-0.5B      | —                             | passes       |
| repeat-mention probe, LLAVA-1.5-7B              | +0.1144 [+0.0519, +0.1734]    | diagnostic   |
| conditioning on types directly                  | +0.002                        | diagnostic   |

Four mention-count-stratified blocks on LLAVA-1.5-7B come out positive, in the direction pre-specified as adverse. We report the failure rather than amend the rule, and we report the diagnostic that indicates why it occurs. Stratifying on repeat mentions alone — the quantity mentions — types, which by construction cannot change any carry outcome — reproduces the effect more strongly (+0.1144) than the mention-count stratification does, while conditioning on types directly gives

+0.002. A variable incapable of affecting the endpoint therefore reproduces the effect better than the variable of interest, which is what one expects if the stratifier is a proxy for something else. It is: every mention stratum above roughly sixteen on LLAVA-1.5-7B is entirely cap-truncated, so high mention count there marks “reached the token cap and looped” rather than verbosity.

**The smallest ratio across strata.** Across all adequately powered blocks, the smallest point estimate of  $\Delta_{\text{ablated}}/\Delta_{\text{primary}}$  is 0.7333 on LLAVA-1.5-7B and 0.9074 on LLAVA-OV-0.5B. These are minima of point estimates, not confidence bounds; the ratio on the full sample is 1.033 [0.922, 1.155]. What we claim is that the image-attributable component of  $J$  is null at our power — not that the image adds nothing detectable, which Appendix D.4 shows directly to be false.

#### B.4 ONE BLOCK THAT RUNS IN OUR FAVOUR AND IS UNEXPLAINED

A type-restricted block on LLAVA-1.5-7B returns  $\Delta_{\text{image}} = -0.0688$  [−0.1343, −0.0057] and is systematic rather than sampling noise: it sits at the 0.27th percentile of same-size random subsets of the same cells. The repeat-mention probe does not reproduce it, no truncation account explains it, and LLAVA-OV-0.5B shows nothing comparable (the corresponding block sits at the 55.5th percentile). A negative  $\Delta_{\text{image}}$  means the image acts in a direction nobody predicted, and it flatters the conclusion of this paper, which is a reason for more suspicion rather than less. The cells this block excludes — those in which the five-caption prefix raised the count of distinct types sharply — return  $\Delta_{\text{image}} = +0.1264$  [+0.0143, +0.2432], so the pooled null combines subgroups of opposite sign; this split was examined after the fact. Both halves are defined by a post-treatment quantity, so neither half is a causal contrast and no re-weighting of these data will settle it; resolving it requires a design that manipulates prefix type-content directly.

#### B.5 GENERATION REGIME

The five-scene condition generates longer continuations and reaches the generation cap more often (share at the budget 0.679 against the one-scene condition’s 0.447), so a control that removes truncation from the comparison is necessary. Requiring that *neither* condition reach the budget keeps 21.5% of cells at 500 images (323 of 1,500) and drops the true-unit count to 75 against a registered floor of 100; the contrast is then +0.4667 [+0.3129, +0.6134] and underpowered by its own rule. Raising the budget from 192 to 384 tokens does not fix this — truncation barely moves (share at the budget 0.448/0.677 at 384 against 0.447/0.679 at 192). Adding images does.

Table 9: The generation-regime control at power: the contrast measured only on cells in which neither condition reached the generation cap, at 3,500 images. On a subset where neither condition reached the budget, greedy decoding makes the 192- and 384-token continuations identical, so that row is a determinism check rather than an independent replication.

| estimate                                                 | $\Delta$ | 95% CI             | width  |
|----------------------------------------------------------|----------|--------------------|--------|
| both conditions untruncated, 3,500 images (primary)      | +0.3460  | [+0.2849, +0.4058] | 0.1209 |
| the same, length-matched                                 | +0.1999  | [+0.1462, +0.2524] | 0.1062 |
| the same, 192-token reconstruction (determinism check)   | +0.3460  | [+0.2849, +0.4058] | 0.1209 |
| the same, alternate bootstrap seed                       | +0.3460  | [+0.2876, +0.4050] | 0.1174 |
| composition control, base-rate tertiles                  | +0.1994  | [+0.1059, +0.2897] | 0.1838 |
| composition control, shared object types                 | +0.3496  | [+0.2923, +0.4168] | 0.1245 |
| all cells, raw, 3,500 images                             | +0.4719  | [+0.4507, +0.4928] | 0.0421 |
| all cells, length-matched, 3,500 images                  | +0.3679  | [+0.3478, +0.3872] | 0.0394 |
| original 500 images only, both untruncated (descriptive) | +0.4663  | [+0.3139, +0.6124] | 0.2985 |

The control passes: interval width 0.1209 against a registered ceiling of 0.130, true units 466/914 against a floor of 100, on 2,089 of 10,500 cells. The untruncated subset is short-continuation by construction (mean 28.9 tokens in the one-scene condition against 40.3 in the five-scene condition), so a positive verdict says the effect exists *within* the regime-matched population and no more.

Two facts bear on this estimate. The confirmatory estimate is 26% below the underpowered 500-image estimate that motivated running it, and the confirmatory interval *excludes* that point estimate

while the underpowered interval comfortably contains the confirmatory one. An arithmetic decomposition of the reported unit counts puts the 3,000 extension images alone at +0.3237; that figure carries no interval and enters no registered decision, and is given because it makes the shrinkage concrete. The frozen original stimulus set replicates cleanly alongside it: the original all-cells contrast of +0.4859 reproduces at +0.4877 [+0.4298, +0.5434] on the regenerated cells, and the regenerated untruncated subset gives +0.4663 on 321 cells at 76 true units against the original’s +0.4667 on 323 cells at 75.

The same control is unevaluated on LLAVA-OV-0.5B, where the untruncated subset reaches 90 true units on 287 of 1,500 cells against the same floor of 100. It points the same way as that architecture’s primary (+0.2830) and may not be read as support. The dose grid’s own regime control is likewise unevaluated: truncation rises monotonically with the dose (share at the budget 0.436  $\rightarrow$  0.655, Spearman  $\rho = +1.0$ ), and the registered all-untruncated control reaches 150 cells with true-unit counts between 29 and 72 against a floor of 100.

## B.6 SEEDS AND RECONSTRUCTIONS

Table 10: Alternate-seed and alternate-budget replications of the primary contrasts.

| quantity                                                            | estimate | 95% CI             |
|---------------------------------------------------------------------|----------|--------------------|
| $\Delta_{\text{primary}}$ , LLAVA-OV-0.5B, alternate bootstrap seed | +0.3780  | [+0.3233, +0.4352] |
| regime control, alternate bootstrap seed                            | +0.3460  | [+0.2876, +0.4050] |
| regime control, 192-token reconstruction                            | +0.3460  | [+0.2849, +0.4058] |
| forced-choice contrast, alternate bootstrap seed                    | −0.0095  | [−0.0312, +0.0118] |

## C SIGNAL-DETECTION DERIVATION

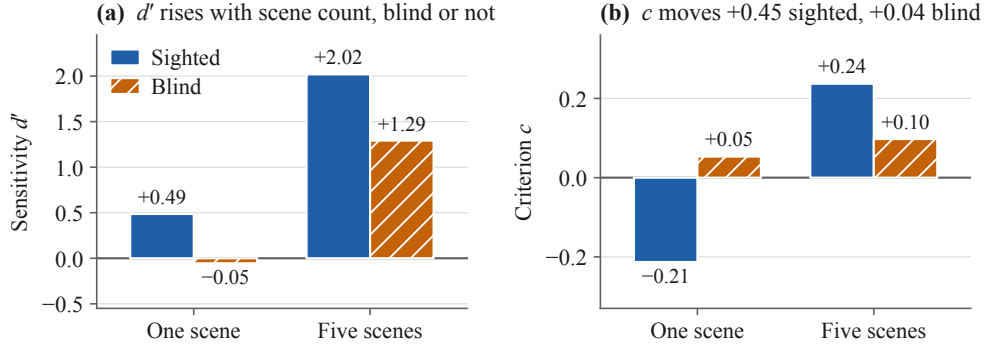


Figure 7: Signal-detection decomposition on LLAVA-1.5-7B. Sensitivity rises with scene count whether or not the image is present; the criterion moves mainly when it is.

This section derives the relationship the main text relies on, and states the assumptions it needs; Figure 7 shows the operating points the derivation is evaluated at. Write  $H$  for the carry-forward rate on prefix objects present in the image and  $F$  for the rate on absent ones. Let  $\Phi$  be the standard normal distribution function,  $\varphi$  its density, and  $Z = \Phi^{-1}$ .

### C.1 THE REPARAMETERISATION

The three summaries are

$$J = H - F, \quad (2)$$

$$d' = Z(H) - Z(F), \quad (3)$$

$$c = -\frac{1}{2} [Z(H) + Z(F)]. \quad (4)$$

Equations 3–4 invert exactly:

$$Z(H) = \frac{d'}{2} - c, \quad Z(F) = -\frac{d'}{2} - c, \quad (5)$$

so that

$$H = \Phi\left(\frac{d'}{2} - c\right), \quad F = \Phi\left(-\frac{d'}{2} - c\right), \quad J(d', c) = \Phi\left(\frac{d'}{2} - c\right) - \Phi\left(-\frac{d'}{2} - c\right). \quad (6)$$

The map  $(H, F) \mapsto (d', c)$  is a bijection from  $(0, 1)^2$  onto  $\mathbb{R}^2$ : two observed rates, two parameters. On a single operating point the equal-variance Gaussian model is therefore *saturated*. Nothing is estimated beyond the two rates, and  $d'$  and  $c$  are re-expressions of them rather than fits — a point we return to in Appendix C.4, because it cuts both ways.

## C.2 $J$ IS NOT INVARIANT TO THE CRITERION

Differentiating equation 6 in  $c$  at fixed  $d'$ , and using the symmetry of  $\varphi$ ,

$$\frac{\partial J}{\partial c} = \varphi\left(\frac{d'}{2} + c\right) - \varphi\left(\frac{d'}{2} - c\right) = -\frac{2}{\sqrt{2\pi}} \exp\left(-\frac{d'^2/4 + c^2}{2}\right) \sinh\left(\frac{cd'}{2}\right). \quad (7)$$

This vanishes if and only if  $cd' = 0$ . For  $d' > 0$  it is strictly negative for  $c > 0$  and strictly positive for  $c < 0$ , so at fixed  $d'$  the statistic  $J$  increases in  $c$  on  $c < 0$ , decreases on  $c > 0$ , and attains a unique interior maximum at  $c = 0$ :

$$J^*(d') = \max_c J(d', c) = 2\Phi\left(\frac{d'}{2}\right) - 1, \quad \text{so} \quad J(d', c) \leq 2\Phi\left(\frac{d'}{2}\right) - 1 \quad (8)$$

with equality if and only if  $c = 0$ . The right-hand side of equation 8 is Youden's index of the equal-variance Gaussian ROC with separation  $d'$ .

Equation 8 is the precise sense in which a present-minus-absent rate both is and is not criterion-free. Youden's index, as a summary of an ROC *curve*, is criterion-free by definition: it is the maximum over operating points, and a maximum does not depend on where on the curve one happens to sit. A present-minus-absent rate measured at a single, uncontrolled operating point equals that maximum only when the operating point happens to be unbiased, and otherwise under-reports it by an amount that is itself a function of the criterion. Free generation supplies no mechanism that pins the operating point at  $c = 0$ , and Appendix D.4 shows that in our data it is not pinned there.

The local behaviour explains why the confusion is durable. Since  $\varphi'(x) = -x\varphi(x)$ ,

$$\left.\frac{\partial^2 J}{\partial c^2}\right|_{c=0} = -d' \varphi\left(\frac{d'}{2}\right), \quad \text{hence} \quad J(d', c) = 2\Phi\left(\frac{d'}{2}\right) - 1 - \frac{1}{2} d' \varphi\left(\frac{d'}{2}\right) c^2 + O(c^4). \quad (9)$$

The leading criterion dependence is quadratic, not linear. A criterion shift confined to a neighbourhood of the unbiased point costs  $J$  only at second order, which is why the statistic is routinely described as bias-corrected. A criterion shift between two points that are not symmetric about zero moves  $J$  at first order, with the slope given by equation 7.

## C.3 THE CONDITIONS UNDER WHICH IT WOULD BE INVARIANT

From equation 7,  $J$  is *exactly* invariant to the criterion in three cases and no others.

1.  $d' = 0$ , where  $J \equiv 0$  for every  $c$ . This is the degenerate case, not a useful one.
2.  $c$  held fixed across the compared conditions. Then  $\Delta J$  is a function of  $\Delta d'$  and of the common  $c$  alone. This is the condition an experiment can actually arrange, and arranging it is what a response-rate control does.
3. Criteria symmetric about the unbiased point,  $c_2 = -c_1$ , since  $J$  is even in  $c$  at fixed  $d'$ .

It is *approximately* invariant whenever  $\frac{1}{2}d'\varphi(d'/2)c^2$  falls below the resolution of interest. The coefficient  $d'\varphi(d'/2)$  rises from 0.193 at  $d' = 0.5$  to 0.484 at  $d' = 2.0$ . At  $d' = 1.5$ , exact evaluation of equation 6 gives a cost of 0.0023 in  $J$  for a criterion displacement of  $|c| = 0.1$  and 0.054 for  $|c| = 0.5$ .

**The complementary exact statement, which needs no distributional assumption.** On the rate scale, if both rates receive a common additive shift  $\delta$ , then  $J$  is exactly unchanged while  $d'$  is not, unless  $H = F$ . On the probit scale the roles reverse: a pure criterion move at fixed  $d'$  is a common additive shift of  $Z(H)$  and  $Z(F)$ , under which  $d'$  is exactly unchanged while  $J$  is not, unless  $c = 0$ . The two summaries are exactly invariant to different perturbations, and neither is invariant to the other's. Which of those two invariances an evaluation wants is a substantive choice about what the evaluation is for; reporting only  $J$  makes that choice silently.

#### C.4 WHERE THE EQUAL-VARIANCE GAUSSIAN ASSUMPTION IS AND IS NOT WARRANTED

The assumption is warranted in three respects and unwarranted in four, and we state both halves because the derivation above is only as good as the model it sits in.

**Warranted.** (i) *Nothing is fitted.* On one operating point the map of equation 6 is a bijection, so  $d'$  and  $c$  consume exactly the information in  $H$  and  $F$ . Every statement above is checkable directly from the two rates, and the values in Table 11 are verified to be so. (ii) *No boundary pathology.* Rates are log-linear corrected,  $(\text{hits} + 0.5)/(n_{\text{signal}} + 1)$ , applied unconditionally to every condition so that none is selectively corrected; no boundary-rate events were recorded, so no  $Z$  is evaluated at an endpoint. (iii) *The qualitative conclusion is link-free.* Under a logit link the same two contrasts read  $+0.4207$  [ $+0.1498$ ,  $+0.6911$ ] on the discriminability coordinate and  $+0.6953$  [ $+0.5448$ ,  $+0.8478$ ] on the criterion coordinate, so sign and exclusion of zero are not probit artefacts. More generally, the argument of Appendix C.3 needs only that the reparameterisation separate a discriminability coordinate from a propensity coordinate; the Gaussian choice fixes which coordinates, not whether the confound exists.

**Not warranted.** (i) *There is no  $z$ -ROC.* Equal variance is testable only with several criterion points per condition. A binary carry outcome at one operating point supplies a single  $(H, F)$  pair, so the model is saturated and the equal-variance assumption carries no degree of freedom against which it could fail. Unequal variance would change the numerical values of  $d'$  and  $c$ ; it would not change the fact that  $J$  depends on both. (ii) *There is no explicit decision threshold.* Carry-forward is not a thresholded judgement on a latent evidence axis. It is an emergent property of autoregressive generation: whether a type named in the prefix reappears anywhere in a continuation produced under a fixed token budget. The latent-variable story is a description of two rates, not a mechanism claim, and should not be read as one. (iii) *Units within an image are not independent.* Object types drawn from one prefix compete for one continuation and one token budget. Deduplication removes repetition *within* a continuation but not this competition. The intervals respect the dependence, since they cluster on the image (Appendix A.6); the Gaussian model's unit-level independence does not. (iv) *The budget binds.* With shares at the budget of 0.447 and 0.679 in the two conditions of LLAVA-1.5-7B, a large fraction of continuations end at the budget, so a unit's carry probability depends on how many other units compete for the remaining budget. A capacity constraint of that kind is not representable as a threshold on a per-unit evidence variable, and it cannot be absorbed into either  $d'$  or  $c$ .

Finally, the cancellation of a response-propensity shift out of  $J$  is *exact* only under an additive shift of both rates, and only *approximate* under an equal-variance Gaussian model of the same shift. The same qualification attaches to the component analysis of Appendix D.4.

#### C.5 OPERATING POINTS IN SIGNAL-DETECTION COORDINATES

Figure 8 places the four cells of the main contrast in  $z$ -coordinates, where a change that moves along a constant- $d'$  contour is a pure threshold shift and one that crosses contours is a change in separability. It is a reading aid for Table 4 and adds no new quantity. In panel (a) both arms cross contours between the one-scene and five-scene conditions ( $d'$  rises 1.53 sighted and 1.35 blind), so the rise is largely a change of separability with or without the image. In panel (b) PAI moves nearly parallel to the contours, a large  $\Delta c$  beside a  $\Delta d'$  whose interval contains zero.

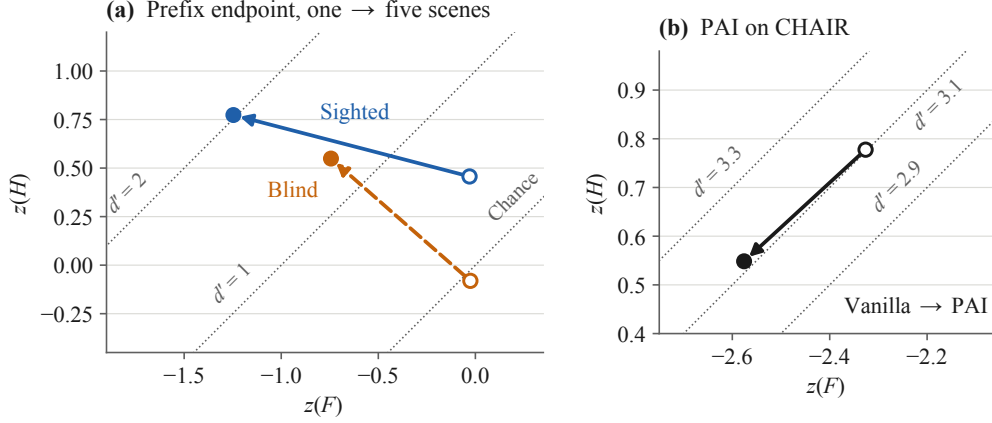


Figure 8: Operating points in  $z$ -coordinates. Dotted lines hold  $d' = z(H) - z(F)$  constant, so height above the chance diagonal is separation and movement parallel to it is a change of threshold. (a) The scene-count rise on LLaVA-1.5-7B, from one scene (open) to five (filled), moves away from the diagonal almost as far without the image as with it (post hoc for these conditions). (b) PAI on standard CHAIR moves almost parallel to the diagonal: on this probit scale the threshold changes and separation does not, although a logit separation measure does rise (Section 3).

#### C.6 NUMERICAL AUDIT AT THE REPORTED OPERATING POINTS

Table 11 evaluates equation 6, equation 7 and equation 8 at the per-condition  $d'$  and  $c$  reported for LLaVA-1.5-7B, and compares the result against the independently measured  $J$ . The reconstruction is not a new measurement, but the identity equation 6 applied to values reported elsewhere in this paper, and it is included because a reviewer can and should check it.

Table 11: Reconstruction of the measured  $J$  from the reported  $d'$  and  $c$  on LLaVA-1.5-7B, via  $J(d', c) = \Phi(d'/2 - c) - \Phi(-d'/2 - c)$ . Residuals are at the resolution of the four-decimal  $d'$  and  $c$  inputs.  $J^*$  is the criterion-free Youden index at the same  $d'$ .

| condition            | $d'$    | $c$     | $J(d', c)$ | meas. $J$ | residual | $J^*(d')$ | $\partial J / \partial c$ |
|----------------------|---------|---------|------------|-----------|----------|-----------|---------------------------|
| sighted, one scene   | +0.4861 | -0.2130 | +0.1878    | +0.18814  | -0.00033 | +0.1920   | +0.039                    |
| sighted, five scenes | +2.0183 | +0.2371 | +0.6736    | +0.67401  | -0.00038 | +0.6871   | -0.113                    |
| blind, one scene     | -0.0549 | +0.0532 | -0.0219    | -0.02191  | +0.00004 | -0.0219   | +0.001                    |
| blind, five scenes   | +1.2918 | +0.0975 | +0.4797    | +0.47997  | -0.00032 | +0.4817   | -0.041                    |

Three things follow, and the third is the one that matters.

First, the reconstruction reproduces all four measured gaps to within  $4 \times 10^{-4}$ , which is the resolution of the quoted inputs. The signal-detection and  $J$  analyses are therefore describing the same four cells, and the reader may treat  $(d', c)$  and  $(H, F)$  as interchangeable coordinates on this data.

Second, the operating points are close to, but not at, the unbiased point. The measured  $J$  falls short of the criterion-free Youden index  $J^*$  by 0.0039, 0.0131, 0.0000 and 0.0017 in the table’s row order (for the blind one-scene cell, where  $d' < 0$ ,  $J^*$  is the most negative value attainable rather than the largest), so at these operating points  $J$  under-reports the discriminability available at the same  $d'$ , by an amount that is itself a function of the criterion. This is the second-order regime of equation 9.

Third, decomposing  $\Delta_{\text{image}}$ . Evaluating equation 6 with the reported  $d'$  values but  $c$  set to zero in all four cells gives contrasts of +0.4951 sighted and +0.5036 blind, so  $\Delta_{\text{image}} = -0.0085$  on the discriminability coordinate alone; the criterion terms supply the remaining  $-0.0075$  of the reported  $-0.0160$ . On these data the *movement* of  $J$  is therefore carried predominantly by  $d'$ , and the criterion coordinate contributes about 0.007 to a contrast of 0.486. This sharpens the diagnosis rather than softening it. The criterion moves substantially in the sighted conditions, from  $-0.213$  to  $+0.237$ , and barely at all in the blind ones, from  $+0.053$  to  $+0.098$ , and the image-attributable component of the

criterion contrast is  $+0.4057$  [ $+0.3157$ ,  $+0.4957$ ] — *larger* than the image-attributable component of discriminability,  $+0.1855$  [ $+0.0252$ ,  $+0.3463$ ]. Both of those image effects are invisible in  $J$ : the criterion effect because  $J$  is stationary in  $c$  near the unbiased point (equation 9), and the discriminability effect because the map  $d' \mapsto 2\Phi(d'/2) - 1$  is concave, its slope  $\varphi(d'/2)$  falling from 0.387 at  $d' = 0.5$  to 0.242 at  $d' = 2.0$ , so the same increment in  $d'$  buys less  $J$  at the higher  $d'$  at which the sighted conditions sit. An endpoint that is stationary in one coordinate by construction and compressive in the other is not a measure of either.

## D ADDITIONAL MODEL RESULTS

Table 12: The rise with and without the image, for the four architectures in this group. The blind rise is positive in every case. The image-attributable component lies within  $\pm 0.10$  on the first three and excludes zero on KOSMOS-2, where removing the image *increases* the rise.  $^\dagger$ QWEN2.5-VL-7B produces a median continuation of four tokens (five on its blind one-scene arm), which leaves a carry-forward endpoint almost nothing to register. Its registration states that it does not constitute a replication, and the main text does not count it. We did not verify that this length reflects the model rather than our prompt and chat template for it, so we do not interpret the row.

| Architecture             | Sighted $\Delta_{\text{primary}}$ | Blind $\Delta_{\text{ablated}}$ | $\Delta_{\text{image}}$                               |
|--------------------------|-----------------------------------|---------------------------------|-------------------------------------------------------|
| LLAVA-1.5-7B             | $+0.4859$                         | $+0.5019$                       | $-0.0160$ [ $-0.0701$ , $+0.0399$ ]                   |
| LLAVA-OV-0.5B            | $+0.3780$                         | $+0.4128$                       | $-0.0348$ [ $-0.0906$ , $+0.0179$ ]                   |
| QWEN2.5-VL-7B $^\dagger$ | $+0.1598$                         | $+0.1441$                       | $+0.0157$ [ $-0.0226$ , $+0.0552$ ]                   |
| KOSMOS-2                 | $+0.2148$                         | $+0.3176$                       | <b><math>-0.1028</math></b> [ $-0.1677$ , $-0.0360$ ] |

### D.1 PER-CONDITION RATES AND CONTRASTS

Table 13: Per-condition carry-forward rates and gaps. Unit counts are in Table 5; the blind conditions share them by construction.

| architecture  | condition            | $H$    | $F$    | $J$                                 |
|---------------|----------------------|--------|--------|-------------------------------------|
| LLAVA-1.5-7B  | sighted, one scene   | 0.6761 | 0.4880 | $+0.1881$ [ $+0.1416$ , $+0.2362$ ] |
| LLAVA-1.5-7B  | sighted, five scenes | 0.7803 | 0.1063 | $+0.6740$ [ $+0.6410$ , $+0.7072$ ] |
| LLAVA-1.5-7B  | blind, one scene     | 0.4678 | 0.4897 | $-0.0219$                           |
| LLAVA-1.5-7B  | blind, five scenes   | 0.7085 | 0.2285 | $+0.4800$ [ $+0.4441$ , $+0.5148$ ] |
| LLAVA-OV-0.5B | sighted, one scene   | 0.5578 | 0.5171 | $+0.0406$                           |
| LLAVA-OV-0.5B | sighted, five scenes | 0.5471 | 0.1284 | $+0.4187$                           |
| LLAVA-OV-0.5B | blind, one scene     | 0.2058 | 0.3230 | $-0.1172$                           |
| LLAVA-OV-0.5B | blind, five scenes   | 0.3922 | 0.0965 | $+0.2956$ [ $+0.2632$ , $+0.3271$ ] |

A model that cannot see the image at all posts a large positive  $J$ :  $+0.4800$  and  $+0.2956$  in the five-scene blind conditions of the two architectures. Table 14 gives the contrasts and their pre-registered controls.

**The doubling check.** The assembler of Appendix A.3 reaches its budget by self-concatenation, and on LLAVA-OV-0.5B the budget is close to twice what five COCO captions fill, so the one-scene condition’s prefixes are frequently doubled text while the five-scene condition’s are not. Doubling is detected on the prefix string itself — whether its leading half reappears verbatim, plus an 8-gram distinctness floor — rather than inferred from the assembler’s counter. Removing every doubled prefix leaves the contrast at  $+0.3733$  [ $+0.3162$ ,  $+0.4310$ ] against a raw  $+0.3780$  [ $+0.3213$ ,  $+0.4333$ ], so LLAVA-OV-0.5B’s result is not a duplication artefact.

**Why the composition controls are load-bearing.** Five source scenes contribute roughly twice the prefix objects of one: a mean of 3.92 distinct object types against 1.91 on LLAVA-1.5-7B. Length

Table 14: The primary contrast and its controls on both architectures. The two point estimates differ (+0.4859 against +0.3780), but their raw intervals overlap narrowly, as do the length-matched ones.

| contrast                                          | $\Delta$ | 95% CI             |
|---------------------------------------------------|----------|--------------------|
| LLAVA-1.5-7B                                      |          |                    |
| primary, raw                                      | +0.4859  | [+0.4283, +0.5417] |
| co-primary, length-matched                        | +0.3558  | [+0.3045, +0.4082] |
| composition control, base-rate tertiles           | +0.3327  | [+0.2212, +0.4367] |
| composition control, shared object types          | +0.4696  | [+0.4153, +0.5261] |
| blind                                             | +0.5019  | [+0.4441, +0.5565] |
| $\Delta_{\text{ablated}}/\Delta_{\text{primary}}$ | 1.033    | [0.922, 1.155]     |
| LLAVA-OV-0.5B                                     |          |                    |
| primary, raw                                      | +0.3780  | [+0.3213, +0.4333] |
| co-primary, length-matched                        | +0.2871  | [+0.2375, +0.3365] |
| object-type-matched                               | +0.3655  | [+0.3129, +0.4276] |
| doubled prefixes removed                          | +0.3733  | [+0.3162, +0.4310] |
| empty continuations removed                       | +0.3905  | [+0.3311, +0.4478] |
| blind                                             | +0.4128  | [+0.3609, +0.4640] |
| $\Delta_{\text{ablated}}/\Delta_{\text{primary}}$ | 1.092    | [0.954, 1.265]     |

matching is likewise not optional, because the five-scene condition generates longer text and reaches the budget far more often (0.679 against 0.447); it costs 27% of the effect and threatens neither its sign nor its power.

**What the composition controls do not fix.** Every composition control in this paper matches on the object *set*, and the set operation discards per-occurrence counts, so these controls fix which object types appear and never how often each is stated. Measured on the frozen prefixes themselves, with no model in the loop (Table 15), total object mentions are nearly conserved across the two conditions while the count of distinct types roughly doubles, so mentions per type falls reciprocally. That quantity is unmatched in every contrast above, and it is part of the reason the contrast is named by its manipulation rather than by a mechanism.

Table 15: Stimulus-side census on the frozen prefixes of LLAVA-1.5-7B, 1,500 cells, no model in the loop.

| per prefix                                      | one scene | five scenes |
|-------------------------------------------------|-----------|-------------|
| total object mentions                           | 4.933     | 4.829       |
| distinct object types                           | 1.907     | 3.920       |
| mentions per type                               | 2.623     | 1.248       |
| fraction of prefixes containing a restated type | 0.902     | 0.539       |

## D.2 THE IMAGE’S CONTRIBUTION WITHIN EACH CONDITION

The blind null concerns the *change* in the image’s contribution between conditions, not whether the image contributes. Table 16 gives the contribution itself: sighted minus blind  $J$  within each condition, from the same run and bootstrap ( $B = 4000$ ) as Table 18. It is large and excludes zero in all four cells, which shows the design detects image effects of this size; it is nearly equal across conditions, which is why  $\Delta_{\text{image}}$  is null. The image’s contribution to  $d'$  also excludes zero in every cell (Table 18) but is *not* equal across conditions, and the image’s contributions to  $H$  and  $F$  separately differ by condition and offset in  $J$  (Table 4).

## D.3 WHICH RATE MOVES

At matched length the false-object carry-forward rate falls from 0.463 to 0.077 — a factor of six — while the true-object rate does not move at a resolution this design can detect. In raw, unmatched

Table 16: The image’s contribution to  $J$  within each condition: sighted minus blind, with 95% clustered bootstrap intervals.  $\Delta_{\text{image}}$  is the five-scene row minus the one-scene row.

| architecture  | condition   | sighted – blind | 95% CI             |
|---------------|-------------|-----------------|--------------------|
| LLAVA-1.5-7B  | one scene   | +0.2100         | [+0.1635, +0.2584] |
| LLAVA-1.5-7B  | five scenes | +0.1940         | [+0.1591, +0.2316] |
| LLAVA-OV-0.5B | one scene   | +0.1579         | [+0.1119, +0.2041] |
| LLAVA-OV-0.5B | five scenes | +0.1230         | [+0.0921, +0.1538] |

Table 17: Decomposition of the contrast into its two rates, at matched length on LLAVA-1.5-7B. Each row is the five-scene value minus the one-scene value.

| half                 | $\Delta$ | 95% CI             |
|----------------------|----------|--------------------|
| $H$                  | −0.0301  | [−0.0760, +0.0153] |
| $F$                  | −0.3859  | [−0.4086, −0.3650] |
| log-odds interaction | +2.2065  | [+1.9565, +2.4609] |

data LLAVA-1.5-7B’s  $H$  also rises (0.6761  $\rightarrow$  0.7803), so on that architecture both halves move before length is held fixed and only the false-object half survives holding it fixed. On LLAVA-OV-0.5B only the false-object half moves in raw data and matched alike:  $H$  is flat across conditions (0.5578 against 0.5471) and  $F$  collapses 0.5171  $\rightarrow$  0.1284, a factor of 4.03 against LLAVA-1.5-7B’s raw 4.59. One consequence is worth isolating: because LLAVA-OV-0.5B’s true-object half does not move, its one-scene  $J$  is +0.0406, so given five captions that all describe the same wrong scene the smaller model carries true and false prefix objects forward at essentially the same rate.

#### D.4 SENSITIVITY BY CONDITION

Table 18: Sensitivity  $d'$  by condition, sighted against blind. The rightmost column is the paired difference within a condition; all four rows exclude zero, so the image is being used in every condition measured. The per-condition table is given for the two LLAVA architectures; the image-attributable component of  $d'$  on all four measured architectures is given in the text below.

| architecture  | condition   | $d'$ sighted | $d'$ blind | sighted – blind      |
|---------------|-------------|--------------|------------|----------------------|
| LLAVA-1.5-7B  | one scene   | +0.486       | −0.055     | +0.54 [+0.42, +0.67] |
| LLAVA-1.5-7B  | five scenes | +2.018       | +1.292     | +0.73 [+0.61, +0.85] |
| LLAVA-OV-0.5B | one scene   | +0.102       | −0.360     | +0.46 [+0.33, +0.61] |
| LLAVA-OV-0.5B | five scenes | +1.252       | +1.028     | +0.22 [+0.14, +0.31] |

Blinding costs the model real discriminability in every condition, and on LLAVA-1.5-7B’s one-scene condition it costs all of it: blind  $d' = -0.055$  [−0.171, +0.063] contains zero, while the blind five-scene condition reaches  $d' = +1.2918$  [+1.1865, +1.3984]. Any reading of this paper as a claim that the model ignores the image is contradicted by this table.

The contrast is a different matter. The one-scene to five-scene increase in  $d'$  on LLAVA-1.5-7B is +1.5322 [+1.36, +1.70] sighted and +1.3467 [+1.19, +1.50] blind, so the image-attributable component is +0.1855 [+0.0252, +0.3463] on LLAVA-1.5-7B and −0.2382 [−0.400, −0.082] on LLAVA-OV-0.5B. Both exclude zero, in opposite directions, and adding the two further architectures, including QWEN2.5-VL-7B, which the main text does not interpret, splits them two–two rather than resolving them. Measured together in a single run, the two LLAVA components and the two further architectures’ are +0.1855 [+0.0271, +0.3481], −0.2382 [−0.4021, −0.0847], +0.2761 [+0.1160, +0.4467] and −0.1881 [−0.3598, −0.0098]; every interval excludes zero and the sign is not stable across architectures. The first two differ from the values above in the third decimal of their interval endpoints because that run used a different bootstrap seed; the point estimates are identical and both runs are released. The criterion behaves as described in Appendix C.6:

on LLAVA-1.5-7B  $c$  moves  $-0.213 \rightarrow +0.237$  sighted and  $+0.053 \rightarrow +0.098$  blind, an image-attributable component of  $+0.4057$  [ $+0.3157, +0.4957$ ].

Two deflations belong beside this table rather than in a footnote. Because  $c = -\frac{1}{2}[Z(H) + Z(F)]$ , the criterion result is substantially a re-expression of the two carry rates already reported: the image-attributable components of both rate contrasts are negative and of comparable size ( $-0.1366$  [ $-0.1898, -0.0824$ ] and  $-0.1206$  [ $-0.1459, -0.0953$ ] on LLAVA-1.5-7B;  $-0.1971$  [ $-0.2512, -0.1444$ ] and  $-0.1623$  [ $-0.1856, -0.1397$ ] on LLAVA-OV-0.5B), and two comparable falls are a criterion rise. And neither  $d'$  nor  $c$  appears in the pre-registration governing these conditions; both are post-hoc for them, while the registered primary remains  $J$ . The main text's central claim — that the rise in  $J$  does not need the image — rests on the registered primary; its signal-detection reading is post hoc and is labelled as such there.

Note also that  $\Delta_{\text{image}} \equiv d_H^{\text{sight}} - d_F^{\text{sight}}$  is an algebraic identity, not a check:  $J$  is itself a difference, so expanding both sides in the eight underlying rates yields the same eight terms and the equality holds for any data whatsoever. It was verified numerically only to confirm that it *is* the identity (residuals  $5.6 \times 10^{-17}$  and  $2.8 \times 10^{-17}$ ). The empirical content is that each component is large and excludes zero, and that the two components are close in magnitude. Nothing forces that closeness; had the image moved  $H$  by 0.14 and  $F$  by 0.02,  $J$  would have registered a large visual effect.

## D.5 THE DOSE-RESPONSE GRID

Table 19: The five-point scene-count grid, holding five caption units at every point. Rank correlation with realised distinct-scene count is  $\rho_S = +1.0000$  [ $+0.900, +1.000$ ].

|                 | $S=1$   | $S=2$   | $S=3$   | $S=4$   | $S=5$   |
|-----------------|---------|---------|---------|---------|---------|
| $J$             | +0.1947 | +0.4739 | +0.5919 | +0.6290 | +0.6738 |
| $H$             | 0.6674  | 0.7335  | 0.7759  | 0.7770  | 0.7808  |
| $F$             | 0.4727  | 0.2596  | 0.1840  | 0.1480  | 0.1069  |
| true units      | 454     | 638     | 723     | 731     | 780     |
| share at budget | 0.436   | 0.569   | 0.636   | 0.642   | 0.655   |

The endpoint contrast is  $+0.4791$  [ $+0.4197, +0.5391$ ], interval width 0.1194 against a registered ceiling of 0.130. The profile is strictly increasing and all four adjacent steps exclude zero, at  $+0.2792, +0.1180, +0.0372$  and  $+0.0448$ : the largest single step carries 58.3% of the total rise, just under the 60% the readout had fixed in advance as the boundary of the threshold outcome; the fourth step's point estimate slightly exceeds the third's, so the curve is front-loaded without a plateau inside the grid. The length-matched profile keeps the same sign ( $+0.3210$ ) and remains strictly increasing, but its largest step carries 60.1% of its rise, just over the same boundary the raw profile sits just under. No registered outcome depends on it.

Two positive controls were available at no cost and both held. The  $S=5$  condition rebuilds bit-for-bit against the five-scene condition of the two-point measurement (1361/1361 cells identical), and  $S=1$  lands at  $+0.1947$  against the two-point measurement's  $+0.1881$  despite being built by a different assembler with no doubling loop, which closes the objection that the two-point contrast is an artefact of assembler asymmetry between its conditions. Of the three registered co-primaries, two are monotone ( $+0.502, +0.481$ ); the base-rate-tertile co-primary returned a width of 0.2295 against its ceiling and is recorded as no-data, and it halves the endpoint to  $+0.249$ , so part of the magnitude of the two-point contrast is attributable to object base-rate composition rather than to dose. We do not quote the endpoint without that qualification.

**Why the curve cannot name the variable.** At a fixed token budget the grid makes an identity exact. Total object mentions per prefix is conserved across the whole grid ( $4.687 \rightarrow 4.576$ , Spearman  $\rho = -0.9$  [ $-1.0, +0.4$ ]), as is the false-object subset ( $3.640 \rightarrow 3.607$ ), while the count of distinct object types doubles ( $1.847 \rightarrow 3.761$ ,  $\rho = +1.0$  [ $+1.0, +1.0$ ]). Mentions per type therefore falls reciprocally ( $2.582 \rightarrow 1.232$ ,  $\rho = -1.0$  [ $-1.0, -1.0$ ]) and the product of the two stays near-constant ( $4.77$  against  $4.63$ ). Distinct-type count and mentions-per-type track  $S$  with identical precision, so no statistic computable on this grid separates them. This diagnostic was added after the

grid’s pre-registration was committed, enters no registered decision, and is reported as a limitation rather than as a result.

The registered eight-unit extension was not run, on a mechanical branch rather than a judgement call: eight units buys units-as-seen  $2.671 \rightarrow 2.702$  and scenes-as-seen  $+0.031$ , against gates of 5.0 and 0.5.

#### D.6 THE FORCED-CHOICE REGISTER

The identical manipulation was re-posed on the identical units as a binary existence probe, in which the object is supplied by the question rather than selected by the model: 37,910 answers over 3,000 rows, 1,500 cells, 500 images and 79 object types, with an unparseable-answer rate of 0.0000 in all eight conditions.

Table 20: The scene-count contrast in a forced-choice register, on the same units as the generative contrast.

| quantity                                          | estimate | 95% CI             |
|---------------------------------------------------|----------|--------------------|
| scene-count contrast, forced choice               | −0.0095  | [−0.0314, +0.0124] |
| scene-count contrast, free generation, same units | +0.4859  | [+0.4283, +0.5417] |
| difference of differences                         | +0.4954  | [+0.4393, +0.5517] |
| log-odds co-primary, generative                   | +2.6116  | [+2.3160, +2.9007] |
| log-odds co-primary, forced choice                | −0.0034  | [−0.6688, +0.5951] |
| probe with no prefix, $J$                         | +0.8334  | [+0.8012, +0.8656] |

Three checks address the obvious objections. The probe works: with no prefix  $J$  is  $+0.8334$ , four times the registered instrument floor, and a single wrong-scene prefix multiplies the false-positive rate from 0.0217 to roughly 0.88 — about  $40\times$  — identically in both conditions, so the register carries the context pathology itself at full strength. The registered compression flag was not triggered, and the ceiling-free log-odds co-primary agrees with the raw one; a later analysis nonetheless found this binary probe saturated once any prefix is present (Appendix H), so these checks do not rule out a ceiling. And it is not a ground-truth artefact: re-derived under segmentation-only and caption-only definitions the cell is invariant, with the largest pairwise spread in the forced-choice contrast at 0.0116 and neither paired difference excluding zero. The realised interval width is 0.0438 against a pre-registered ceiling of 0.150, so the null is  $3.4\times$  narrower than the design required rather than a failure of power.

One qualification, recorded because it narrows the claim. An exploratory re-read of the same rows finds the forced-choice readout is, once any prefix exists, close to a deterministic function of prefix membership: naming an object in the context moves the false-positive rate from 0.022 to 0.871 — the pooled yes-rate itself runs 0.095 to 0.895 — and nothing else we varied moves it further. The bound may therefore belong to this probe as much as to forced choice in general. Establishing which would require a probe whose no-prefix and with-prefix yes-rates are comparable — a closed set including unmentioned distractors, or a readout that is not thresholded. Both were built and run, and Appendix H reports them: the closed set saturates further rather than less, and the graded readout, which is demonstrably not pinned, returns the same null.

#### D.7 THE BASE-RATE REFERENCE

With no prefix at all, 90.3% (LLAVA-1.5-7B) and 85.0% (LLAVA-OV-0.5B) of the objects scored as “true” already appear in the model’s own unprompted caption, giving a no-prefix  $J$  of  $+0.871$  and  $+0.816$ . Most of  $H$  is therefore available without reading the prefix, which is the base-rate enrichment that an earlier pre-registration predicted in writing, and it is the reason the true-object half has little room to move.

## E PRE-REGISTRATION AND DEVIATIONS

### E.1 SUMMARY OF REGISTERED ANALYSES

Table 22 lists every pre-registered analysis reported in this paper, one row per registration, with its registered primary, its outcome and where it is reported. Decision rules, amendments and full outcomes are given in the entries below and in the sections cited.

**What “registered” means here.** A registration is a written plan, with its decision rules and the reading of each outcome, committed to the authors’ git repository before the first generation job it governs; amendments are appended, dated and committed before the data they concern. The timestamps are the authors’ own commit history, not a third-party registry’s, so they document order rather than prove it. Table 21 gives each registration’s first commit. One plan was not committed: the de-saturated forced-choice registration (row 13) was written to the compute cluster, where its copy is byte-identical to the local file and predates the first calibration row by seven minutes. Random seeds encode the date an analysis was fixed, so they give the same timeline.

Table 21: First commit of each registration (short hash and date). Row numbers follow Tables 22 and 23. Later amendments are separate, dated commits. <sup>a</sup>Not committed; the date is that of its byte-identical copy on the compute cluster. <sup>b</sup>The mismatched-image arm was added by Amendment 2 (463fe01, 2026-09-19), before its data existed.

| Row | Registered analysis                     | Commit  | Date       |
|-----|-----------------------------------------|---------|------------|
| 1   | Scene count, LLAVA-1.5-7B               | 9c77354 | 2026-09-09 |
| 2   | Scene count, LLAVA-OV-0.5B              | 7b8b4a2 | 2026-09-11 |
| 3   | Grey-field baseline, both LLAVA         | 90c0043 | 2026-09-12 |
| 4   | QWEN2.5-VL-7B                           | b7d900d | 2026-09-11 |
| 5   | KOSMOS-2                                | cdee4fe | 2026-09-13 |
| 6   | Dose grid                               | 2fa4a23 | 2026-09-11 |
| 7   | Regime control at power                 | 74e98bc | 2026-09-09 |
| 8   | Mention-rate control                    | 23ef3ce | 2026-09-14 |
| 9   | Repetition                              | 725723d | 2026-09-11 |
| 10  | Agreement                               | 3adf8ac | 2026-09-11 |
| 11  | Natural prose                           | 5701136 | 2026-09-12 |
| 12  | Second register (forced choice)         | f009160 | 2026-09-11 |
| 13  | De-saturated forced choice <sup>a</sup> | none    | 2026-09-15 |
| 14  | Distractor units                        | 8fe97a7 | 2026-09-15 |
| 15  | PAI, standard CHAIR                     | ce3ed30 | 2026-09-15 |
| 16  | VCD, standard CHAIR                     | 28a4caa | 2026-09-15 |
| 17  | Ablation ladder <sup>b</sup>            | f186bcf | 2026-09-19 |
| 18  | POPE with an ablated arm                | 90e428d | 2026-09-19 |
| 19  | QWEN3-VL-8B                             | 7a30268 | 2026-09-19 |

Nine departures from the registrations bear on how the table is read.

- The KOSMOS-2 stop gate returned “degenerate continuations” under its registered thresholds; two thresholds were amended before any endpoint for that model existed, and both rules are released (the stop-gate disclosure in Appendix E.2; Section 4.3).
- The second mitigation method was dropped from the first method’s registration by Amendment 1 and then run under a separate registration with a within-harness positive control; two of the reasons Amendment 1 gave were corrected before that experiment generated anything (Appendix G.9).
- On LLAVA-1.5-7B, four mention-count strata of the mention-rate control show a positive component, the direction pre-specified as adverse, and the rule was not amended (Appendix B.3).
- One rung of the registered type-matching ladder on LLAVA-1.5-7B shows a negative component whose interval excludes zero, and it is unexplained (Appendix B.4).

Table 22: Registered analyses reported in this paper and their outcomes. Quoted phrases are registered outcome labels; other outcomes use the wording of the section cited. Rows are grouped by topic and do not follow the numbering of Appendix E.2.

| Analysis                                  | Registered primary                                                                                                         | Outcome                                                                                                                                                                                    | Reported in        |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 1. Scene count, LLAVA-1.5-7B              | five-scene minus one-scene $J$ ; length-matched co-primary; composition and regime controls                                | positive; both composition controls hold; regime control underpowered (row 7)                                                                                                              | App. D.1, B.5      |
| 2. Scene count, LLAVA-OV-0.5B             | the same contrast, evaluated independently                                                                                 | positive; point estimate below row 1's, intervals overlapping narrowly; regime control unevaluated                                                                                         | App. D.1, B.5      |
| 3. Blind baseline, both LLAVA             | $\Delta_{\text{image}}$ per architecture, behind an attention gate                                                         | registered outcome held on both; $\Delta_{\text{image}}$ null on both; zero- $J$ flag "fails"                                                                                              | Sec. 4.3; App. D   |
| 4. Contrast and blind arms, QWEN2.5-VL-7B | scene-count contrast with blind arms, behind an attention gate                                                             | "image-inert", outranking "replicates"; not counted as a replication                                                                                                                       | Sec. 4.3; App. D   |
| 5. Contrast and blind arms, KOSMOS-2      | the same, with $\Delta_{\text{image}}$ as headline; five-condition stop gate; image-inert outcome requires a null at power | stop gate "degenerate continuations" as registered, amended before any endpoint; readout mechanically "replicates", licence text set aside; $\Delta_{\text{image}}$ excludes zero downward | Sec. 4.3; App. D   |
| 6. Dose grid                              | rank correlation and endpoint contrast over one to five scenes                                                             | monotone; base-rate-tertile co-primary no-data; eight-unit branch not run                                                                                                                  | App. D.5           |
| 7. Regime control at power                | contrast on cells where neither condition reached the budget, on an enlarged sample                                        | positive; below the underpowered estimate that motivated it                                                                                                                                | App. B.5           |
| 8. Mention-rate control                   | $\Delta_{\text{image}}$ at fixed distinct object types, blockwise; adverse direction pre-specified                         | type-matched estimate null on both; rule passes on LLAVA-OV-0.5B and fails on LLAVA-1.5-7B                                                                                                 | App. B.3, B.4      |
| 9. Repetition                             | stated-once against repetition-retained prefixes at a fixed object set; reading conditional on $\rho$                      | unresolved on the width rule; repetition reading cannot be returned; one manipulation gate failed                                                                                          | App. E.2           |
| 10. Agreement                             | contradicting one-scene condition against its anchor, with an edit-matched control                                         | no-data; edit control null; type-count confound found afterwards                                                                                                                           | App. E.2           |
| 11. Natural prose                         | the same manipulation on natural-prose units; units-seen gate                                                              | "construct-specific" (with "unit count unmatched"); null at power, unit-standardised control also null; lead result specific to assembled captions                                         | Sec. 4.2; App. E.2 |

Table 23: Registered analyses and their outcomes, continued from Table 22.

| Analysis                          | Registered primary                                                                                               | Outcome                                                                                                                                                          | Reported in         |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| 12. Second register               | the same contrast as a forced-choice existence probe on the same units                                           | bounded, at power                                                                                                                                                | App. H, D.6         |
| 13. De-saturated forced choice    | select an off-ceiling probe on arm-pooled calibration, then measure it; graded readout reported in every branch  | “still saturated”; contrast not computed; graded evidence post-hoc                                                                                               | App. H              |
| 14. Distractor units              | the contrast on objects the prefix never names; binary and graded readouts gated separately                      | “distractor probe saturated” (binary primary over its ceiling); graded co-primary null but not accepted as confirmation under the registration’s acceptance rule | App. H              |
| 15. PAI, standard CHAIR           | decomposition, blinding and truncation-curve components; positive-control and liveness gates                     | “cannot resolve” at the prescribed $\alpha$ : “link-sensitive”, “blind denominator too thin”, “cannot resolve”                                                   | Sec. 3; App. G.8    |
| 16. VCD, standard CHAIR           | separate registration; within-harness positive control; mode-matched baselines                                   | “no gain”; “no gain to decompose” on all three arms                                                                                                              | App. G.9            |
| 17. Ablation ladder, LLAVA-1.5-7B | $\Delta_{\text{image}}$ under ten ablations on both endpoints; severity, geometry and collapse screens           | seven graded rungs “ladder null”; substitution “image contributes” (after Amendment 3); on CHAIR, substitution “gain is image-attributable”                      | Sec. 4.4, 3; App. I |
| 18. POPE with a blind arm         | $\Delta_{\text{image}}$ of PAI’s gain on official POPE under grey and a mismatched image; fifteen blocking gates | attention-only arm “partial”; released default no gain; grey “blind arm degenerate”                                                                              | Sec. 3; App. J      |
| 19. QWEN3-VL-8B                   | the scene-count contrast with a grey arm, behind a harness gate calibrated on LLAVA-1.5-7B                       | gate “run endpoint”; registered outcome “image-attributable”                                                                                                     | Sec. 4.3; App. K    |

- The natural-prose registration’s reading required retitling the paper once the lead result proved specific to assembled captions. The title under which these experiments were registered was replaced, and the present one makes no claim about the construction (Appendix E.2).
- Amendment 3 of the ablation-ladder registration replaced a comparative collapse screen with absolute thresholds, before any ladder verdict was read. It changed one verdict, the substitution arm’s, from “confounded by collapse” to “image contributes” (Appendix I).
- Both untruncated-regime ablation arms, the extension of both ablations to the 3,500-image sample, and the mismatched-image arm on QWEN3-VL-8B were added after the corresponding experiments were registered and analysed; they are post hoc (Section 4.4; Appendix L).
- The  $J$  endpoint and the aggregation across cuts were adopted in the presence of data, and  $d'$  and  $c$  are post-hoc for the conditions they decompose (Appendix A.5; Section 4.4).
- In the distractor-unit analysis, the first scoring run reported the non-verdict “distractor probe saturated” as its headline; a later selector change reported the graded co-primary’s “distractor null confirmed” instead. Every number is identical between the two runs. Because the registration accepts a null only when the binary gates pass, the paper reports the registered outcome as “distractor probe saturated” and the graded null as supporting evidence. Both result files are kept (Appendix H).

## E.2 REGISTERED ANALYSES IN DETAIL

Each entry below states the decision rule as it was fixed before the data it governs existed, and then the realised outcome. Pre-registrations are released with the paper, with their amendments. Every amendment was committed before the run whose endpoint it governs. One requires a sharper statement: KOSMOS-2’s feasibility gate was amended after that gate’s own probe had reported its failures, though before any endpoint existed and on calibration against already-frozen earlier families. We treat correcting a mis-specified feasibility gate before any endpoint exists as admissible, and relaxing a readout threshold after a result as not; the entry below gives both rules and both outcomes so a reader can weigh the distinction independently.

1. **Blind and no-prefix baselines.** *Rule:*  $\Delta_{\text{image}} = \Delta_{\text{primary}} - \Delta_{\text{ablated}}$  is the primary, evaluated on each architecture independently, with a 0.14 interval-width ceiling, a floor of 100 true units, an attention gate that must pass before any outcome can be declared, and a standing flag on the assertion that a model which does not consult the image has  $J$  of zero. *Outcome:* the registered outcome held independently on both architectures.  $\Delta_{\text{ablated}} +0.5019 [+0.4441, +0.5565]$  and  $+0.4128 [+0.3609, +0.4640]$ ;  $\Delta_{\text{image}} -0.0160 [-0.0708, +0.0400]$  and  $-0.0348 [-0.0905, +0.0192]$ , both null;  $\Delta_{\text{ablated}}/\Delta_{\text{primary}} 1.033 [0.922, 1.155]$  and  $1.092 [0.954, 1.265]$ . The attention gate passed. The standing flag reads “fails” on both architectures. This is the control that was registered as a confirmation and returned a refutation, under a grey field on truncated generations; Appendix L shows that answer changes with the ablation and the regime.
2. **Scene-count dose grid.** *Rule:* five caption units at every grid point,  $S \in \{1, \dots, 5\}$ , registered decisions fixed in advance with a 0.13 interval-width ceiling and a 0.60 step-fraction boundary separating a graded reading from a threshold reading. *Outcome:* monotone.  $\rho_S = +1.0000 [+0.900, +1.000]$ ; endpoint  $+0.4791 [+0.4197, +0.5391]$  at width 0.1194; all four adjacent steps exclude zero; largest step 58.3% of the rise. Two of three co-primaries concur; the base-rate-tertile co-primary returned no-data at width 0.2295 and halves the endpoint to  $+0.249$ . The registered eight-unit branch was not run on a mechanical gate (Appendix D.5).
3. **Generation-regime control, promoted to primary and run at power.** *Rule:* the contrast restricted to cells in which neither condition reached the generation cap is the primary; sample raised to 3,500 images with the original 500 nested inside; the floor of 100 true units and the 0.13 width ceiling carried over unchanged; a sensitivity analysis on the original 500 pre-declared as descriptive and expected to be underpowered. *Outcome:* positive.  $+0.3460 [+0.2849, +0.4058]$  at width 0.1209, true units 466/914, on 2,089 of 10,500

cells; length divergence flag was not triggered; both composition controls remain positive inside the regime-matched subset. The magnitude rule returns no outcome in either direction under this rule, and the confirmatory estimate is 26% below the underpowered estimate that motivated the run (Appendix B.5).

4. **Second-register test.** *Rule:* two registered decisions — the contrast crosses the register, or it is bounded to the generative one — both pre-committed, together with the *reading* of the bounded outcome: that it appears in the abstract and that every claim in the paper scopes to generative registers. Interval ceiling 0.150; floor on true units. *Outcome:* bounded, at power and by a wide margin.  $-0.0095$   $[-0.0314, +0.0124]$  at width 0.0438, true units 528/892; difference of differences  $+0.4954$   $[+0.4393, +0.5517]$ . The compression flag was not triggered; the outcome is invariant across all three ground-truth constructions. The pre-committed reading is executed (Appendix D.6).
5. **Mention-rate control.** *Rule:* a control holding distinct object types fixed, with the adverse direction pre-specified, evaluated blockwise. *Outcome:* the type-matched estimate is null on both architectures. LLAVA-OV-0.5B passes on all fourteen powered blocks; the first fails on five of sixteen: four mention-count strata in the pre-specified adverse direction, and one type-matched block in the opposite direction. The failure is reported and the rule was not amended (Appendix B.3).
6. **Repetition, at a fixed object set.** *Rule:* a condition in which each prefix object type is stated once against one in which repetition is retained, at an exactly matched prefix-object set, plus a model-free condition, three manipulation gates and a 0.13 width ceiling; the reading that repetition is the carrier was made conditional in advance on the lower bound of  $\rho = \Delta_{\text{rep}}/\Delta_{\text{total}}$  reaching 0.5. *Outcome:* unresolved on the width rule —  $+0.1181$   $[+0.0397, +0.1957]$  at width 0.1559 over the ceiling, with the true-unit floor cleared threefold. The conditional fixed in advance is what settles the question:  $\rho = 0.230$   $[0.075, 0.389]$ , the whole interval below 0.5, so the repetition reading could not have been returned even at full power. Two manipulation gates passed and one failed; by that pre-registration’s own manipulation-check clause the failure makes the affected control uninterpretable, and it is excluded from every registered decision rather than quoted.
7. **Agreement, at a fixed scene count.** *Rule:* four conditions in one generation job — a contradicting one-scene condition, an edit-matched agreeing control, and both anchors — with registered decisions, subsets and a 0.13 primary-only width ceiling fixed in advance, and a precedence rule making an underpowered member of a pair decide the cell. *Outcome:* no-data. The primary is  $+0.2418$   $[+0.1663, +0.3194]$  at width 0.1531, short of the ceiling by 0.0231, while clearing the true-unit floor more than fourfold. Excluding zero was not treated as a result and the ceiling was not relaxed. The edit control is null ( $-0.0334$   $[-0.1034, +0.0390]$ ), so the edit is a genuine no-op. A confound the pre-registration did not anticipate was found afterwards and is reported against that registration: the contradicting condition carries 4.232 distinct prefix object types against the one-scene condition’s 2.022, so the primary bundles an agreement change with a type-count change, and composition controls that match object sets do not remove it.
8. **Natural prose at matched scene count.** *Rule:* the same manipulation on natural-prose units (Localized Narratives descriptions) rather than assembled COCO captions, at a matched budget, with a gate on units-seen and a pre-committed consequence if it fails. *Outcome:* primary null at power —  $+0.0062$   $[-0.0368, +0.0515]$  at width 0.0883 against a 0.14 ceiling, true units 560/640 — with every co-primary and specification agreeing, including the regime-restricted estimate ( $+0.0076$   $[-0.0366, +0.0543]$ ). The units-seen gate failed (1.577 against 1.062), so the registered prefix “unit count unmatched” applies and the unit-standardised control is required; it is also null ( $-0.0547$   $[-0.1398, +0.0286]$ ) and returns the same outcome. The registered outcome is therefore “construct-specific”, whose pre-committed reading is that the paper’s lead result is specific to the assembled-caption construction; the main text states this (Section 4.2), together with the earlier registered five-source natural-prose run, which returned “negative at power” on its contrast defined as single-source minus five-source ( $-0.0750$   $[-0.1299, -0.0185]$ ; length-matched  $-0.0222$   $[-0.0719, +0.0301]$ ), that is, five non-redundant sources scored higher ( $J = +0.4407$   $[+0.3971, +0.4838]$ ). The same registration recorded before the data that its manipulation

realises about half the headline’s dose (1.0 against about 2.0 scenes touched, rather than 1.0 against 2.83 on that registration’s own metric; the dose grid’s scenes-as-seen measure gives 2.67, Appendix A), so its null is weaker evidence than a null at full dose. The same reading also required that the paper be retitled; it was, and the present title makes no claim about the construction. Both natural-prose conditions sit at  $J$  of +0.4221 and +0.4283, against the assembled one-scene anchor’s +0.188; natural-prose continuations rarely reach the token budget, so passage construction and generation regime are confounded in that comparison.

9. **Two further architectures with different visual interfaces.** *Rules:* two separate registrations, with different rules. The QWEN2.5-VL-7B registration was committed after a checkpoint fetch and a plumbing smoke test but before any score was computed; its “image-inert” outcome is returned when  $\Delta_{\text{ablated}}$ ’s lower bound is above zero and  $\Delta_{\text{image}}$ ’s interval contains zero, with no power requirement. The KOSMOS-2 registration was committed before any artefact for that model existed, with registered decisions for replication, non-replication, reversal and image-inertness, five stop-gate conditions and an attention gate; its image-inert outcome requires  $\Delta_{\text{image}}$  to be null *at power* against a ceiling derived for that contrast, routing a too-wide interval to no-data. *Outcome:* run on two architectures outside the LLAVA lineage. QWEN2.5-VL-7B returns an image-attributable component of +0.016 [−0.023, +0.055], inside the reference margin. Its registration returns “image-inert” under the rule fixed at its §6.1 —  $\Delta_{\text{blind}}$ ’s lower bound above zero and  $\Delta_{\text{image}}$  containing zero — and that cell outranks the “replicates” the primary alone would have produced, with the registration recording that it must not be written as a third-architecture replication. Its median continuation of four tokens is a *post-hoc* confound, not a registered condition: no length floor appears in that registration, and the campaign record declines to formalise one because any threshold excluding this model would also reject LLAVA-OV. KOSMOS-2 returns −0.103 [−0.168, −0.036], which excludes zero and falls outside the reference margin. Its §6.2 readout mechanically returns “replicates”, but that readout keyed reversal to the primary’s sign rather than the headline estimand’s, so its licence text is false for these data and we set it aside; the substantive finding is that  $\Delta_{\text{image}}$  excludes zero *downward*. Its stop gate is treated in the stop-gate disclosure immediately below. KOSMOS-2 is reported in the main text; the third is named there but not interpreted, and its numbers are here and in Appendix D.

*Stop-gate disclosure for KOSMOS-2.* Under the rule as registered, the stop gate returned “degenerate continuations” and the generation reported above would not have run. Two of five conditions failed: a units-inside floor (2.665 against a floor of 3.0) and a degeneracy ceiling (0.383 in the worst arm against a ceiling of 0.20). Both thresholds were amended, in a dated amendment committed before any endpoint for that architecture existed — no ground truth was loaded, no object extracted and no  $J$  computed at that point — and the justification was calibration against the already-frozen earlier families. Measured on those: the units floor also rejects LLAVA-1.5-7B at 2.832 and the third at 2.987, and the degeneracy ceiling also rejects LLAVA-1.5-7B’s arms at 0.678 and 0.715 and the second’s at 0.535 and 0.653. Both thresholds therefore reject this paper’s own lead result. The clause that triggered was additionally found to be tracking the rate at which generation reached the token cap rather than collapse: its two hazard sub-clauses triggered zero times in all four arms. We regard correcting a mis-specified feasibility gate before any endpoint exists as admissible and relaxing a readout threshold after a result as not, but that distinction is ours, and both the original rule and the amended one are released with the paper. The registered readout for this architecture is a further complication we do not paper over: its rule keyed reversal to the primary’s sign rather than the headline estimand’s, so it mechanically returns “replicates” while the estimand it was meant to govern excludes zero downward. We set that licence text aside and report the estimand. It also tightens a rule the QWEN2.5-VL-7B registration used: that earlier registration returned an image-inert outcome on an interval that merely contained zero, with no power requirement. That does not change the reported result, whose realised width of 0.0780 would pass the tighter ceiling.

10. **A published mitigation method, decomposed on standard CHAIR.** *Rule:* three components fixed before any generation — a decomposition component with a  $\pm 0.15$  equivalence margin on  $\Delta d'$  and a logit companion required to agree with it, a blinding component with a thin-denominator guard on the blind arm, and a truncation-curve component with a  $\pm 0.03$

margin on the matched- $F$  hit rate — together with a positive control on published vanilla CHAIR levels that had to pass before any contrast was read, and a blocking gate requiring the intervention to be demonstrably live. *Outcome*: “cannot resolve” at the  $\alpha$  the method’s own paper prescribes for this model. The positive control and every blocking gate passed. The decomposition component returns “link-sensitive”, because the probit and logit companions disagree; the blinding component returns “blind denominator too thin”, the guard triggering exactly as written; and the truncation component returns “cannot resolve” on an interval that straddles its margin, which is an underpowered result and not a pass. Two amendments are on the record, each committed before the endpoint it could have touched. **Amendment 1** dropped a second candidate method from *this* experiment, on the grounds that it has no published CHAIR evaluation to reproduce and that its released code cannot be run greedily, which is the mode this experiment’s other arms use. That method was subsequently run under a separate registration, with a within-harness positive control in place of an external one and with both decoding modes run against mode-matched baselines (Appendix G.9). Amendment 1’s *conclusion* — that the arm did not belong in this experiment — stands. Two of the reasons it gave were found to be inaccurate on re-verification and are corrected in Appendix G.9, before any generation for that experiment existed. **Amendment 2** derived the degenerate-corner constant of the log-linear correction and added a permuted-ground-truth secondary after the blind arm was seen to degenerate. That secondary is reported in Appendix G.7.

11. **A de-saturated forced-choice readout.** *Rule*: a two-stage design — calibrate candidate probes on an 80-image subset that writes *only* arm-pooled quantities, so the scene-count contrast is not computable at selection time; select on pooled  $F$  alone; then measure the selected probe — with three off-ceiling conditions fixed in advance ( $F \in [0.20, 0.80]$ ,  $H \leq 0.90$ , yes-rate  $\in [0.15, 0.85]$ ), an instrument gate requiring the selected probe to show a no-prefix  $J$  of at least  $+0.2$ , and a standing requirement that the graded readout of the published prompt be reported in every branch whatever the selection returns. *Outcome*: “still saturated”. Every candidate probe failed all three off-ceiling conditions, including the closed candidate set built specifically to de-saturate, which saturated further rather than less, to a yes-rate of 1.0000. Under the registered rule the selected probe’s contrast was therefore not computed and not interpreted, and that is enforced in the scorer rather than by intention. The standing diagnostic still reports: the graded readout of the published prompt gives  $+0.0008 [-0.0142, +0.0166]$ . Because the ceiling gate is defined on the binary rate and triggered, the interior-mass evidence that the graded readout is genuinely unpinned is *post-hoc*, and is labelled as such wherever it appears (Appendix H).

**A scoring error that the registration’s own gate caught, and what it changed.** The two-alternative variant of the de-saturation design counterbalances which side the unit object is presented on. Its first scoring pass scored the raw “answered A” rate instead of “chose the unit object”, which that registration defines explicitly. The effect was to turn the readout into something close to a coin flip, which is exactly why it appeared to be off ceiling — pooled yes-rate 0.5581 — and also why the registered instrument gate triggered on it: its no-prefix  $J$  was  $-0.3185 [-0.3602, -0.2752]$  against a floor of  $+0.2$ , anti-diagnostic, because under the wrong orientation “answered A” is pinned near the coin for true units while for false units it follows the model’s side bias. The gate did exactly the job it exists to do. The side is a deterministic function of the image, the derangement index and the object, so the correction is a re-scoring at the identical seed with no regeneration, and the superseded scoring is retained in the released record.

Stated against our own interest: the selection stage ran on the mis-oriented statistic, so the two-alternative variant was selected as the only in-band candidate. Correctly oriented, that variant’s  $F$  is about 0.93, so no candidate would have been in band, the no-candidate-in-band flag would have raised, and the primary would instead have been the “answer only from the image” variant with the two-alternative variant as its fallback. The registered outcome is “still saturated” either way, because both sit outside the band (Table 34), so the conclusion is invariant to the selection error — but the selection itself was made on a wrong number, and that is recorded here rather than quietly repaired. One further deviation belongs with it: the scorer evaluates the ceiling gate before the instrument gate, while the registration orders the instrument gate first. Both fail for that variant, so the outcome, no verdict, is the same under either ordering.

**Outcomes against interest.** Four registered tests returned against the hypothesis they were attached to and were honoured as written: the blind baseline, which refuted the paper’s own lead reading; the mention-rate control, which records a failure on five of sixteen powered blocks on LLAVA-1.5-7B; the repetition conditional, which could only ever have gone against the reading it was attached to; and the natural-prose condition, whose null is reported at full strength. One control confirmed the quantity it was built to test and simultaneously shrank the estimate that motivated building it, and both halves are reported. Two registered analyses failed a width ceiling that was not relaxed after the fact. Two later registrations returned non-affirmative outcomes and are reported at their registered strength rather than reframed: the mitigation decomposition returns “cannot resolve” on its combined primary, and the de-saturation design returns no verdict because its own ceiling gate triggered.

**What was not pre-registered, and is marked as such.** Three elements of the specification were selected in the presence of data and are labelled wherever they appear. The  $J$  endpoint itself was adopted after CHAIR and a per-mention hallucination rate were found structurally defective on these conditions (Appendix A.4). The aggregation across cuts was likewise not pre-specified. And  $d'$  and  $c$  are post-hoc for these conditions: the pre-registration governing them names neither statistic, its registered primary is  $J$ , and  $J$  is null. The main text’s central claim rests on the registered primary; its signal-detection reading is post hoc and labelled as such.

**Operational choices fixed before data but not by an external document.** The clean/errorful classification rule, the pairing and density-matching rule, the monotonicity threshold, and the sentence-boundary construction were left open by the candidate specification and set by us before the data existed. They are pre-data but not pre-specified, and are distinguished from the registered rules above.

## F REPRODUCIBILITY

### F.1 WHAT CAN BE CHECKED, AND HOW

Every pre-registration is released with the paper, together with its dated amendments and its resolution. The machine-readable result artefacts are released with their integrity metadata, and with a script that re-derives every table cell in this paper from them and fails on disagreement. The analysis runs on CPU from the released artefacts, so every number in this paper can be checked without a GPU or the original checkpoints. Regenerating the stimuli and the generations requires a single consumer GPU and the five public checkpoints of Appendix A.1.

The verification gate of Appendix A.7 is the operative reproducibility instrument rather than a formality: no analysis in this paper reports a new quantity until it has first reproduced previously published per-condition rates to within  $5 \times 10^{-3}$ , and the realised margins are three to four orders of magnitude inside that tolerance.

### F.2 BOOTSTRAP IMPLEMENTATION CHECK

All reported intervals were recomputed with image-level paired resampling with replacement, multiplicity preserved and resample size asserted on every replicate. An implementation error affecting a subset of auxiliary intervals was identified during verification, and all affected estimates were recomputed under the registered procedure; no primary conclusion changed. The multiplicity assertion described in Appendix A.6 is the standing check that closes this class of defect, and it is active in every analysis reported here.

### F.3 KNOWN LIMITS OF THE RELEASED RECORD

We list these rather than leave them to be discovered.

- **Intervals on the cross-construction differences were not retrieved.** The six paired differences between ground-truth constructions in Table 6 are quoted as point estimates. Each

is a difference of independently intervalled estimates, which is what it is, but the paired intervals themselves were computed and not retrieved. Nothing in the reported null depends on them.

- **The inference library version is recorded, not enforced.** Every run used `transformers` 5.14.1, but the released environment does not pin it, and three behaviours we rely on are version-sensitive: attention-interface registration, the `dtype` argument at checkpoint load, and processor-side image-token expansion. The COCO file hashes of the downloaded images are not recorded.
- **One numerical-realisation caveat.** Greedy decoding in `bfloat16` admits near-ties, so a rebuild under a different batch composition can differ on a small number of tokens. Measured on the regenerated stimulus set, this moves the primary contrast from  $+0.4859$  to  $+0.4877$ , which bounds the effect for these conditions.
- **The exploratory re-reads of Appendix D.6 are not pre-registered and enter no registered decision.** Their degeneracy threshold and scoring window are inherited rather than registered as endpoints, their windowed and whole-continuation versions disagree materially, and their stratifications carry no intervals and no multiplicity correction.

#### F.4 REPORTING CONVENTIONS

Intervals are 95% percentile bootstrap intervals throughout, written  $[\text{lower}, \text{upper}]$  next to the point estimate. A contrast is written as the five-scene value minus the one-scene value, so a positive contrast means  $J$  is larger when the prefix’s five captions describe five scenes. The image-attributable component is written  $\Delta_{\text{image}} = \Delta_{\text{primary}} - \Delta_{\text{ablated}}$ , so a positive value means the sighted contrast exceeds the blind one. Rates are fractions in  $[0, 1]$ ; the literature commonly quotes the same quantities as percentages.

### G DECOMPOSING PUBLISHED METHODS ON STANDARD CHAIR

Section 3 applies the decomposition of Section 2 to a published training-free mitigation method scored with standard CHAIR on free COCO captioning, rather than to this paper’s own prefix endpoint; Appendix G.9 reports that a second method had no gain to decompose. Sections G.1–G.8 cover the first method: the arms and decoding, the CHAIR port and its validation, the construction of  $H$ ,  $F$ ,  $J$ ,  $d'$  and  $c$ , the integrity gates, the full decomposition with intervals, the link-sensitivity analysis, and the degeneracy of the blind arm. Appendix G.9 covers the second, including why it was dropped from the first experiment’s registration and then run under its own.

The pre-registered primary returns “cannot resolve”. It is a conjunction of three components fixed before any of these data existed, none of which returned an affirmative outcome (Appendix G.8). What survives every instrument is weaker than a resolved endpoint, and is the decomposition reading Section 3 takes from *this* method (it also reports the permuted-ground-truth secondary of Appendix G.7): the change in  $J$  is *dominated* by the criterion, with a small separation component that cannot be excluded.

#### G.1 METHOD, ARMS AND DECODING

The method is PAI (Liu et al., 2024b), a training-free intervention that amplifies attention to the image tokens during decoding, reimplemented here as an attention hook and gated as described in Appendix G.4. The model is LLAVA-1.5-7B of Appendix A.1, `llava-1.5-7b-hf` in `bfloat16` on a single consumer GPU. Decoding is greedy (`do_sample=False`) at the 512-token budget and under the prompt that the method’s own published protocol prescribes for COCO captioning, not this paper’s prefix prompt.

Six arms are run on the same 500 images:

- `vanilla`: no intervention ( $\alpha = 0.0$ ).
- `pai02`:  $\alpha = 0.2$ ,  $\gamma = 1.1$  — the value shipped as the released repository’s default and used by its documented commands.

- `pai05`:  $\alpha = 0.5$ ,  $\gamma = 1.1$  — the value the method’s paper prescribes for this model, and the only arm with a published CHAIR effect to decompose.
- `vanilla_blind`, `pai02_blind`, `pai05_blind`: the identical three arms with the image replaced by a flat mid-grey (127, 127, 127) rectangle at each image’s own pixel dimensions, the same blinding constant used throughout this paper.

Images are 500 COCO val2017 images. This is a documented deviation from the published protocol, which draws from val2014 under its own fixed seed; it is one of three deviations fixed in advance, the others being the `llava-hf` checkpoint port rather than the original release, and the CHAIR port of Appendix G.2.

Intervals are paired image-level bootstraps clustered on the image and paired across arms,  $B = 8000$ , with resample size asserted on every replicate as in Appendix A.6. A *gain* is written so that a positive value means CHAIR fell:  $\text{CHAIR}(\text{vanilla}) - \text{CHAIR}(\text{method})$ . Every other contrast is written method minus `vanilla`.

## G.2 THE CHAIR PORT, ITS POSITIVE CONTROL, AND WHAT ITS DEVIATIONS TOUCH

The scorer is the port of Appendix A.4: the 80-category synonym table verbatim, with a regex `[a-z]+` word tokenizer substituted for the NLTK tokenizer and a rule-based singularizer substituted for `pattern.singularize`.

**A direct cross-check against the reference scorer was not run, and is recorded as no-data rather than as a clean check.** The reference implementation depends on the `pattern` package and several NLTK corpora that were unavailable in the analysis environment. The two analyses below therefore *bound* the deviation from the other side; they do not verify the port against its reference.

**Positive control.** Three published LLAVA-1.5-7B vanilla CHAIR values at this protocol agree closely: 46.2/13.8 and 46.6/13.4 (Liu et al., 2024b), and 46.4 (Li et al., 2025b). The tolerance registered in advance was  $\text{CHAIR}_s \in [36.2, 56.2]$  and  $\text{CHAIR}_i \in [9.8, 17.8]$ . Measured:  $\text{CHAIR}_s = 45.00$  and  $\text{CHAIR}_i = 12.78$ , so the largest deviation from any published value is 1.6 points of  $\text{CHAIR}_s$  and 1.0 of  $\text{CHAIR}_i$ , far inside the registered band. A second registered component required the  $\alpha = 0.5$  effect on  $\text{CHAIR}_s$  to reach at least 10.0 with its interval excluding zero; measured +16.00 [+11.20, +20.80]. Both components pass. Truncation at the token budget is 0.000 on the vanilla arm, so the generation-budget confound is absent by measurement rather than by assumption.

Passing this control licenses one claim and no more: *the harness measures something that behaves like CHAIR at the published level*. It does not license treating absolute rates from this port as equal to the reference implementation’s to the decimal.

**What the two substitutions actually touch.** Of the 3,827 category-mention occurrences in the vanilla arm, 686 (17.9%) reach a mention only through the rule-based singularizer, so that substitution is doing real work. Disabling it entirely — a bound, not a plausible alternative scorer — moves  $\text{CHAIR}_i$  by 0.156 and  $\text{CHAIR}_s$  by 0.000; changing hyphen and apostrophe handling moves them by 0.228 and 0.000.

Table 24: Bounding the CHAIR port’s two substitutions on the vanilla arm. `nosing` disables singularization entirely; `hyphen` changes hyphen and apostrophe handling. Neither is offered as an alternative scorer; both bound how many decisions the substitutions can touch.

| scorer variant | $\text{CHAIR}_s$ | $\text{CHAIR}_i$ | mention occurrences |
|----------------|------------------|------------------|---------------------|
| base           | 45.00            | 12.78            | 3,827               |
| hyphen         | 45.00            | 13.01            | 3,760               |
| nosing         | 45.00            | 12.93            | 3,804               |

### G.3 HOW $H$ , $F$ , $J$ , $d'$ AND $c$ ARE FORMED

The scored unit is the (image, COCO category) pair: 500 images  $\times$  80 categories = 40,000 units per arm. A unit is *mentioned* if the arm’s caption names that category under the scorer above, and *present* if the category lies in the image’s ground-truth set  $G(i)$ , built by the union construction of Appendix A.4. Pooled over images,  $H$  is the mention rate over present units and  $F$  the mention rate over absent units. The three reported coordinates follow exactly as in Appendix C, on those same two rates and with nothing fitted:

$$J = H - F, \quad d' = Z(H) - Z(F), \quad c = -\frac{1}{2} [Z(H) + Z(F)].$$

The standard CHAIR statistics come from the same mention records:  $\text{CHAIR}_s$  is the proportion of captions containing at least one absent category, and  $\text{CHAIR}_i$  the proportion of category-mention occurrences that are absent.

Rates entering  $Z(\cdot)$  carry a log-linear (add-one-half) correction. On the three sighted arms its effect on  $d'$  is at most 0.0013 and the registered materiality flag is not triggered. On an arm that mentions nothing it is the entire quantity, which is why the blind rows of Table 25 carry no  $d'$  or  $c$ ; Appendix G.7 gives that case.

Table 25: Per-arm levels.  $\alpha$  is the intervention strength;  $\gamma = 1.1$  on both intervened arms.  $\text{CHAIR}_i$  is undefined on the blind arms because its denominator is empty, and their  $d'$  and  $c$  are arithmetic of the log-linear correction rather than measurements, so both are omitted here (Appendix G.7). Blind  $H$ ,  $F$  and  $J$  are raw rates: no caption names any of the 80 categories (the log-linear correction would print 0.0003 for  $H$ ).

| arm           | $\alpha$ | $\text{CHAIR}_s$ | $\text{CHAIR}_i$ | $H$    | $F$    | $J$     | $d'$    | $c$     |
|---------------|----------|------------------|------------------|--------|--------|---------|---------|---------|
| vanilla       | 0.0      | 45.00            | 12.78            | 0.7816 | 0.0100 | +0.7716 | +3.1032 | +0.7741 |
| pai02         | 0.2      | 48.40            | 13.04            | 0.7951 | 0.0107 | +0.7845 | +3.1263 | +0.7388 |
| pai05         | 0.5      | 29.00            | 7.21             | 0.7083 | 0.0050 | +0.7032 | +3.1212 | +1.0123 |
| vanilla_blind | 0.0      | 0.00             | undef.           | 0.0000 | 0.0000 | 0.0000  | —       | —       |
| pai02_blind   | 0.2      | 0.00             | undef.           | 0.0000 | 0.0000 | 0.0000  | —       | —       |
| pai05_blind   | 0.5      | 0.00             | undef.           | 0.0000 | 0.0000 | 0.0000  | —       | —       |

Table 26: Generation profile of the same six arms. Columns are mean generated tokens, truncation rate at the budget, mean category-mention occurrences per caption, mean distinct categories named per caption, distinct-8-gram fraction, and mean image-attention mass during decoding. The intervention is live in the blind arms too: image-attention mass rises with  $\alpha$  there as it does in the sighted arms, so the blind degeneracy of Appendix G.7 is not a dead hook.

| arm           | tokens | trunc. | mentions | distinct | 8-gram | attn. mass |
|---------------|--------|--------|----------|----------|--------|------------|
| vanilla       | 112.5  | 0.000  | 7.654    | 3.304    | 0.9990 | 0.1108     |
| pai02         | 114.2  | 0.000  | 7.714    | 3.398    | 0.9990 | 0.1459     |
| pai05         | 128.4  | 0.016  | 9.316    | 2.684    | 0.9440 | 0.2412     |
| vanilla_blind | 66.0   | 0.000  | 0.000    | 0.000    | 1.0000 | 0.0777     |
| pai02_blind   | 61.0   | 0.000  | 0.000    | 0.000    | 1.0000 | 0.1152     |
| pai05_blind   | 70.0   | 0.000  | 0.000    | 0.000    | 0.9048 | 0.2076     |

A registered informativeness guard would have triggered had the intervened arm bought its CHAIR gain by saying less. It is not triggered: at  $\alpha = 0.5$  captions are *longer* (112.5  $\rightarrow$  128.4 tokens) and carry *more* category-mention occurrences (7.654  $\rightarrow$  9.316), while the number of distinct categories named falls (3.304  $\rightarrow$  2.684) and repetition rises (distinct-8-gram 0.9990  $\rightarrow$  0.9440; 95th-percentile maximum token run 2  $\rightarrow$  3).

#### G.4 INTEGRITY GATES

Gates were run at the full 512-token budget on 16 images before the endpoint arms. Every blocking gate passed. The gate that matters most for reading anything below is the liveness gate: without it, a decomposition of “the intervention” could be a decomposition of nothing.

Table 27: Integrity gates for the decomposition. All blocking gates passed. The liveness gate is the one that establishes the intervention is actually changing the computation; it gates *this* reimplementa-tion and makes no claim about any released implementation.

| gate                                   | result                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| signal-detection instru-<br>ment       | Pass. Round-trip $(d', c) \rightarrow (H, F) \rightarrow (d', c)$ worst absolute error $1.7 \times 10^{-14}$ ; $Z(0.975)$ exact; the degenerate corners return the constants derived in advance ( $+0.609175$ measured against $+0.609175$ expected).                                                                                                                                                                                           |
| intervention    liveness<br>(blocking) | Pass. Determinism control ( $\alpha = 0$ run twice) differs on a fraction 0.0000 of captions. $\alpha = 0$ against $\alpha = 0.5$ differs on 1.0000, and $\alpha = 0$ against $\alpha = 0.2$ on 1.0000, against a floor of 0.90. Mean first divergence at token 12.1. Image-attention mass is monotone in $\alpha$ : $0.1066 < 0.1411 < 0.2380$ . Attention edits executed: 0 at $\alpha = 0$ , 547 at $\alpha = 0.2$ , 665 at $\alpha = 0.5$ . |
| causality                              | Pass. Residual post-softmax mass on strictly-future keys is exactly 0.0 when the kernel is handed no attention mask.                                                                                                                                                                                                                                                                                                                            |
| decode loop                            | Pass. 0/16 mismatches against the library’s own greedy generation through the same kernel at 512 tokens.                                                                                                                                                                                                                                                                                                                                        |
| kernel                                 | 0/16 mismatches against the stock attention implementation. Reported, not blocking: all arms share the same kernel, so a reduction-order difference is common to every arm and cancels in the paired contrasts.                                                                                                                                                                                                                                 |
| copy drift                             | Pass. 0 mismatches against the already-gated generator at $\alpha = 0.0$ and $\alpha = 0.5$ .                                                                                                                                                                                                                                                                                                                                                   |
| image block<br>blind arm               | Pass. Every row carries exactly 576 image tokens.<br>Pass. The blind pixel tensor is bit-identical across all images, and 16/16 blind captions differ from their sighted counterparts.                                                                                                                                                                                                                                                          |
| mask leak                              | 0. Across 376,033 attention calls the interface was handed an explicit mask 0 times and the causal repair was applied 2,689 times, so the repair is load-bearing.                                                                                                                                                                                                                                                                               |

Arm liveness at full scale confirms the same thing on the endpoint run: the fraction of captions differing is 0.998 between `vanilla` and `pai02` and 1.000 for every other arm pair, including each sighted arm against its blind counterpart and the blind arms against each other.

#### G.5 THE DECOMPOSITION

Three readings follow, and the order matters.

**At the repository default there is no gain to decompose.**  $\Delta\text{CHAIR}_i$  is  $-0.26 [-1.42, +0.91]$  and  $\text{CHAIR}_s$  moves the wrong way,  $45.00 \rightarrow 48.40$ . The registered outcome for that arm is “no gain to decompose”, and nothing is decomposed from it. The two settings are far apart on this benchmark:  $\text{CHAIR}_s$  is 48.40 at the repository default and 29.00 at the value the method’s own paper prescribes for this model, against a vanilla 45.00.

**At the paper-prescribed  $\alpha$ , CHAIR and  $J$  move in opposite directions.**  $\text{CHAIR}_i$  improves by 5.56 and  $\text{CHAIR}_s$  by 16.00, while  $J = H - F$  gets slightly worse,  $\Delta J = -0.0683 [-0.0871, -0.0486]$ , and  $d'$  does not move. Both rates fall,  $H$  by 0.0733 and  $F$  by 0.0050.

**The change in  $J$  is dominated by the criterion, and a small separation component cannot be excluded.** Counterfactually moving only  $c$  reproduces  $-0.0716 [-0.0844, -0.0583]$ ; moving only  $d'$  gives  $+0.0029 [-0.0092, +0.0150]$ . The criterion term is about 25 times the separation term in magnitude. That is the whole of what this measurement establishes about the split. It is *not* the stronger statement that the gain lives in  $c$  and not in  $d'$ : all three separation instruments have a positive point estimate, and Appendix G.6 shows why they cannot be collapsed into one verdict.

Table 28: The full decomposition, each quantity as method minus `vanilla` with a 95% interval, except the two CHAIR gains, which are written so that positive means CHAIR fell. “criterion only” and “separation only” are the counterfactual changes in  $J$  obtained by moving  $c$  alone and  $d'$  alone through Equation 6. The matched- $F$  row uses the truncation curve of Table 29; it is undefined at  $\alpha = 0.2$ , whose  $F$  of 0.0107 exceeds the largest  $F$  vanilla attains at any token budget (0.0100).

| quantity                                   | $\alpha = 0.5$ (paper-prescribed) | $\alpha = 0.2$ (repository default) |
|--------------------------------------------|-----------------------------------|-------------------------------------|
| CHAIR <sub>i</sub> gain                    | +5.56 [+3.88, +7.10]              | −0.26 [−1.42, +0.91]                |
| CHAIR <sub>s</sub> gain                    | +16.00 [+11.20, +20.80]           | −3.40 [−7.20, +0.40]                |
| $\Delta H$                                 | −0.0733 [−0.0922, −0.0534]        | +0.0136 [+0.0000, +0.0270]          |
| $\Delta F$                                 | −0.0050 [−0.0061, −0.0039]        | +0.0007 [−0.0003, +0.0016]          |
| $\Delta J$                                 | −0.0683 [−0.0871, −0.0486]        | +0.0129 [−0.0008, +0.0264]          |
| $\Delta d'$ (probit separation)            | +0.0181 [−0.0577, +0.0938]        | +0.0231 [−0.0374, +0.0822]          |
| $\Delta c$ (threshold)                     | +0.2383 [+0.1955, +0.2807]        | −0.0353 [−0.0636, −0.0070]          |
| $\Delta \log \text{OR}$ (logit separation) | +0.3037 [+0.1349, +0.4761]        | +0.0176 [−0.1135, +0.1453]          |
| $\Delta H$ at matched $F$                  | +0.0245 [−0.0037, +0.0511]        | undefined                           |
| $\Delta J$ , criterion only                | −0.0716 [−0.0844, −0.0583]        | +0.0093 [+0.0018, +0.0168]          |
| $\Delta J$ , separation only               | +0.0029 [−0.0092, +0.0150]        | +0.0037 [−0.0060, +0.0132]          |
| CHAIR <sub>i</sub> gain, blind             | undefined                         | undefined                           |
| sighted gain minus blind gain              | undefined                         | undefined                           |

**The matched- $F$  instrument.** Truncating the vanilla arm’s own captions at a shorter token budget lowers mention propensity without changing the model, the image or the decoding, so it traces a purely criterion-driven trajectory through  $(F, H)$  against which a method can be placed. Table 29 gives it. Reading  $H$  off that curve at the intervened arm’s own  $F$  and subtracting gives  $\Delta H$  at matched  $F = +0.0245$  [−0.0037, +0.0511], against an equivalence margin of  $\pm 0.03$  fixed in advance — the interval straddles the margin, so the registered outcome is “cannot resolve”, which is an underpowered result and not a pass. On the same construction, vanilla truncated to the token budget at which its false-alarm rate matches the intervened arm’s gives CHAIR<sub>i</sub> = 8.04 [6.91, 9.27] against the intervened arm’s 7.21, a difference of −0.83 [−1.80, +0.38]. The asymmetry in how that is read was fixed in advance: lying *on* this curve is strong evidence of a criterion move, while lying *above* it is weaker evidence of a discrimination gain, because a method can sit on a better but still purely criterion-driven trajectory.

Table 29: The vanilla arm’s own truncation curve: the same captions scored at shorter token budgets.  $F$  is monotone in the window. This is the instrument behind the matched- $F$  row of Table 28.

| window (tokens) | $F$    | $H$    | CHAIR <sub>s</sub> | CHAIR <sub>i</sub> | mentions/cap. |
|-----------------|--------|--------|--------------------|--------------------|---------------|
| 8               | 0.0001 | 0.1925 | 0.8                | 1.23               | 0.65          |
| 16              | 0.0004 | 0.3620 | 3.2                | 2.61               | 1.30          |
| 24              | 0.0008 | 0.4482 | 6.2                | 3.24               | 1.98          |
| 32              | 0.0011 | 0.4988 | 8.6                | 3.73               | 2.52          |
| 48              | 0.0019 | 0.5696 | 13.0               | 4.81               | 3.58          |
| 64              | 0.0031 | 0.6275 | 20.2               | 5.87               | 4.60          |
| 96              | 0.0069 | 0.7384 | 36.8               | 10.15              | 6.44          |
| 128             | 0.0096 | 0.7754 | 44.0               | 12.55              | 7.43          |
| 192             | 0.0100 | 0.7816 | 45.0               | 12.78              | 7.65          |
| 256             | 0.0100 | 0.7816 | 45.0               | 12.78              | 7.65          |
| 384             | 0.0100 | 0.7816 | 45.0               | 12.78              | 7.65          |
| 512             | 0.0100 | 0.7816 | 45.0               | 12.78              | 7.65          |

## G.6 LINK SENSITIVITY, AND THE MECHANICAL REASON THE LINKS DISAGREE

The three separation instruments all have a positive point estimate at  $\alpha = 0.5$ ; they differ only in whether the interval clears zero. Writing “two of three say the separation did not change” would overstate the evidence, so we do not.

Table 30: The three separation instruments at  $\alpha = 0.5$ , with  $\Delta d' = \Delta Z(H) - \Delta Z(F)$  and  $\Delta \log \text{OR}$  its logit companion. One clean null, one positive, and one inconclusive that leans in the same direction as the positive one. The matched- $F$  contrast is uninformative, not affirmative.

| instrument                     | $\Delta$ separation        | reads as                                                                    |
|--------------------------------|----------------------------|-----------------------------------------------------------------------------|
| probit, $\Delta d'$            | +0.0181 [−0.0577, +0.0938] | contains 0; the whole interval sits inside the registered $\pm 0.15$ margin |
| logit, $\Delta \log \text{OR}$ | +0.3037 [+0.1349, +0.4761] | excludes 0                                                                  |
| matched- $F$ , $\Delta H$      | +0.0245 [−0.0037, +0.0511] | contains 0, leaning positive                                                |

**Why they disagree.** Both rates fell:  $H$  from 0.7816 to 0.7083 and  $F$  from 0.0100 to 0.0050. Whether that is “the same latent move in both” depends entirely on how the link stretches a probability into latent units, and the probit and the logit stretch very differently at these two operating points. Evaluated at the midpoint of the two arms’ rates:

Table 31: Local stretch of the two links,  $d(\text{link})/dp$ , evaluated at the midpoint of the two arms’ rates. The logit amplifies movement at the low rate relative to movement at the mid rate by a further factor of about 1.6 compared with the probit.

| operating rate     | probit | logit  | ratio |
|--------------------|--------|--------|-------|
| $\bar{H} = 0.745$  | 3.11   | 5.26   | 1.69  |
| $\bar{F} = 0.0075$ | 48.12  | 133.81 | 2.78  |

Because  $2.78/1.69 \approx 1.64$ , the same pair of proportional falls decomposes differently under the two links:

$$\Delta Z(H) = -0.2292, \quad \Delta Z(F) = -0.2473,$$

against, with  $L(p) = \log\{p/(1-p)\}$ ,

$$\Delta L(H) = -0.3880, \quad \Delta L(F) = -0.6917.$$

Under the probit the two shifts are nearly equal, so  $d'$  is unchanged; under the logit  $F$  falls far more than  $H$ , so separation rises.

**This is not a units artefact.** Dividing  $\Delta \log \text{OR}$  by the usual mid-scale ratio of 1.70 gives +0.1787 [+0.0793, +0.2800], which still excludes zero and still exits the  $\pm 0.15$  margin at the top. So on a logistic evidence model there is a small but real separation gain, and on a Gaussian one there is none. At a *single* operating point these two readings are observationally equivalent: the same  $(H, F)$  pair is consistent with both, and nothing in this design distinguishes them. Only a multi-point ROC, obtained by sweeping a decoding knob, identifies the latent distribution, and no such sweep was run here.

Two properties of this flag were fixed in advance and are recorded rather than buried. The equivalence margin on  $\Delta d'$  is  $\pm 0.15$ , a judgement call; a reader who thinks the right margin is 0.05 should read the interval rather than the label, which is why every interval is printed. And the logit companion is applied at the same *numeric* margin as the probit one, which is stricter on the logit scale by roughly a factor of 1.7, so the link-sensitivity flag triggers more readily than a scale-matched margin would.

### G.7 THE BLIND ARM DEGENERATES, AND WHY THAT IS A PROPERTY OF THE ENDPOINT

The blind arm was registered and run, and it does not yield a contrast. This is a fact about the *ablation*, not about CHAIR: every graded degradation we later ran leaves the denominator non-empty and does yield one (Section 3).

Across all 500 images the blind vanilla arm emits a single distinct caption, and that caption mentions none of the 80 COCO categories: mean category-mention occurrences 0.00, with 0/500 captions containing any. It is not a truncation or a decoding failure — mean length is 66 tokens, so the model talks at length, about nothing in the category vocabulary — and it is not a dead intervention, since image-attention mass still rises monotonically with  $\alpha$  across the blind arms (Table 26) and every blind arm pair differs on 100% of captions.

Two statistics break, and they break differently. CHAIR<sub>i</sub>’s denominator *is* the model’s own mention set, which here is empty, so CHAIR<sub>i</sub> is 0/0 and genuinely undefined. CHAIR<sub>s</sub> is defined and equals 0.00: no caption contains a hallucinated object because no caption contains an object at all. One statistic is undefined; the other is floored. Both are uninformative, and every blind contrast in Table 28 is therefore reported as undefined rather than as a null.

**The blind arms’  $d'$  and  $c$  are correction artefact, not measurement.** With zero mentions both rates collapse to the log-linear correction’s  $0.5/(n+1)$ , and because the present and absent classes have very different sizes the correction manufactures a spurious positive separation —  $d' = +0.781$  and  $c = +3.815$  on all three blind arms, identically. That constant was derived from the correction before any generation existed and the instrument gate of Table 27 asserts the corresponding degenerate corner. It is never reported as a measurement, and it must not be confused with the permuted-ground-truth value below, which is numerically close and conceptually unrelated.

**The underlying property is an endogenous denominator, and it bites before degeneracy does.** The difficulty is not only that the support can empty. It is that CHAIR’s denominator is a function of the arm: even a blind arm that still named objects would give a CHAIR<sub>i</sub> computed over a *different* item set from the sighted arm, so the sighted-blind contrast would conflate a change in rate with a change in composition. Total collapse is the extreme case of a confound present whenever the ablation moves mention counts at all. An endpoint whose candidate set is supplied externally — as this paper’s prefix endpoint is — does not carry that exposure, which is why the blind control of Section 4.3 was runnable there.

**Scope.** This is one model against one ablation. A different model may well name objects under a grey field, and a weaker ablation — noise, blur, shuffled patches — would likely leave the support non-empty. The defensible claim is that *this* ablation degenerates on *this* model and that any self-selected-support endpoint is exposed to that failure, not that CHAIR cannot be blinded in general. Swapping to a different ablation after seeing the registered one degenerate was available and was not done.

**The secondary that blinds the evaluation instead.** Because blinding the generation degenerates, a secondary was added that blinds the *evaluation*: each already-generated sighted caption is scored against a different image’s ground truth, over 20 fixed-point-free derangements averaged inside each bootstrap replicate. It was fixed with its own decision rule before it was computed, costs no generation, and is never promoted over a primary. It returns “gain is image-specific”: the  $\alpha = 0.5$  arm’s CHAIR<sub>i</sub> advantage under permuted ground truth is +0.22 [−0.94, +1.33], so scoring the same captions against the wrong image removes the advantage. The gain is therefore tied to this image’s content even though it buys no measurable  $\Delta d'$ . Independently, vanilla’s own  $d'$  falls from +3.1032 on true ground truth to +0.7432 under permutation: a caption naming common objects achieves  $d' \approx 0.74$  with no reference to the image at all, which is category base-rate bookkeeping rather than grounding. This secondary answers “is the gain image-specific?”, the question a mismatched-image control asks; on this prompt-only endpoint the two are algebraically the same test (Appendix I). It does not say what a uniform-field blind arm would return, since that arm has no denominator.

Table 32: The three registered components and the combined outcome, per arm. The combined primary is “cannot resolve” at the paper-prescribed  $\alpha$ . Anyone reading this as a confirmation is misreading it.

| registered component | $\alpha = 0.5$               | $\alpha = 0.2$               |
|----------------------|------------------------------|------------------------------|
| decomposition        | “link-sensitive”             | “no gain to decompose”       |
| blinding             | “blind denominator too thin” | “blind denominator too thin” |
| truncation curve     | “cannot resolve”             | “extrapolation”              |
| combined             | “cannot resolve”             | “no gain to decompose”       |

## G.8 REGISTERED OUTCOMES, AND WHAT THIS MEASUREMENT CANNOT DECIDE

The following hold whichever way the measurement came out and were fixed before the verdict.

- **It cannot show that any published CHAIR number is wrong.** It measures a reimplement of one method in this harness on this 500-image draw. The published numbers were produced on a different checkpoint port, a different image list and a different CHAIR implementation.
- **It cannot say what any released implementation does.** The liveness gate establishes that  $\alpha$  is live *here*. If a released hook were inert and this one responds, those are two different facts and are not conflated.
- **One operating point per arm is not an ROC.**  $d'$  from a single  $(H, F)$  pair assumes equal-variance Gaussian evidence. The logit companion tests sensitivity to that assumption and the truncation curve is an assumption-light cross-check, but neither turns one point into a curve.
- **Two intervention strengths of one method, one model, one prompt, one decoding mode.** The second method, and the second decoding mode, are Appendix G.9. Nothing here generalises to every training-free method, and Section 3 does not say it does.

## G.9 A SECOND METHOD, BRIEFLY

A second method, VCD (Leng et al., 2024), was separated from the PAI registration by its Amendment 1 and run under a registration of its own. It returned no CHAIR gain to decompose:  $\text{CHAIR}_i$  worsens under greedy decoding and is unchanged under the method’s sampling protocol. We report it as a registered null, not as a claim about VCD.

**Why the harness is not the likely explanation.** The same CHAIR port reproduces the unmodified model’s published  $\text{CHAIR}_s$  and  $\text{CHAIR}_i$  exactly (45.00 and 12.78), the same harness reproduces PAI’s published POPE gain (+1.10 accuracy points against +1.06), and its CHAIR recoding recovers the recall drop Jung et al. (2025) report for PAI. VCD’s own paper evaluates POPE, MME and LLaVA-Bench, with sampling as its main protocol, and reports no CHAIR figure, so there is no published CHAIR gain that this harness failed to reproduce. The worsening appears only under greedy decoding, which is not VCD’s protocol; under its sampling protocol the estimate is a null. Yin et al. (2025) report, separately, that contrastive decoding’s POPE gains reflect a shift in response tendency.

**Arms and settings.** Amplification 2.0 and adaptive-plausibility cutoff 0.1, as the method’s paper states. The forward noise process is the repository’s rather than the paper’s — a declared and load-bearing deviation, since under the paper’s own linear schedule the cumulative signal coefficient is  $3.7 \times 10^{-12}$  at  $T = 500$ , so no  $T$  in its ablation grid would be distinguishable from any other, while the repository’s sigmoid schedule retains 0.863 and makes  $T$  a live dose. Three arms: `vcd` (greedy,  $T = 500$ , primary), `vcd_t999` (greedy, sensitivity) and `vcd_s` (sampled, the published protocol), the first two against the parent experiment’s greedy `vanilla` and the third against a sampled `vanilla_s`; no contrast crosses decoding modes.

**Gates.** The positive control passed exactly: re-scoring the parent experiment’s own `vanilla` rows through this port returns  $\text{CHAIR}_s = 45.00$  and  $\text{CHAIR}_i = 12.78$ , its published figures. Liveness passed on all four of its clauses — determinism with the method held fixed 0.000; method off versus on 1.000 against a floor of 0.90; method off versus the parent experiment’s `vanilla` 0.000, which must be zero; and the noise dose,  $T = 0$  versus  $T = 500, 1.000$ . Because a contrastive stream that silently failed to diverge would look like a working one, the second stream is checked *during* decoding rather than assumed: the mean per-step KL between the two streams is 0.0084 against a floor of  $10^{-3}$  at  $T = 500$  and rises to 0.1704 at  $T = 999$ , so the contrast strengthens with the dose as it must. Degeneracy screens pass in every arm (unique-token ratio  $\geq 0.9990$ ; 95th-percentile maximum token run 2).

Table 33: The second method, all three arms, each quantity as method minus its own mode-matched baseline. The CHAIR gains are written so that *positive means CHAIR fell*, matching Table 28; both greedy arms are therefore negative, meaning CHAIR rose.  $\Delta d'$  is the probit separation term and  $\Delta c$  the threshold term.

| quantity                        | vcd, $T=500$ (primary)                   | vcd_t999 (sensitivity)     | vcd_s (published)          |
|---------------------------------|------------------------------------------|----------------------------|----------------------------|
| decoding baseline               | greedy<br>vanilla                        | greedy<br>vanilla          | sampled<br>vanilla_s       |
| $\text{CHAIR}_i$ gain           | -1.95 [-3.26, -0.65]                     | -3.48 [-4.87, -2.09]       | +0.72 [-0.94, +2.38]       |
| $\text{CHAIR}_s$ gain           | -6.60 [-10.80, -2.40]                    | -11.20 [-15.60, -6.80]     | -2.20 [-6.80, +2.60]       |
| $\Delta J$                      | -0.0054 [-0.0206, +0.0099]               | +0.0208 [+0.0046, +0.0365] | +0.0322 [+0.0122, +0.0518] |
| $\Delta d'$ (separation)        | -0.0717 [-0.1369, -0.0062]               | -0.0541 [-0.1233, +0.0141] | +0.0873 [+0.0151, +0.1584] |
| $\Delta c$ (threshold)          | -0.0234 [-0.0548, +0.0082]               | -0.1158 [-0.1502, -0.0816] | -0.0585 [-0.0983, -0.0186] |
| $\Delta J$ , criterion only     | +0.0062 [-0.0021, +0.0146]               | +0.0291 [+0.0203, +0.0379] | +0.0170 [+0.0054, +0.0288] |
| $\Delta J$ , separation only    | -0.0117 [-0.0222, -0.0010]               | -0.0088 [-0.0200, +0.0023] | +0.0155 [+0.0027, +0.0283] |
| criterion:separation ratio      | 0.53                                     | 3.30                       | 1.10                       |
| $\Delta d'$ inside $\pm 0.15$ ? | yes                                      | yes                        | no (upper +0.158)          |
| registered outcome              | "no gain to decompose" on all three arms |                            |                            |

**Registered verdict: “no gain”.** Two of the three arms have  $\text{CHAIR}_i$  gains whose intervals exclude zero *on the wrong side*: at both of the method’s published noise settings, under greedy decoding, CHAIR gets worse. The third — the method’s own published sampling protocol — is a null whose interval spans zero. There is no gain here to decompose, and the registered primary returns “no gain to decompose” on all three arms and “no gain” overall. The decomposition columns are therefore reported for completeness and carry no claim about where a gain lives, because there is no gain.

**The one arm that reports against the section’s reading.** The sampled arm’s separation term is +0.0873 [+0.0151, +0.1584], the only positive  $\Delta d'$  in either experiment whose interval excludes zero (the greedy primary’s -0.0717 [-0.1369, -0.0062] also excludes zero, in the other direction) and the only one that leaves the  $\pm 0.15$  margin, and it is also the only arm whose  $\text{CHAIR}_i$  gain is not negative. It decomposes a gain this experiment’s own registration classes as immaterial (+0.72, interval spanning zero), so it cannot be read as a separation-driven improvement; but it is the single result in this appendix pointing away from “the separation term never moves.”

#### What this experiment cannot decide, fixed before the verdict.

- **It is not evidence that the method does not work.** It is evidence about a reimplementations of its published equations, on one model, one prompt, one image draw, with the repository’s noise process. Because VCD publishes no CHAIR figure, no external number anchors any arm, and that absence is carried as a limit rather than solved.
- **The greedy arms are not the authors’ protocol.** They exist to be commensurable with the parent experiment. The published-protocol arm is the sampled one, and it is a null, not a loss.
- **Fifteen registered verdicts across the two experiments.** Three primaries for each of the five intervened arms (three here, two in the first experiment), every interval printed so a reader can apply their own family-wise adjustment. No verdict was selected after the fact.

## H FORCED-CHOICE READOUTS

**The registered probe.** Posed as a pre-registered yes/no probe on the identical units, the scene-count contrast is  $-0.01$   $[-0.03, +0.01]$  against  $+0.49$  in free generation (Table 20); on that outcome the registration committed us to confine every claim to free generation, and we do. The probe saturates and two non-saturating graded readouts are also null, as the rest of this appendix shows.

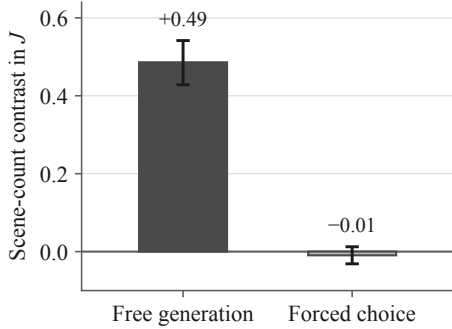


Figure 9: The scene-count contrast in free generation, where the model selects which objects to name, and as a forced-choice probe, where the question supplies the object; 95% intervals.

Table 34: Every probe fails all three off-ceiling conditions fixed in advance (pooled over arms): C1  $F \in [0.20, 0.80]$ , C2  $H \leq 0.90$ , C3 yes-rate  $\in [0.15, 0.85]$ . Image-only: “answer only from the image”; mixed: adds “some are, some are not”. Closed-set rows: calibration,  $n = 478$ .

| probe                        | $H$    | $F$    | yes-rate |
|------------------------------|--------|--------|----------|
| published probe              | 0.9732 | 0.8796 | 0.8949   |
| published probe, re-run here | 0.9739 | 0.8791 | 0.8945   |
| image-only prompt            | 0.9711 | 0.8622 | 0.8799   |
| two-alternative choice       | 0.9437 | 0.9347 | 0.9362   |
| closed candidate set         | 1.0000 | 1.0000 | 1.0000   |
| closed candidate set, mixed  | 1.0000 | 0.9949 | 0.9958   |

**The objection.** The registered forced-choice probe moves the scene-count contrast by  $+0.4859$  under free generation and by  $-0.0095$   $[-0.0314, +0.0124]$  under forced choice on identical units (Figure 9). But the published forced-choice probe is saturated: its false-positive rate rises from 0.022 with no prefix to 0.8714 and 0.8835 under the one- and five-scene prefixes, and it fails every off-ceiling condition fixed in advance (Table 34). Its null could mean that forced choice removes the selection step carrying the effect, or that the probe would return a null for anything.

**Changing the probe does not de-saturate it.** We pre-registered four candidate probes, each read in one forward pass as a probability  $p_{\text{yes}}$  over two fixed answer-token groups and thresholded for the binary answer. All four stay at ceiling, the closed candidate set with never-mentioned distractors hardest, at a yes-rate of 1.0000. A post-hoc diagnostic on the two-alternative variant shows why: it chooses the unit object at 0.9362 when the prefix named it and 0.4990 when it did not, so the binary format measures prefix membership. The registered outcome is “still saturated”, which returns no verdict.

**The graded readout of the published probe.** Registered to be reported in every branch, it gives  $+0.0008$   $[-0.0142, +0.0166]$ , width 0.0309 against 0.0425 for the binary answer, and it is not pinned: between 61% and 74% of each arm’s units have  $p_{\text{yes}}$  inside  $(0.05, 0.95)$ . That evidence is post-hoc, because the registered ceiling gate was defined on the binary rate and triggered.

**Scoring objects the prefix never names.** A second pre-registered analysis scores distractors, objects the prefix never mentions, with the two-alternative probe; prefix membership is constant across them and cannot pin the readout. The registration names the binary readout as primary, gated by the conditions of Table 34, and the graded readout as co-primary, gated on the graded quantity itself (interior fraction at least 0.50,  $p > 0.99$  fraction at most 0.10). Both gates were fixed before any five-scene row existed (the earlier run had produced one-scene rows only), and a failure of one does not block the other. Free generation on these units moves by  $+0.3655$   $[+0.2987, +0.4303]$ , so there is an effect to find. The binary primary missed its ceiling gate ( $H = 0.9041$  against a ceiling of 0.90), so the registered outcome is “distractor probe saturated”, which is not a verdict. The graded co-primary passed its own gate: the forced-choice register moves by  $-0.0057$   $[-0.0153, +0.0036]$ , or  $-0.006 \pm 0.010$  once widened in quadrature by run-to-run replay noise, and the readout is live on the same units and scale, registering ground-truth presence at  $+0.2981$  (0.7966 against 0.4985).

The registration reports the two readouts independently, but its pre-committed reading accepts a null verdict only if the binary gates pass. They did not, so the graded null is supporting evidence and not a registered confirmation.

**Corrections.** The first scoring run reported “distractor probe saturated” as the headline cell. A later change to the headline selector, reasoning from the clause that the two readouts are gated independently, reported the co-primary’s “distractor null confirmed” instead; every number is identical between the two runs, and both result files are kept. That change conflicts with the registration’s pre-committed acceptance rule for a null, which requires the binary gates to pass. We follow the stricter clause: the registered outcome is “distractor probe saturated”. Separately, a replay gate shown to reject a perfect re-run of the earlier probe was replaced before endpoint generation by one requiring exact agreement away from the decision boundary.

**Scope.** Both graded analyses are consistent with a forced-choice null that does not depend on prefix-membership pinning, since this one scores units where pinning cannot operate. Neither is a registered confirmation: the earlier graded readout is post hoc, and this one is excluded from confirmation by its registration’s own acceptance rule. No off-ceiling binary probe was obtained.

## I THE ABLATION LADDER

Section 4.4 reports that the answer to “does the rise need the image?” depends on the ablation. This appendix gives the ten ablations, how severe each one is, and what each returns.

**Construction.** Every ablation is applied to the canonical  $336 \times 336$  image the processor produces, so geometry is identical across arms; a gate asserts 32/32 byte-identical geometry against the unablated path. `noise $k$`  adds uniform pixel noise of amplitude  $k$ ; `blur $k$`  applies a Gaussian blur of radius  $k$ ; `lowres16` downsamples to  $16 \times 16$  and back; `shuffle24` permutes  $24 \times 24$  patches; `grey` and `black` replace the image with a uniform field; `mismatch` substitutes a different real COCO image under a derangement (seed 20260919, asserted bijective and self-pair-free). Prefix token ids are bit-identical across all arms.  $n = 3000$  cells per rung on the prefix endpoint (1500 per condition), 500 images on CHAIR.

**Severity.** We define severity as  $1 - \cos$  between the mean-pooled image features the model’s own vision pathway produces for the clean and the ablated image. It is a property of the ablation, measured before any endpoint is read. It measures *displacement in feature space*, not destruction of task-relevant content, and we have not validated it against a downstream task: two real scenes share much of their mean-pooled structure, while a uniform field is out of distribution for the encoder, so the substitution can displace features less than a blank while destroying far more of what the endpoint scores. Claims below are therefore about displacement, and the substantive point is the four named counterexamples rather than the ordering as a whole.

**Reading.** The seven graded pixel ablations pool to  $-0.006$   $[-0.037, +0.025]$ ; the substitution returns  $+0.160$   $[+0.105, +0.213]$ , and the paired difference against grey is  $+0.176$   $[+0.119, +0.230]$ . As blind-to-sighted ratios the same contrast is  $1.033$   $[0.921, 1.160]$  under grey against  $0.671$   $[0.573, 0.774]$  under the substitution; these are the two figures the main text compares. Over the nine pixel ablations severity and  $\Delta_{\text{image}}$  have a rank correlation of  $+0.10$ ; with  $n = 9$  that is uninformative and we draw no conclusion from it. What does the work is the four named counterexamples — `lowres16`, `black`, `grey` and `blur16` — each displacing the image further than the substitution and each returning an interval containing zero. The contribution sits in the hit rate ( $\Delta H = +0.149$   $[+0.097, +0.201]$  against  $\Delta F = -0.010$   $[-0.034, +0.013]$ ): with the true image  $H$  rises  $0.676 \rightarrow 0.780$  across the contrast, with a substituted one it *falls*  $0.644 \rightarrow 0.599$ . Grey hides this by collapsing the one-scene  $H$  to  $0.468$ , preserving the difference while destroying the level. Cross-condition length matching gives  $+0.138$   $[+0.096, +0.181]$ .

**The substitution arm is not degenerate.** Pooled over both conditions its median continuation is 384 tokens against grey’s 60, its empty rate  $0.017$  against  $0.025$ , its  $\leq 3$ -token rate  $0.082$  against  $0.124$ , and it produces 2848 distinct continuations of 3000 against grey’s 2693. It is healthier than the arm it is compared with.

**Registration and one amendment that mattered.** The ladder was pre-registered before any rung was generated, with three dated amendments. Amendment 3 replaced a *comparative* collapse screen (“no more degenerate than the reference arms”) with absolute thresholds, because in the five-scene condition both reference arms sit at the 384-token generation budget, so a bar of  $\min(\text{sighted}, \text{grey})$  equals the ceiling and flags every rung that terminates naturally; the diversity ratios fall mechanically with length and have the same fault. The amendment was written before any ladder verdict was read and it changed exactly one verdict — the substitution arm, from “confounded by collapse” to “image contributes”, on an 8-gram ratio of 0.567 against 0.575. Under the original screen, the substitution result would have been discarded by a screen whose form was wrong.

**Positive controls.** The scorer reproduces the published  $\Delta_{\text{primary}} = +0.4859$ ,  $\Delta_{\text{ablated}} = +0.5019$  and  $\Delta_{\text{image}} = -0.0160$  to  $3 \times 10^{-5}$ , and a re-generation of the grey arm is 3000/3000 token-identical to the published one with  $\Delta_{\text{image}}$  reproducing to 0.000000.

**On CHAIR.** Under grey, and equally under flat black, the model names none of the 80 categories in any of 500 captions, so  $\text{CHAIR}_i$  is 0/0. Every graded degradation leaves the denominator non-empty: 500, 500, 498, 489, 378, 405, 450 and 500 captions of 500 retain a category mention under `noise16/32/64`, `blur4`, `blur16`, `lowres16`, `shuffle24` and `mismatch` respectively. Against a sighted PAI gain of +5.56 in  $\text{CHAIR}_i$  (this experiment’s resample gives [+3.88, +7.18], the PAI experiment’s [+3.88, +7.10]), the image-attributable part is +5.16 [+2.89, +7.42] under substitution. Under `blur16` it is +8.70 [+2.55, +14.96], more than the whole gain, so PAI raises  $\text{CHAIR}_i$  on the 378 blurred captions that keep a mention; with that denominator shrinking, we do not use the blur arm as evidence. The three noise rungs retain the gain but fail the registered severity floor and are not quoted as evidence. On this prompt-only endpoint the substitution arm is algebraically the registered permuted-ground-truth secondary evaluated at a single derangement; the secondary averages twenty.

**Not done.** LLAVA-OV-0.5B was not run on the ladder: its processor uses AnyRes multi-crop, so the canonical geometry the geometry gate licenses does not transfer and the gate would need redesigning first.

## J POPE WITH A BLIND ARM

**Data.** Official POPE files from RUCAIBox/POPE, all three MS-COCO splits (`random`, `popular`, `adversarial`), 3000 questions each for 9000 total over 500 distinct images, exactly 1500 yes and 1500 no per split, with per-file sha256 re-asserted at every load. Because the splits are balanced,  $H$  is the reported recall and  $F = H + 1 - 2 \text{acc}$  exactly; the scorer asserts this identity on every cell and it holds to  $2.2 \times 10^{-16}$  across 48 cells.

**A snapshot trap.** LLAVA-1.5’s own evaluation documentation points at a different POPE snapshot whose `random` split has 2910 questions at 1500 yes / 1410 no. An all-“yes” responder scores 51.55 there against 50.00 on the balanced file, so the balanced identity above does not hold for numbers computed on that snapshot. We use the current balanced files throughout.

**Arms.** `vanilla` ( $\alpha = 0$ ,  $\gamma = 1$ , kernel- and loop-matched rather than a separate code path), `pai_full` ( $\alpha = 0.5$ ,  $\gamma = 1.1$ ; the released `-use-attn -use-cfg`), `pai_attn` ( $\alpha = 0.5$ ,  $\gamma = 1.0$ ; attention only, amplifying the prefill row), and `pai_both`, a constructed configuration that is not a released one and is labelled as such wherever it appears. All arms are greedy; no contrast compares across decoding modes. Ablations: `sighted`, `grey`, and a mismatched real image, plus a severity ladder (`noise25/50/100`, `blur4`, `blur16`, `lowres`, `pshuffle`).

**Result.** On `pai_attn` — the configuration that reproduces PAI’s published POPE number, +1.10 accuracy points against their +1.06 — the gain is *partly* attributable to the image, at the registered label “partial” defined below:  $\Delta_{\text{sighted}}(J) = +0.0220$  [+0.0107, +0.0336],  $\Delta_{\text{ablated}}(J) = +0.0013$  [−0.0102, +0.0124], and  $\Delta_{\text{image}}(J) = +0.0207$  [+0.0042, +0.0371]; in accuracy +1.033 [+0.21, +1.86].  $B = 4000$ , paired, clustered on the 500 images. The registered label

is “partial” rather than “image-attributable”, because the lower bound  $+0.0042$  falls below the pre-registered half-gain line of  $0.0110$ , so the rule cannot certify that more than half the gain is image-attributable. `pai_both` agrees ( $+0.0238$  [ $+0.0078$ ,  $+0.0398$ ]).

**The released default measures nothing.** `pai_full` gives  $\Delta_{\text{image}} = +0.000$ . Its attention stage is guarded so that the conditional prefill pass is never amplified, and on POPE the answer is the first generated token; `pai_both` and `pai_full` differ only in whether the prefill last row is touched, and that difference is worth  $+1.112$  accuracy points against  $+0.056$  for the decoding-distribution stage. The whole published gain lives in one attention row.

**The grey arm returns no data.** Under grey the model answers “no” to all 9000 questions, fluently and correctly — a grey field contains none of the queried objects — so  $H = F = 0$  by construction,  $\Delta_{\text{abladed}}$  is mechanically zero, and an unguarded reading would have returned “image-attributable” while measuring nothing. The registration forbade that reading in advance: the arm is “blind arm degenerate” and  $\Delta_{\text{image}}$  is undefined, not null. The severity ladder shows the same dissociation as Appendix I: `blur16` retains  $J = +0.19$  while a sharp photograph of another scene retains  $+0.04$ , and `pshuffle` dissociates the rates ( $H$   $0.883 \rightarrow 0.857$  while  $F$   $0.184 \rightarrow 0.330$ ).

**Gates and controls.** Fifteen blocking gates passed before any endpoint was read, including per-item token identity across arms with zero padding, exactly 576 image tokens per row, copy-drift token-identity against the original kernel, 0/4 decode mismatches against Hugging Face `generate`, post-softmax mass on strictly-future keys exactly 0.0, and determinism 0/288. Positive control: our `vanilla` sighted arm scores 84.91 accuracy against PAI’s published 84.76.

**PAI’s own published table, recoded.** On POPE’s balanced splits the two coordinate systems interconvert exactly:  $\text{acc} = (H + 1 - F)/2$  and  $F1 = 2H/(1 + H + F)$ , so  $F = H + 1 - 2\text{acc}$  and  $H = F1(1 - \text{acc})/(1 - F1)$ . PAI’s Table 2 (Liu et al., 2024b) reports accuracy and F1 for LLaVA-1.5 under single-turn greedy decoding; recoded, it gives Table 35. The recoding is derived, not measured: it carries no interval, inherits the equal-variance assumption behind  $d'$  and  $c$ , and is approximate because F1, unlike accuracy, does not average exactly over the three splits. It is a cross-check on Table 1, which shows the same ordering,  $\Delta c$  well above  $\Delta d'$ .

Table 35: PAI’s published POPE result (Liu et al., 2024b, Table 2), recoded into rates and signal-detection coordinates.

| Arm     | Acc   | F1    | $H$    | $F$    | $J$    | $d'$   | $c$    |
|---------|-------|-------|--------|--------|--------|--------|--------|
| Vanilla | 84.76 | 85.51 | 0.899  | 0.204  | +0.695 | +2.105 | −0.226 |
| PAI     | 85.82 | 85.97 | 0.869  | 0.153  | +0.716 | +2.147 | −0.048 |
| Change  | +1.06 | +0.46 | −0.031 | −0.052 | +0.021 | +0.042 | +0.178 |

**Registration.** Pre-registered before the first generation job, with two dated amendments (the mismatched-image co-primary, and a frame-equivalence fix after a rounding discrepancy was found and corrected from 2.221 to 0.030 maximum absolute pixel difference).

## K QWEN3-VL-8B

**Checkpoint and environment.** `Qwen/Qwen3-VL-8B-Instruct` (Qwen Team, 2025b), revision `0c351dd0`, 8,767,123,696 parameters, `bfloat16`, one RTX 4090. This experiment ran under a different interpreter from the other architectures (`torch 2.11.0`, same `transformers 5.14.1`) because the base interpreter has neither PIL nor torchvision, so no vision-language processor loads in it.

**Template and cut.** The prompt template is the processor’s own `apply_chat_template` output, frozen to disk and byte-compared by every later stage. It carries *no* system turn, unlike QWEN2.5-VL’s. The realised census cut is 0.205. Free-running continuations hit the 256-token

cap on 99.2% of images, so the per-cell budget is near-constant rather than image-keyed; this cut is therefore not comparable to the 0.230 and 0.460 of the other architectures.

**Harness gate, calibrated against our own lead model.** Before any endpoint was generated, six gates ran. The decisive one exploits greedy determinism: cut the model’s own caption at the census cut, splice it back as an assistant-turn prefill, and require it to re-emit the remainder. A broken template cannot pass this; a merely terse model can. QWEN3-VL-8B scores a median 16-gram agreement of 1.000, mean 0.958, with 92% of rows above threshold. The identical test run on LLAVA-1.5-7B in the same job scores 1.000, 0.945 and 92%. The bar therefore comes from this paper’s own lead model rather than from an invented threshold, and the verdict was “run endpoint”; the endpoint arms were held and released only on that verdict.

**Result.**  $\Delta_{\text{primary}} = +0.5828 [+0.5337, +0.6342]$ ,  $\Delta_{\text{ablated}} = +0.5231 [+0.4708, +0.5764]$ ,  $\Delta_{\text{image}} = +0.0598 [+0.0235, +0.0963]$ ,  $n = 1500$  cells per arm,  $B = 4000$  clustered on the image with one joint resample across all four arms. The interval excludes zero, so on this architecture part of the rise does need the image — but it lies entirely inside the  $\pm 0.10$  reference band, so about 90% of the rise still survives blinding. The contribution is confined to the five-scene condition ( $+0.0612 [+0.0392, +0.0841]$ ) and absent in the one-scene condition ( $+0.0014 [-0.0262, +0.0285]$ ).

**Not the QWEN2.5-VL failure.** The degeneracy screen passes on all four arms: median continuation 384 tokens in every arm, empty rates 0.000 to 0.003,  $\leq 3$ -token rates approximately zero. Supporting contrasts: length-matched  $\Delta_{\text{primary}} = +0.576$  and  $\Delta_{\text{ablated}} = +0.511$ ; object-matched  $+0.565$ ; empty-removed  $+0.584$ ; alternate seed  $+0.583$ .

**A screen we now know is insufficient.** Our degeneracy screens test length and short-output rate, not repetition. Measured after the fact, 90% and 83% of this model’s cells fall below a 0.5 distinct-8-gram ratio — but so do 67.5% of the lead model’s five-scene cells, so looping is a property of the design rather than of this architecture. The sighted and blind arms nonetheless loop at different rates (0.827 against 0.911) in exactly the condition carrying  $\Delta_{\text{image}}$ , and no looping-matched contrast was run. QWEN2.5-VL’s apparently clean ratio of 1.0000 is an artefact: four-token continuations contain no 8-gram.

**A mismatched-image arm (post hoc).** After the registered experiment, we ran the substitution of Appendix I on this model: the same 1,500 cells and prefixes, each image replaced by another of the 500 under the derangement with seed 20260919 (asserted bijective and self-pair-free, 3,000 rows). Every hard gate of the registered scorer passed; its grey-flag gate, which requires the ablated arm to be grey, was inverted to require that no arm is grey and that every substituted row was generated from a different image. The scorer’s positive controls reproduced both reference families ( $+0.48587$  against  $+0.4859$ ;  $+0.15981$  against  $+0.1598$ ), and the sighted contrast reproduces  $+0.583$ . The substituted contrast is  $+0.518 [+0.464, +0.571]$ , so  $\Delta_{\text{image}} = +0.065 [+0.031, +0.100]$  (ratio 0.89), against  $+0.060 [+0.023, +0.096]$  (ratio 0.90) under grey. The two ablations agree on this model. Its one-scene  $J$  is  $+0.049$  sighted and  $+0.050$  substituted ( $H$  0.83,  $F$  0.78): the model repeats most passage objects in the one-scene condition, with or without the correct image, so there is no one-scene level for a blank to destroy, and the image’s contribution sits entirely in the five-scene condition ( $+0.065 [+0.042, +0.087]$ ). The regime factor cannot be tested here: 94% of continuations reach the budget and only 14 cells are untruncated.

**Registration.** Pre-registered before the checkpoint existed locally, fixing the gate thresholds, the census rule, the arms, the estimand, and the text to be published for every outcome including the two adverse ones. The estimator was validated before any QWEN3-VL row existed by reproducing two published families from their frozen rows:  $+0.48587$  against a published  $+0.4859$ , and  $+0.15981$  against  $+0.1598$ .

## L BOTH ABLATIONS ON THE 3,500-IMAGE SAMPLE

Appendix B.5 controls the generation regime on the sighted arms by restricting to cells where neither condition reaches the 384-token budget, which drops the contrast from  $+0.472$  on all 10,500 cells

of the 3,500-image sample to +0.346. That control had no ablated arm. This appendix supplies two, the grey field and the mismatched real image of Appendix I, each run on all 10,500 cells, so every cell of the  $2 \times 2$  in Figure 1(b) is computed on one population. The whole extension is post hoc.

**Construction.** Each ablated generator is the sighted generator with one line changed. For grey, the target image is replaced by a flat mid-grey field of its own pixel dimensions, through the same mirrored code path used for the published blind arms. For the substitution, each image is replaced by another image of the same block under a derangement (seed 20260919, asserted bijective and self-pair-free, 875 images per block). Prefix token ids, cell construction, sort order and batch size are untouched. Gates ran before any row was kept, on all four blocks: for grey, a mirror gate requiring the blind code path with the substitution *off* to reproduce the sighted generator token-for-token (**32/32** identical in every block); for both, a prefix-drift gate against the published cells (**0** of 3,000 published prefixes drift) and a census gate on the cell md5s (**0** of 5,250 per block). Each arm is 21,000 rows. The derangement is reconstructed at scoring time and checked against the text: under the substitution, 54.6% of the objects a continuation names are in the substitute image’s ground truth and 18.7% in the target’s, against 19.0% and 54.1% sighted.

**Cell sets and positive controls.** “All cells” is every cell with both conditions in all three arms (10,500 cells, 3,500 images). “Untruncated” applies the *registered* regime rule to the sighted arms only — both sighted continuations below the budget — giving 2,089 cells on 1,649 images. On that set the sighted contrast is +0.3460, reproducing the registered value in Table 9 to four decimals, and the two ablated estimates reproduce their first scoring runs to four decimals. The rule does not constrain the ablated arms: inside the untruncated cells grey continuations still reach the budget in 2% (one scene) and 27% (five scenes) of cases, and substituted ones in 11% and 27%. On all cells the three arms reach it at similar rates (sighted 48% and 69%, substituted 47% and 69%), except grey’s one-scene arm (25%).

Table 36: The one-to-five-scene contrast under each ablation on the 3,500-image sample. The one- and five-scene columns give  $J = H - F$  in each condition. Intervals are  $B = 4000$  bootstrap replicates clustered on the image. Ratio is the ablated contrast over the sighted one; its intervals are percentile intervals of the replicate ratios.

| Cells                  | Arm        | One scene | Five scenes | Contrast | $\Delta_{\text{image}}$        | Ratio             |
|------------------------|------------|-----------|-------------|----------|--------------------------------|-------------------|
| All<br>(10,500)        | Sighted    | +0.176    | +0.647      | +0.472   | —                              | —                 |
|                        | Grey field | −0.028    | +0.447      | +0.475   | −0.003 [−0.025, +0.019]        | 1.01 [0.96, 1.05] |
|                        | Mismatched | +0.099    | +0.418      | +0.319   | +0.153 [+0.132, +0.175]        | 0.68 [0.64, 0.72] |
| Untruncated<br>(2,089) | Sighted    | +0.105    | +0.451      | +0.346   | —                              | —                 |
|                        | Grey field | −0.069    | +0.177      | +0.246   | +0.100 [+0.041, +0.163]        | 0.71 [0.56, 0.87] |
|                        | Mismatched | +0.057    | +0.221      | +0.164   | <b>+0.182</b> [+0.122, +0.245] | 0.47 [0.33, 0.62] |

**Result.** On all cells the 3,500-image sample reproduces both registered-sample answers — grey 1.01 against 1.03, substitution 0.68 against 0.67 — and 3,000 of its images played no part in either. Restricting to untruncated cells moves both (Table 36). Under grey the untruncated one-scene level falls below zero (−0.069 against +0.105 sighted), the level collapse the blank produces on the registered sample. The substitution keeps the one-scene level near the sighted one (+0.057), and it is the five-scene gain that shrinks.

**The substitute image is not information-free.** A real image contains common objects, and the passage’s present objects are enriched for them: 46–48% of present units on all cells (41–42% untruncated) name an object that the substitute image also contains, against 10–12% of absent units. For those units the substituted arm still has visual evidence. Excluding them from every arm raises  $\Delta_{\text{image}}$  to +0.241 [+0.211, +0.270] on all cells (ratio 0.46 [0.40, 0.52]) and to +0.266 [+0.192, +0.342] on untruncated cells (ratio 0.22 [0.02, 0.39]). The overlap therefore biases the substitution towards finding no image contribution, and the estimates in Table 36 are conservative. The exclusion changes the population scored, so we report it as evidence on the direction of the bias, not as a replacement estimate.

**What this does not settle.** The substitution uses one derangement. The untruncated cells are those that terminate naturally under *both* sighted conditions, a population that is easier to caption than the sample as a whole, and the rule does not force the ablated arms to terminate.